# Global Financial Integration and Productivity: The Role of Risk Sharing*

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**Abstract**

This paper studies how integration into global capital markets affects aggregate productivity through risk sharing and the allocation of capital. We develop a small open economy model where heterogeneous capitalists choose between producing and investing. Foreign participation in domestic equity markets shifts the country's aggregate risk abroad and lowers the hurdle rate for production. This raises investment and output but lowers aggregate TFP by drawing less productive capitalists into production. Greater domestic financing capacity instead raises TFP by allowing productive firms to expand and displace marginal producers. We validate these predictions using emerging-market equity liberalizations in the late 1980s and 1990s, which were followed by equity price booms, more listings, and slower TFP growth where booms were larger. The calibrated model shows that liberalization alone lowers TFP and raises output only temporarily, while liberalization with deeper domestic financing raises both.

*Keywords:* financial integration, cost of capital, productivity, small open economy.
*JEL codes:* F36, F41, F43, G15, O16, O47.

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# 1 Introduction

A central prediction of international macroeconomics is that integrating into global capital markets raises both investment and productivity through risk sharing. When foreign investors bear part of a country's aggregate risk, the cost of capital falls and investment rises. Better risk sharing also allows entrepreneurs to adopt riskier but more productive technologies and improves the allocation of capital across firms. The empirical evidence broadly supports the predictions for the cost of capital, investment, and output growth, but the evidence on productivity is mixed. In some countries, TFP growth rose after financial integration, while in others growth came from the accumulation of capital and labor with no change in TFP. Existing international macro-finance frameworks offer little explanation for why a similar fall in the cost of capital raises productivity in some countries but not in others.

This paper studies how global financial integration affects aggregate productivity through risk sharing and the selection of producers. Because the cost of capital is the hurdle rate that firms use as the threshold for production, improvements in risk sharing that reduce the cost of capital alter the set of firms in operation, and consequently the average productivity of firms in the economy. The model features two main channels that operate on this margin of producer selection. More foreign participation in domestic equity markets lowers the risk premium and the hurdle rate, drawing less productive firms into production. In contrast, an improved domestic financing capacity allows productive firms to operate at a larger scale, which pushes up the labor cost of production and crowds out the least productive firms. We first develop a small open economy model in which capitalists choose between producing and investing, firms raise outside equity from domestic and foreign investors, and equity prices determine who produces in equilibrium. We validate the model's predictions using the equity market liberalizations of emerging market economies in the late 1980s and 1990s, documenting three facts. Following liberalization, (i) foreign equity inflows increased; (ii) domestic equity prices rose sharply and more companies listed; and (iii) countries with larger increases in prices or listings experienced weaker aggregate TFP growth in subsequent years. Finally, we calibrate the model to the liberalization episodes and find that liberalization alone raises output temporarily but lowers aggregate TFP persistently, whereas liberalization combined with an expansion of domestic financing capacity raises both output and aggregate TFP.

To understand how financial integration affects productivity, we develop a small open economy model of occupational choice in which financial frictions and equity pricing jointly determine aggregate TFP. The model combines heterogeneous capitalists who face financing constraints (Lucas, 1978; Evans and Jovanovic, 1989; Buera et al., 2011; Moll, 2014) with portfolio choice and equilibrium equity pricing under limited foreign participation (Stulz, 1981; Errunza and Losq, 1985). In each period, capitalists draw an idiosyncratic productivity and choose whether to produce or invest. Producers combine capital and labor and finance capital with their own wealth and outside equity. However, the outside equity they can raise relative to their wealth is limited, which we refer to as financing capacity. Investors hold a portfolio of domestic equity and a safe bond that pays an exogenous world interest rate. Domestic equity is a claim on firms' operating profits, which are

exposed to a common country-level shock, and on their undepreciated capital. Foreign investors, who are diversified across countries, can hold up to a given share of domestic equity. Aggregate TFP is determined by the productivity of the capitalists who choose to produce.

Equity prices determine the productivity threshold that separates producers from investors. Capitalists choose the occupation with higher expected utility, considering both the returns and risk of each occupation. A producer's return rises with their productivity, as well as with the price at which they can sell outside equity, as higher valuations allow the producer to raise the same amount of funds while giving up a smaller share of the firm's payoffs. An investor earns the return on domestic equity and the world interest rate. In equilibrium, capitalists above a productivity threshold produce and those below it invest. Because domestic investors are the marginal holders of domestic equity, their willingness to bear the country's aggregate risk sets the risk premium embedded in equity prices. For a given world interest rate, a lower premium raises firm valuations and lowers the productivity threshold. We model an equity market liberalization as an exogenous increase in the share of domestic equity held by foreign investors. Because foreign investors diversify the country's risk across their global portfolios, domestic investors bear less aggregate risk. Consequently, they require a lower premium and equity prices rise. Capitalists who previously invested now choose to produce, expanding the capital stock and output. As the entrants are less productive than incumbents, the average productivity among producers falls. This selection effect links changes in equity prices and the number of producers around liberalizations to subsequent changes in aggregate productivity.

The model implies that better international risk sharing need not improve the allocation of capital when financing constraints limit producers' scale. With financing capacity unchanged, a lower cost of capital expands investment through entry of less productive capitalists rather than through expansion by the incumbent producers. As a result, capital inflow raises output on impact but lowers aggregate TFP. Entry also creates a pecuniary externality, as capitalists take wages as given and do not internalize that having more producers raises wages and lowers the profits of the incumbent producers. Income shifts toward hand-to-mouth workers, who do not save, while foreign shareholders receive a part of the equity returns that would otherwise accrue to domestic capitalists. When these transfers outweigh the capitalists' gains from additional production, domestic saving falls. This results in weaker wealth accumulation, and can leave long-run output below its pre-liberalization level. In contrast, relaxing financing constraints raises both output and productivity. When producers can operate more capital relative to their own wealth, expanding firms bid up wages for a fixed pool of workers, and the least productive producers exit. Greater financing capacity can therefore offset or reverse the productivity loss from improved risk sharing.

We validate these predictions by empirically examining equity market liberalizations, in which countries opened their domestic equity markets to foreign investors. Among various forms of financial integration, equity market liberalization operates most directly through equity pricing and risk sharing, whereas FDI, for example, affects productivity mainly through technology diffusion. Using official liberalization dates for 28 emerging market economies between 1986 and 1997 (Bekaert

and Harvey, 2000; Bekaert et al., 2005) and a staggered difference-in-differences design (Callaway and Sant'Anna, 2021), we document three facts that map into the model. First, liberalization raised foreign ownership of domestic equity. Four years after liberalization, foreign holdings of domestic portfolio equity were 2.8 percent of GDP higher than in countries that had not yet liberalized. This reflects higher foreign holdings of domestic equity rather than changes in residents' foreign holdings. Second, consistent with a lower risk premium and increased entry of producers, domestic equity prices rose 67 percent around liberalization, and listed companies per capita rose 51 percent over the following five years. Third, consistent with the selection effect, countries with larger increases in equity prices or listings experienced slower TFP growth. A 10 percent larger price increase is associated with 0.18 percentage points lower annual TFP growth over the following decade. Taken together, these facts match the key model mechanisms. In response to financial integration, foreign investors absorb domestic risk, equity prices rise, new producers enter, and productivity falls.

In the data, liberalizations coincided with expansions in domestic financing capacity. In the model, these two forces have counteracting effects on aggregate productivity. Liberalization alone lowers aggregate TFP by drawing less productive capitalists into production, whereas greater financing capacity raises aggregate TFP by allowing productive firms to expand. We use the calibrated model to separate the two forces, comparing a scenario with liberalization alone with a liberalization scenario accompanied by the expansion in financing capacity observed in these episodes. With financing capacity fixed, liberalization raises output by 0.3 percent on impact as foreign capital finances the entrants, but lowers aggregate TFP by 0.15 percent. Over time, part of the premium that domestic capitalists previously earned accrues to foreign shareholders. The resulting decline in domestic wealth deepens the TFP loss to 0.18 percent and lowers long-run output, so the initial output gain is temporary. When liberalization is combined with the financing expansion, aggregate TFP rises by 0.28 percent and output by 1.7 percent on impact, and the gains persist at 0.23 and 1.4 percent in the long run. Reallocation toward productive firms more than offsets the selection effect. When a liberalization and a financing expansion lower the risk premium by the same amount, the former lowers TFP by 0.15 percent while the latter raises it by 0.37 percent. A fall in the cost of capital is therefore not a sufficient statistic for its effect on productivity. Its effect depends on whether the fall reflects foreign investors absorbing domestic risk or productive firms experiencing an expansion in financing capacity. Consequently, opening to foreign investors raises productivity when it is combined with reforms that deepen domestic financial markets.

**Related literature and contribution.** This paper is related to three strands of literature. First, we contribute to the literature on financial integration and economic growth. The aggregate gains from international capital mobility appear small (Gourinchas and Jeanne, 2006), and capital has tended to flow toward countries with slower productivity growth (Prasad et al., 2007; Gourinchas and Jeanne, 2013). Models with financial frictions attribute this pattern to cross-country differences in financial development (Mendoza et al., 2009; Song et al., 2011; Buera and Shin, 2017). Theory also predicts that diversification allows investors to hold riskier and more productive projects

(Obstfeld, 1994; Acemoglu and Zilibotti, 1997). Empirical studies of equity market liberalizations find that they lowered the cost of equity (Bekaert and Harvey, 2000; Henry, 2000a; Chari and Henry, 2004), attracted equity inflows (Bekaert et al., 2002), and raised investment and growth (Henry, 2000b; Chari and Henry, 2008; Bekaert et al., 2005), but their effects on productivity are mixed and depend on domestic financial development (Bonfiglioli, 2008; Levchenko et al., 2009; Bekaert et al., 2011a; Kose et al., 2009). Firm-level studies find that liberalizations improved the allocation of capital among incumbent firms (Larrain and Stumpner, 2017; Varela, 2018; Bau and Matray, 2023) and that growth came through the expansion of existing firms rather than through entry (Gupta and Yuan, 2009). This evidence shows that domestic financial development is a key determinant of the gains from integration. However, there is no conclusive evidence on the specific channels underlying this result. The most closely related paper is Aoki et al. (2010), who show that the effects of capital account liberalization depend on domestic financial development when integration operates through debt and the interest rate. Our paper focuses on a similar channel, but with a focus on risk sharing. Foreign ownership of domestic equity lowers the risk premium and draws less productive capitalists into production. In contrast, greater domestic financing capacity allows productive firms to expand and displace marginal producers. The interaction between these two forces explains why the productivity gains from liberalization depend on domestic financial development.

Second, the paper relates to the literature on financial frictions, misallocation, and aggregate productivity. Distortions in the allocation of capital across producers lower aggregate TFP (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009; Bartelsman et al., 2013), and financial frictions are a leading source of such distortions (Banerjee and Moll, 2010; Buera et al., 2011; Midrigan and Xu, 2014; Moll, 2014). Our model builds on the occupational choice framework of Lucas (1978), Evans and Jovanovic (1989), and Buera et al. (2011), and our financing capacity corresponds to the collateral constraint in Evans and Jovanovic (1989) and Buera et al. (2011), applied to outside equity. A separate literature emphasizes selection on the extensive margin (Hopenhayn, 1992; Melitz, 2003), and firm-level evidence indicates that the cost of finance affects both the number and the productivity of entrants (Bertrand et al., 2007; Kerr and Nanda, 2009; Ateş and Saffie, 2021; Saffie et al., 2026). In particular, Fama and French (2004) show that the surge in new listings after 1980 was concentrated among firms with weaker profitability and lower survival rates, and link it to a decline in the cost of equity. Most of this work studies how frictions distort the allocation of capital among firms facing a given cost of funds, although recent work shows that risk premia account for a large share of the dispersion in the marginal product of capital (David et al., 2022). Our paper shows that the pricing of aggregate risk, not only how risk is distributed across firms, affects aggregate TFP by determining which firms choose to produce. To do so, we embed equilibrium equity pricing under limited foreign participation, as in models of international asset pricing with segmented markets (Stulz, 1981; Errunza and Losq, 1985), in an occupational choice economy.

Third, the paper relates to work on interest rates, asset prices, and productivity. Low interest rates and lender forbearance can reduce productivity by keeping unproductive firms alive (Caballero

et al., 2008; Schivardi et al., 2022; Acharya et al., 2022), by raising market concentration (Liu et al., 2022), and by shifting resources toward less productive firms and sectors (Reis, 2013; Benigno et al., 2015; Gopinath et al., 2017). Asriyan et al. (2025) study a mechanism close to ours, where a fall in the interest rate draws less productive entrepreneurs into investment and, by raising the price of capital goods, crowds out more productive entrepreneurs. These papers operate through the safe rate or the terms of credit, and do not focus on the risk premium. Real yields on government bonds have fallen over recent decades (Del Negro et al., 2019), while the cost of equity has not (Caballero et al., 2017; Farhi and Gourio, 2018). Equity markets also remain segmented and home-biased, whereas safe assets trade in a deep global market (Bekaert et al., 2011b; Coeurdacier and Rey, 2013; Maggiori et al., 2020). Relative to this work, we show that the price of risk is a distinct determinant of producer selection. In our model, the hurdle rate for production maps one-to-one into an equity price and equals the user cost of capital marked up by the price of risk. This margin matters most in an open economy, where the risk-free rate is set globally. In a closed economy, a lower risk premium would raise the demand for risky assets and push up the risk-free rate, largely offsetting the effect on the hurdle rate. Equity market liberalization therefore affects the hurdle for production through a margin distinct from the fall in safe rates emphasized in existing work.

**Roadmap.** The rest of the paper proceeds as follows. Section 2 presents the model, and Section 3 derives its main results under log utility. Section 4 discusses the key mechanisms of the model. Section 5 documents the stylized facts, and Section 6 presents the quantitative analysis. Section 7 concludes.

## 2 The Model

In this section, we develop a small open economy model to study how financial integration shapes aggregate productivity through risk sharing and selection of producers. The model builds on the occupational choice framework of Buera et al. (2011) and Buera and Moll (2015) where capitalists can decide to be a producer or an investor, depending on their idiosyncratically drawn productivity level. We make two key modifications. First, we introduce aggregate risk that is not fully diversifiable, and second, we add portfolio choice and equilibrium equity pricing that determines the cost of capital and the occupational choice of capitalists. A capitalist produces when running a firm yields higher expected utility than holding an optimal financial asset portfolio yields. The equity prices determine the productivity threshold that separates producers from investors. Foreign investors affect equity prices by holding at most a share $\omega$ of domestic equity, and we model the opening of the equity market as a rise in $\omega$ at a given world rate.

### 2.1 Environment, timing, and the aggregate state

**Capitalists and workers.** Time is discrete, $t = 0, 1, 2, \ldots$, and a single good is used for consumption and as capital. The economy has a unit mass of capitalists, indexed by $i$, with CRRA

preferences and each maximizing expected discounted utility over consumption streams

$$\mathbb{E}_0 \sum_{t \geq 0} \beta^t \frac{c_{i,t}^{1-\gamma}}{1-\gamma},$$

where $\gamma \geq 1$ is relative risk aversion and $\beta \in (0,1)$ the discount factor. Each period a capitalist chooses an occupation $o_{i,t} \in \{P, I\}$. A producer ($P$) runs a firm. An investor ($I$) holds the two traded assets, a domestic equity index and a world bond. Workers are hand-to-mouth and supply one unit of labor inelastically, consume the wage income, and do not hold any assets.

**Productivity and aggregate risk.** At the start of every period each capitalist draws a productivity $z_{i,t}$ from a Pareto distribution with tail parameter $\eta > 1$,

$$\Pr(z_{i,t} \leq z) = \Phi(z) \equiv 1 - z^{-\eta}, \qquad z \geq 1, \tag{1}$$

independently across capitalists and over time. There is aggregate risk that affects all producers and is modeled as a multiplicative productivity shock $\varepsilon_t$ that is independent of the idiosyncratic productivity $z_{i,t}$ and independent over time. Thus, the effective productivity of a producer is $\tilde{z}_{i,t} = e^{\varepsilon_t} z_{i,t}$, where $\varepsilon_t \sim \mathcal{N}(\mu, \sigma^2)$, with $\sigma > 0$. The distribution of the aggregate shock is known to capitalists when they make their occupational choice, but the realization happens along with the production process.

**Timing.** Within period $t$ (Figure 1), each capitalist, knowing its draw and the prices of the period, chooses an occupation before the shock. Producers then sell outside equity $E_{i,t}$, and investors choose the share $\theta_t$ of their wealth held in the index (Section 2.2). After the shock $\varepsilon_t$, the labor market clears at the wage $w_t$, and each capitalist earns the gross return $R^o_{i,t}$ of its occupation and splits the cash on hand $m_{i,t} = R^o_{i,t} a_{i,t}$ between consumption and next period's net worth, $c_{i,t} + a_{i,t+1} = m_{i,t}$. Unless stated otherwise, $\mathbb{E}$ denotes the expectation over $\varepsilon_t$ under its physical probabilities, given the information available when occupations are chosen. We drop time subscripts when no confusion arises; the financing capacity $\xi$, the degree of equity-market integration $\omega$, and the world rate $R^{f*}$ (Section 2.2) are constant.

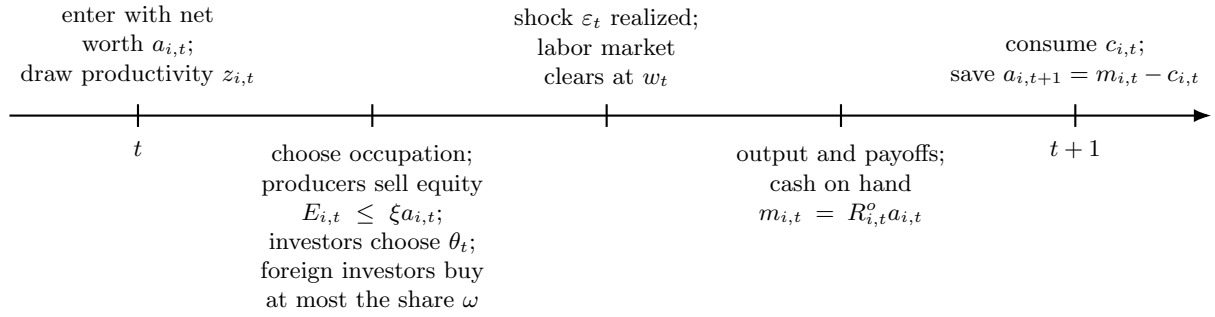



Figure 1: Timing within period $t$

## 2.2 Production, financing, and financial assets

Producers run firms with constant returns to scale and finance them with their own net worth and with outside equity. Investors hold that equity through a domestic index and trade a world bond, and foreign investors may hold part of the index.

**Technology and labor.** A producer with productivity $z_{i,t}$ who operates capital $k_{i,t}$ and hires labor $l_{i,t}$ produces

$$y_{i,t} = f(\tilde{z}_{i,t}, k_{i,t}, l_{i,t}) = (\tilde{z}_{i,t} k_{i,t})^{\alpha} \, l_{i,t}^{1-\alpha}, \qquad \alpha \in (0,1). \tag{2}$$

Labor is hired after the shock at the competitive wage $w_t$; thus the wage is ex-ante stochastic and depends on the shock. The operating profit of a producer is

$$\Pi_{i,t}(\tilde{z}_{i,t}, k_{i,t}) = \max_{l_{i,t}} (\tilde{z}_{i,t} k_{i,t})^{\alpha} \, l_{i,t}^{1-\alpha} - w_t l_{i,t}.$$

The optimal choice of labor makes operating profit linear in capital:

$$\Pi_{i,t}(\tilde{z}_{i,t}, k_{i,t}) = \pi_t \, \tilde{z}_{i,t} k_{i,t}, \qquad \pi_t \equiv \alpha \left( \frac{1-\alpha}{w(\varepsilon_t)} \right)^{(1-\alpha)/\alpha}. \tag{3}$$

Consider each unit of capital invested in a firm operated by a producer with productivity $z$. Capital depreciates at the rate $\delta \in (0,1)$, and the return on a unit of capital is the sum of its operating profit and its undepreciated capital

$$R^K(z_{i,t}) = \pi_t \, e^{\varepsilon_t} z_{i,t} + 1 - \delta = z_{i,t} \, D_t + 1 - \delta \tag{4}$$

where the aggregate factor $D_t \equiv \pi_t e^{\varepsilon_t}$ is common across all producers.

**Financing.** A producer finances capital with own net worth $a_{i,t}$ and by selling outside equity. Let $q(z_{i,t})$ denote the market value of a unit of capital run by a producer with productivity $z_{i,t}$, and thus his firm with capital $k_{i,t}$ would be worth $q(z_{i,t}) \, k_{i,t}$. The firm is valued with the domestic investors' one-period stochastic discount factor (SDF) $M^{\Lambda}$,

$$q(z_{i,t}) = \mathbb{E}\big[ M^{\Lambda} R^K(z_{i,t}) \big]. \tag{5}$$

The producer raises additional capital via equity $E_{i,t}$ by selling the fraction $1 - \iota_{i,t}$ of the firm. In other words, the capital raised today is determined by the value of the firm and the fraction of the firm sold:

$$E_{i,t} = (1 - \iota_{i,t}) \, q(z_{i,t}) \, k_{i,t} \tag{6}$$

and the total operating capital is $k_{i,t} = a_{i,t} + E_{i,t}$. The equity issuance is limited by the country-level financing constraint

$$E_{i,t} \leq \xi a_{i,t}, \tag{7}$$

where the financing capacity $\xi \geq 0$ is the amount of funds a producer can raise per unit of own net worth. We think of $\xi$ as summarizing country-level characteristics such as investor protection and institutional quality that affect the depth of domestic financial markets. A higher $\xi$ lets producers operate at a larger scale.

The producer keeps the fraction $\iota_{i,t}$ of the firm's payoff, so the gross return on his net worth is

$$R^P(z_{i,t}) = \frac{\iota_{i,t}\, R^K(z_{i,t})\, k_{i,t}}{a_{i,t}} = \left[1 + \frac{E_{i,t}}{a_{i,t}} \left(1 - \frac{1}{q(z_{i,t})}\right)\right] R^K(z_{i,t}). \tag{8}$$

Since $R^K(z_{i,t}) > 0$ in every state, a producer raises as much equity as the financing constraint allows when $q(z_{i,t}) > 1$ and issues no outside equity when $q(z_{i,t}) < 1$. Proposition 2.3 shows that in equilibrium, if a capitalist chooses to produce, the value of a unit of capital is at least one, $q(z_{i,t}) \geq 1$, and each producer issues equity until the financing constraint binds, $E_{i,t} = \xi a_{i,t}$. This implies that we can write the return on a producer's net worth as

$$R^P(z_{i,t}) = R^K(z_{i,t}) + \xi \left[R^K(z_{i,t}) - \frac{R^K(z_{i,t})}{q(z_{i,t})}\right]. \tag{9}$$

On the one hand, the return on a producer's net worth is raised by the financing capacity $\xi$. On the other hand, raising equity is costly: outside shareholders earn the return $R^K(z_{i,t})/q(z_{i,t})$ per unit they pay, and we refer to this ratio as the firm's cost of equity.

**The equity index.** Producers sell their outside equity to domestic and foreign investors through a domestic equity index, which pools all the outside equity claims. In other words, the domestic equity index is a claim to the aggregate operating profits of all producers in the country. The total payoff and price of the equity index are

$$\Pi^e = \int_{i\in\mathcal{P}} (1 - \iota_{i,t})\, k_{i,t}\, R^K(z_{i,t})\, di = \int_{i\in\mathcal{P}} E_{i,t}\, \frac{R^K(z_{i,t})}{q(z_{i,t})}\, di,$$
$$Q^e = \int_{i\in\mathcal{P}} (1 - \iota_{i,t})\, q(z_{i,t})\, k_{i,t}\, di = \int_{i\in\mathcal{P}} E_{i,t}\, di,$$

where $\mathcal{P}$ is the set of producers.

**Proposition 2.1** (Equity index return)**.** *Let $\xi > 0$. Suppose every producer sells equity in proportion to net worth, $E_{i,t} = \xi a_{i,t}$, and equity is valued with the SDF $M^\Lambda$. Then the return on the equity index is the average cost of equity of producers, and it is priced by the investors' SDF:*

$$R^e(\varepsilon_t) = \frac{\Pi^e(\varepsilon_t)}{P^e} = \mathbb{E}_z\left[\frac{R^K(z_{i,t}, \varepsilon_t)}{q(z_{i,t})} \;\middle|\; i \in \mathcal{P}\right], \qquad \mathbb{E}_\varepsilon\left[M^\Lambda R^e(\varepsilon_t)\right] = 1, \tag{10}$$

*where $\mathbb{E}_z$ averages over the cross section of producers' productivity distribution and $\mathbb{E}_\varepsilon$ over the aggregate shock.*

The proof is in Appendix A.3.

**The world bond and investors.** Domestic investors allocate their wealth between the domestic equity index and the world risk-free bond. The world bond is a one-period safe asset that pays the gross world rate $R^{f*}$, which is exogenous to the small open economy. The domestic equity index is a risky asset that pays the gross return $R^e(\varepsilon_t)$ of Proposition 2.1. Investors choose the share $\theta_t$ of their wealth to invest in the domestic equity index and invest the rest, $1-\theta$, in the world bond. The gross return on an investor's net worth is thus

$$R^I(\varepsilon_t) = R^{f*} + \theta\big(R^e(\varepsilon_t) - R^{f*}\big)$$

where $\theta$ is chosen to maximize expected utility. The optimal portfolio allocation is determined in the same way as in the standard consumption-based asset pricing model, where the SDF is connected to the marginal utility of investors' consumption, and the first-order condition is

$$\mathbb{E}\left[M^\Lambda(\theta^*)\left(R^e(\theta_t^*) - R_t^{f*}\right)\right] = 0. \tag{11}$$

**Foreign investors.** Foreign investors hold diversified world portfolios in which the country's aggregate shock is diversifiable, so they value a claim to a risky domestic payoff at its expected value under the physical probabilities, discounted at the world risk-free rate. Their holdings are capped: they may hold at most the share $\omega_t \in [0,1)$ of the market value of the index, which they buy at the domestic price. We consider $\omega_t$ the degree of equity-market integration, and an opening of the equity market is a rise in $\omega_t$ at a given world rate. Foreign holdings are voluntary and we assume that foreign holdings are non-negative.

## 2.3 Dynamic occupational choices

**The capitalist's problem.** A capitalist chooses an occupation and a portfolio before the shock is realized and then consumes afterwards. Given fixed labor supply, labor cost of production is determined by both the aggregate shock and capitalists' aggregate wealth. This implies that the aggregate wealth of capitalists is the state variable which affects capitalists' occupational choice. Let $W'$ denote next period's aggregate wealth, which is a function of current wealth $W$ and of the shock $\varepsilon$, and let $W' = G(W,\varepsilon)$ denote the equilibrium law of motion for aggregate capitalist wealth. It maps beginning-of-period wealth $W$ and the current aggregate shock $\varepsilon$ into next period's beginning-of-period wealth. Each capitalist takes this transition as given; in equilibrium, $G(W,\varepsilon)$ equals aggregate saving after the shock and before the start of the next period. A capitalist with

net worth $a$ and productivity $z$ solves

$$\begin{aligned} V(a,z;W) &= \max_{o\in\{P,I\}} \max_{\mathbf{x}} \mathbb{E}_{\varepsilon}\Big[U\big(R^{o}(z,\varepsilon;\mathbf{x})\,a;\, G(W,\varepsilon)\big)\Big], \\ U(m;W') &= \max_{0\le a'\le m} \Big\{u(m-a') + \beta\,\mathbb{E}_{z'}\big[V(a',z';W')\big]\Big\}, \end{aligned} \tag{12}$$

where $u(c) = c^{1-\gamma}/(1-\gamma)$, $\mathbb{E}_{z'}$ is the expectation over next period's draw of productivity level and $\mathbf{x}$ is the vector of financial or production choices given an occupation. $U(m;W')$ is the value function evaluated at end-of-period with cash on hand $m$ when next period's aggregate wealth is $W'$.

The next proposition shows that the occupational choice can be expressed in terms of comparing certainty equivalents of the returns of the two occupations, and that the occupational choice does not depend on net worth.

**Proposition 2.2** (Occupational choice rule)**.** *Assume $\gamma \geq 1$, and $\beta\big(R^{f*}\big)^{1-\gamma} < 1$.*

*(a) If $\gamma > 1$, the value of end-of-period cash on hand $m$, when next period's aggregate wealth is $W'$, is $U(m;W') = \Lambda(W')\,m^{1-\gamma}/(1-\gamma)$, with $\Lambda(W') = \big(1-s(W')\big)^{-\gamma}$, where every capitalist saves the fraction $s(W')$ of cash on hand. When occupations and portfolios are chosen, $W'$ depends on the shock, so $\Lambda_t \equiv \Lambda(W_{t+1})$ is a random variable common to all capitalists. For a random variable $Y$ let $\mathbb{E}^{\Lambda}[Y] \equiv \mathbb{E}[\Lambda_t Y]/\mathbb{E}[\Lambda_t]$, and for a positive random gross return $R$ let*

$$\mathrm{CE}^{\Lambda}(R) \equiv \big(\mathbb{E}^{\Lambda}\left[R^{1-\gamma}\right]\big)^{1/(1-\gamma)}. \tag{13}$$

*Then the value of occupation $o \in \{P,I\}$ to a capitalist with net worth $a$ and productivity $z$ is*

$$V^{o}(a,z;W) = \mathbb{E}[\Lambda_t]\,\frac{\mathrm{CE}^{\Lambda}\big(R^{o}(z)\big)^{1-\gamma}}{1-\gamma}\,a^{1-\gamma}, \tag{14}$$

*where $R^{o}(z)$ embeds the financial choice that maximizes $\mathrm{CE}^{\Lambda}\big(R^{o}(z;x)\big)$ over $x$, and the capitalist produces if and only if*

$$\mathrm{CE}^{\Lambda}\big(R^{P}(z)\big) \geq \mathrm{CE}^{\Lambda}\big(R^{I}\big). \tag{15}$$

*(b) The occupational choice is independent of individual net worth, and has no option value. All capitalists save the same fraction of cash on hand from their end-of-period consumption-saving problem.*

*(c) With log utility, $U(m;W') = \ln m/(1-\beta) + h(W')$, every capitalist saves the fraction $\beta$, the weights drop out, and the capitalist produces if and only if $\mathbb{E}\big[\ln R^{P}(z)\big] \geq \mathbb{E}\big[\ln R^{I}\big]$.*

The proof is in Appendix A.1.

Since productivity is redrawn every period and the aggregate shock is i.i.d., today's occupation choice leaves tomorrow's opportunities unchanged and does not have any option value.

**Investors' portfolio and the SDF.** An investor chooses the equity share $\theta$ that maximizes $\mathrm{CE}^{\Lambda}(R^I)$, with $R^I$ given in Section 2.2. With $\gamma > 1$ this is the same as maximizing the expected weighted utility $\mathbb{E}\big[\Lambda_t (R^I)^{1-\gamma}\big]/(1-\gamma)$. The objective is the same for every investor and strictly concave in $\theta$, so the optimal share is unique and common to all investors. The investors' stochastic discount factor (SDF) is thus

$$M^{\Lambda} \equiv \frac{\Lambda_t \, (R^I)^{-\gamma}}{\mathbb{E}\left[\Lambda_t \, (R^I)^{1-\gamma}\right]}. \tag{16}$$

The interior first-order condition of portfolio choice implies standard asset pricing relations for the domestic equity index and the world bond.

$$\mathbb{E}\big[M^{\Lambda}\big] = \frac{1}{R^{f*}}, \qquad \mathbb{E}\big[M^{\Lambda} R^e\big] = 1.$$

## 2.4 Asset prices and occupational selection

Given the investors' SDF, we can value every firm or risky equity claim in the economy, and express every return in terms of two prices, the world risk-free rate and the price of the aggregate risk factor $D(\varepsilon)$.

**The price of the aggregate risk factor.** A unit of capital run by a producer with productivity $z$ pays $R^K(z) = zD + 1 - \delta$ by (4). It is valued with the SDF (16) and the value of a unit of capital operated by a producer with productivity $z$ is therefore

$$q(z) = \mathbb{E}\left[M^{\Lambda} R^K(z)\right] = z\,p + \frac{1-\delta}{R^{f*}}, \qquad p \equiv \mathbb{E}\left[M^{\Lambda} D(\varepsilon)\right], \tag{17}$$

where $p$ is the price of the aggregate risk factor $D(\varepsilon)$. Its return is $R^D \equiv D/p$, and we define the risk premium

$$\lambda \equiv \frac{\mathbb{E}[R^D]}{R^{f*}} = \frac{\mathbb{E}[D]}{p\,R^{f*}} \tag{18}$$

The expected excess return on this aggregate risk is $\mathbb{E}[R^D] - R^{f*} = R^{f*}(\lambda - 1)$. Thus, the pair $(R^{f*}, \lambda)$ is the cost of capital of this economy: the world bond market sets the risk-free rate, while domestic investors set the risk premium.

**Risky shares and exposure.** The value (17) of a unit of capital has a risky part, $zp$, the value of its operating profit, and a safe part, $(1-\delta)/R^{f*}$, the value of its undepreciated capital. Let

$$\psi(z) \equiv \frac{z\,p}{q(z)} \in (0,1) \tag{19}$$

denote the firm's risky share. Dividing (4) by (17), the cost of equity of a producer with productivity $z$ is

$$\frac{R^K(z)}{q(z)} = \big(1 - \psi(z)\big) R^{f*} + \psi(z)\, R^D, \tag{20}$$

so each firm's equity is a portfolio of the world bond and the claim to the aggregate risk factor $D$ with weight $\psi(z)$, and a more productive firm has a larger risky share. The index return is the weighted average of producers' costs of equity, so by (20)

$$R^e = \big(1-\bar{\psi}\big)R^{f*} + \bar{\psi}\, R^D, \qquad \bar{\psi} \equiv \frac{\int_{\mathcal{P}} E_i\, \psi(z_i)\, di}{\int_{\mathcal{P}} E_i\, di}. \tag{21}$$

The expected excess return on the index is

$$\mathbb{E}\left[R^e\right] - R^{f*} = \bar{\psi}\left(\mathbb{E}[R^D] - R^{f*}\right), \tag{22}$$

and investors, who hold the share $\theta$ of their wealth in the index, earn

$$R^I = R^{f*} + \vartheta\left(R^D - R^{f*}\right), \qquad \vartheta \equiv \theta\bar{\psi}, \tag{23}$$

so their wealth is exposed to the aggregate factor in the proportion $\vartheta$.

Since every payoff in the economy combines the world bond and the claim to the aggregate risk factor $D$, every asset or portfolio can be summarized by its exposure to the aggregate factor. An asset position with exposure $x$ allocates the fraction $x$ in the claim to $D$ and the rest in the bond, and it pays $R^{f*}g_x$, where $g_x \equiv 1 - x + x\lambda X$, where $X \equiv e^{\alpha\varepsilon}/\mathbb{E}[e^{\alpha\varepsilon}]$. Define

$$\mathcal{C}^{\Lambda}(x;\lambda) \equiv \mathrm{CE}^{\Lambda}\big(g_x\big), \qquad x \in [0,1], \tag{24}$$

the certainty equivalent of an asset claim with exposure $x$ to aggregate risk, and write $\mathcal{C}^{\Lambda}(x)$ for the certainty equivalent given the equilibrium risk premium. Since $\mathrm{CE}^{\Lambda}$ is homogeneous of degree one, Equation (23) gives investors $\mathrm{CE}^{\Lambda}(R^I) = R^{f*}\,\mathcal{C}^{\Lambda}(\vartheta)$.

**Assumption A0 (Interior exposure).** *Investors' exposure to the aggregate factor is below one,* $\vartheta = \theta\bar{\psi} < 1$.

This assumption imposes that domestic investors cannot borrow beyond their natural borrowing limit. Their portfolio return is $R^I = (1-\vartheta)R^{f*} + \vartheta R^D$, and since the aggregate shock is lognormally distributed, it can take values arbitrarily close to zero, which implies $R^D$ can be arbitrarily close to zero after a large adverse shock. At $\vartheta > 1$, sufficiently adverse shocks would make investors' net worth negative. The assumption still allows borrowing abroad to finance equity holdings greater than investors' own wealth, since equity includes the safe payoff from undepreciated capital, and the risky share satisfies $\bar{\psi} < 1$. Thus an equity share $\theta > 1$ in the domestic investor portfolio allocation can remain consistent with $\vartheta = \theta\bar{\psi} < 1$.

**The productivity cutoff.** Let $z_q \equiv \big[1-(1-\delta)/R^{f*}\big]/p$ denote the productivity at which a unit of capital is worth its replacement cost, $q(z_q) = 1$ by (17).

**Proposition 2.3** (The productivity cutoff, the occupational wedge, and the cutoff identity)**.** *Let* $\gamma \geq 1$, $\xi > 0$, *and* $R^{f*} > 1-\delta$. *Suppose that a positive mass of capitalists invests, that investors*

*hold an interior exposure, $\vartheta < 1$ (Assumption A0), that every firm is valued with the investors' SDF, so that (17) holds for every $z$, and that the continuation weights $\Lambda_t$ are a positive, bounded function of the shock. Then:*

*(a)* Cutoff rule. *The certainty equivalent of producing, $\mathrm{CE}^{\Lambda}\big(R^P(z)\big)$, is continuous, strictly increasing and unbounded in $z$, and there is a cutoff $\bar{z} > 1$ such that the capitalists with $z \geq \bar{z}$ choose to be producers, and the marginal producer is indifferent between producing and investing:*

$$\mathrm{CE}^{\Lambda}\big(R^P(\bar{z})\big) = \mathrm{CE}^{\Lambda}\big(R^I\big). \tag{25}$$

*(b)* Occupational wedge. *The marginal producer's capital is worth its replacement cost plus the occupational wedge $\Delta \equiv q(\bar{z}) - 1$, which satisfies*

$$\Delta = \frac{1}{1+\xi}\left[\frac{\mathcal{C}^{\Lambda}(\vartheta)}{\mathcal{C}^{\Lambda}\big(\psi(\bar{z})\big)} - 1\right] \geq 0, \tag{26}$$

*with $\Delta = 0$ if and only if $\psi(\bar{z}) = \vartheta$. Hence every producer has $q(z) \geq 1$, and raising the equity capital to the full financing capacity, $E = \xi a$, is optimal for every producer.*

*(c)* Cutoff identity. *The marginal producer's risky share is*

$$\psi(\bar{z}) = 1 - \frac{1-\delta}{R^{f*}(1+\Delta)}, \tag{27}$$

*and the cutoff satisfies*

$$\underbrace{\bar{z}\,\mathbb{E}[D]}_{\text{marginal producer's expected operating payoff}} = \underbrace{\lambda}_{\text{risk premium}} \times \Big( \underbrace{R^{f*} + \delta - 1}_{\text{user cost}} + \underbrace{R^{f*}\Delta}_{\text{wedge}} \Big). \tag{28}$$

The proof is in Appendix A.4.

The wedge exists because investors are able to choose their optimal exposure to aggregate risk via portfolio choices, while producers can only hold their own firm and bear the firm's risky profit share $\psi(z)$. The wedge therefore compensates the marginal producer for a mismatch between the exposure to aggregate risk that running a firm imposes and the exposure that he would have chosen if investing in financial assets. An investor chooses his exposure and holds $\vartheta$, the maximizer of $\mathcal{C}^{\Lambda}(\cdot\,;\lambda)$ at the given risk premium. A producer holds his whole net worth in his own firm, and outside equity is a pro-rata claim, so the producer's stake carries the firm's risky share $\psi(\bar{z})$ regardless of the fraction of the firm he sells. Thus, the wedge is positive if the marginal producer's exposure deviates from investors' optimal exposure to aggregate risk: higher exposure entails unwanted risk, while lower exposure sacrifices part of the risk premium. It therefore rises with deviations in either direction and is not just a compensation solely for bearing additional risk. Nor does it compensate for idiosyncratic risk, since productivity is known when occupations are chosen.

## 2.5 Aggregation and equilibrium

Since the productivity draws are i.i.d., every producer's scale is proportional to his net worth, and every investor holds the same portfolio, the equilibrium can be expressed in terms of aggregate wealth and the productivity cutoff without tracking the underlying wealth distribution across capitalists.

**Aggregation.** The cutoff divides capitalists into investors, of mass $\Phi(\bar{z}) = 1-\bar{z}^{-\eta}$, and producers, of mass $\bar{z}^{-\eta}$. Net worth is independent of the current draw, so investors' and producers' aggregate wealth are in proportion to their population mass,

$$W^I = \Phi(\bar{z})\, W, \qquad W^P = \bar{z}^{-\eta}\, W. \tag{29}$$

where $W = \int_i a_i\, di$ is the total aggregate wealth of all capitalists. Each producer operates $k = (1+\xi)a$, so the aggregate and productivity-weighted capital stocks are

$$K = (1+\xi)\,\bar{z}^{-\eta}\, W, \qquad \tilde{K} \equiv \int_{i\in\mathcal{P}} z_i k_i\, di = \mathbb{E}_z\left[z \,|\, z \geq \bar{z}\right] K = \frac{\eta}{\eta-1}\,\bar{z}\, K, \tag{30}$$

and TFP, the Solow residual $Y/(e^{\alpha\varepsilon}K^\alpha)$ of the aggregate output in (32), is the average productivity of producers raised to the capital share,

$$\text{TFP} \equiv \Big(\mathbb{E}_z\left[z \,|\, z \geq \bar{z}\right]\Big)^\alpha = \Big(\frac{\eta}{\eta-1}\,\bar{z}\Big)^\alpha. \tag{31}$$

With aggregate labor equal to one, the aggregate output takes the Cobb–Douglas form

$$Y = e^{\alpha\varepsilon}\,\text{TFP}\,K^\alpha = \big(e^\varepsilon \tilde{K}\big)^\alpha, \tag{32}$$

labor income is $(1-\alpha)Y$, aggregate operating profit is $D\tilde{K} = \alpha Y$, and

$$w = (1-\alpha)\big(e^\varepsilon \tilde{K}\big)^\alpha, \qquad D = \alpha\,\tilde{K}^{\alpha-1} e^{\alpha\varepsilon}. \tag{33}$$

**The cutoff.** Aggregation turns the cutoff identity (28) into an equation in $\bar{z}$. Let

$$\kappa(\xi, W) \equiv \alpha\left[(1+\xi)\,\frac{\eta}{\eta-1}\,W\right]^{\alpha-1} \mathbb{E}\left[e^{\alpha\varepsilon}\right], \qquad e_D \equiv (\eta-1)(1-\alpha).$$

By (30), (31), and (33), expected operating profit per unit of efficiency capital is $\mathbb{E}[D] = \kappa(\xi, W)\,\bar{z}^{e_D}$, and substituting it into the cutoff identity of Proposition 2.3(c) gives

$$\bar{z} = \left[\frac{\lambda\big(R^{f*}(1+\Delta)+\delta-1\big)}{\kappa(\xi, W)}\right]^{1/(1+e_D)} \tag{34}$$

or, in logarithms,

$$(1+e_D)\ln\bar{z} = \underbrace{\ln\left(R^{f*}(1+\Delta)+\delta-1\right)}_{\text{safe-rate term with wedge}} + \underbrace{\ln\lambda}_{\text{risk premium term}} - \ln\kappa(\xi,W). \tag{35}$$

Conditional on other parameters, the cutoff rises with the risk premium $\lambda$, the occupational wedge $\Delta$, and the safe rate $R^{f*}$. The cutoff is also higher when the financing capacity $\xi$ is higher or aggregate wealth $W$ is larger, because both lower expected operating profit per unit of efficiency capital by raising the labor costs and therefore screen out less productive producers.

**Equity-market clearing and external positions.** When foreign investors hold the share $\omega$ of the index, the equity market clearing, $\theta W^I = (1-\omega)\,\xi W^P$, implies

$$\theta(\bar{z},\omega) = (1-\omega)\,\xi\,\frac{1-\Phi(\bar{z})}{\Phi(\bar{z})} = \frac{(1-\omega)\,\xi}{\bar{z}^{\eta}-1}. \tag{36}$$

When the foreign holding cap is slack, the foreign share $\omega_F$ replaces $\omega$ accordingly. Domestic investors hold the rest of their wealth in the world bond, so the country's net foreign bond position is

$$\text{NFB} \equiv (1-\theta)\,W^I, \tag{37}$$

and is negative when domestic investors borrow abroad. Foreign holdings of domestic equity are a liability of the country, meaning the net foreign equity position is $\text{NFE} \equiv -\omega\xi W^P$. Given $W^I + W^P = W$ and $K = (1+\xi)W^P$, combining with Equation (36) gives the country's net foreign asset position

$$\text{NFA} \equiv \text{NFB} + \text{NFE} = W - K. \tag{38}$$

**Wealth dynamics.** Next period's aggregate wealth is the fraction $s_t$ of aggregate cash on hand that capitalists save. Producers' cash on hand is the payoff of all capital less what outside shareholders receive

$$\text{Producers' cash on hand} = \int_{i\in\mathcal{P}} \iota_{i,t} R^K(z_{i,t}) k_{i,t}\,di = \alpha Y_t + (1-\delta)K_t - \Pi^e_t \tag{39}$$

where $\Pi^e_t = \xi \int_{\mathcal{P}} a_i \frac{R^K(z_i)}{q(z_i)}\,di$ is the payoff of the equity index. Domestic investors' cash on hand is the sum of the bond return and the payoff of the equity index, i.e.

$$\text{Investors' cash on hand} = (1-\theta_t)W^I_t R^{f*} + (1-\omega)\Pi^e_t \tag{40}$$

As both producers and investors follow the same saving policy, the aggregate wealth dynamics is

$$W_{t+1} = s_t H(W_t), \qquad H(W_t) \equiv \alpha Y(W_t) + (1-\delta)K(W_t) + R^{f*}\text{NFB}(W_t) - \omega\,\Pi^e(W_t) \tag{41}$$

where $H_t$ is the aggregate cash on hand of capitalists and $\omega\Pi_t^e$ is the payoff that foreign shareholders take out of the country (Appendix A.2).

**Recursive equilibrium.** Given the model primitives and policy parameters, a recursive equilibrium with aggregate wealth $W$ as the aggregate state variable consists of value functions $V$ and $U$, individual decision rules, price functions, a common saving policy $S(H)$, and a wealth transition $W' = G(W, \varepsilon)$, such that at every wealth level:

(i) Given prices and the aggregate wealth transition, capitalists choose occupations, financing, portfolios, and saving optimally, as described by the Bellman equations, and producers hire labor optimally. After the shock, each capitalist saves the fraction $S(H)$ of cash on hand.

(ii) Domestic equity is priced at domestic investors' marginal valuations, and foreign investors choose their holdings optimally subject to the foreign ownership cap.

(iii) Equity and labor markets clear.

(iv) The aggregate wealth transition equals aggregate saving:

$$G(W, \varepsilon) = S\big(H(W, \varepsilon)\big)\, H(W, \varepsilon),$$

where $H(W, \varepsilon)$ is aggregate cash on hand generated by the equilibrium allocation.

# 3 Theoretical Results with Log Utility

In this section, we illustrate the mechanism through which financial integration affects the occupational choice and the aggregate productivity under the assumption of log utility. The general case with CRRA follows the same mechanism but it loses the analytical tractability. In the quantitative analysis of Section 6, we will calibrate the model with general CRRA utility. We consider two policy parameters that financial integration can affect: the degree of equity-market integration $\omega$ and the financing capacity $\xi$. The equity market liberalization is considered as a direct rise in $\omega$, while we also consider the indirect effect of financial integration through the rise in financing capacity $\xi$ that accompanied the equity market liberalization.

## 3.1 Equilibrium characterization

With log utility, the capitalists make the occupational choices to maximize expected log returns on net wealth. Conditional on aggregate wealth $W$, the equilibrium therefore reduces to two equations, occupational indifference and equity-market clearing, with two endogenous variables, the productivity cutoff $\bar{z}$ and the risk premium $\lambda$. Throughout this subsection, we fix aggregate wealth $W > 0$, and restrict $\omega \in [0, 1)$, and $R^{f*} > 1-\delta$, and consider equilibria with interior investor exposure.

An investor chooses exposure to aggregate risk according to

$$\vartheta^*(\lambda) \equiv \arg\max_{\vartheta\in[0,1]} \mathbb{E}[\ln(1-\vartheta+\vartheta\lambda X)], \qquad X \equiv \frac{D(\varepsilon)}{\mathbb{E}[D(\varepsilon)]}.$$

The optimal portfolio choice depends on the risk premium and the distribution of the normalized aggregate shock. Likewise, the occupational wedge is uniquely determined by the risk premium and financing capacity, and we denote the wedge by $\Delta(\lambda, R^{f*}; \xi)$. It is the compensation that makes the marginal producer willing to hold its own firm's exposure instead of the optimal exposure as an investor. Appendix B.2 establishes the representation of $\Delta$. The equilibrium conditions are reduced to the following two equations in $(\bar{z}, \lambda)$:

$$\kappa(\xi, W)\,\bar{z}^{\,1+e_D} = \lambda\left[R^{f*}\big(1+\Delta(\lambda, R^{f*};\xi)\big)+\delta-1\right], \quad \text{(Entry)}$$

$$\vartheta^*(\lambda) = \frac{(1-\omega)\xi}{\bar{z}^{\eta}-1}\,\bar{\psi}\left(\frac{\nu(\bar{z})}{\lambda}\right), \quad \text{(Pricing)}$$

where $e_D = (\eta-1)(1-\alpha)$ and $\nu(\bar{z}) \equiv \kappa(\xi, W)\bar{z}^{1+e_D}/(1-\delta)$, and $\bar{\psi}(\nu/\lambda)$ is the equity index's exposure to the aggregate risk factor $D$. Formally, $\bar{\psi}(h) = \mathbb{E}_\rho[h\rho/(1+h\rho)]$, where $\rho = z/\bar{z}$ among producers follows a Pareto distribution with lower bound one and tail parameter $\eta$.

**Occupational indifference.** The marginal producer is indifferent between producing and investing. Equation (Entry) equates the marginal producer's expected marginal productivity of capital to its risk-adjusted cost of capital. It shows a positive relation between the cutoff and the risk premium: a higher premium raises the productivity required to make producing as attractive as investing.

**Equity-market clearing.** The risk premium and the occupational cutoff need to adjust to make the demand for equity consistent with the supply. The left-hand side of Equation (Pricing) is investors' desired exposure to the aggregate risk factor $D$, and the right-hand side is the exposure they have by holding the equity index asset that foreign investors do not purchase. Desired exposure rises with the risk premium, while the exposure required by equity market clearing falls with the risk premium.

An interior equilibrium is a pair $(\bar{z}, \lambda)$ satisfying (Entry) and (Pricing), with $\bar{z} > 1$ and $0 < \vartheta^*(\lambda) < 1$, and the following proposition establishes the existence and uniqueness of the static equilibrium.

**Proposition 3.1** (Existence and uniqueness)**.** *Under log utility, an equilibrium with interior investor exposure exists if and only if $R^{f*} > R_{\text{crit}}(\omega; W)$, and if it exists, it is unique.*

The proof is in Appendix B.3. $R_{\text{crit}}(\omega; W)$ specifies a lower bound of the risk-free rate that ensures the natural borrowing limit of domestic investors is not violated, as imposed in Assumption A0.

### 3.2 Comparative statics

We next examine how aggregate TFP adjusts in response to equity-market liberalization and changes in domestic financing capacity through the adjustment of the occupational cutoff. With log utility, we can specify the comparative statics period by period without considering the wealth dynamics. The effect of equity market liberalization (change in $\omega$) or changes in the domestic financing capacity (change in $\xi$) on the occupational cutoff $\bar{z}$ consists of two components: a direct effect and an indirect effect through the risk premium.

We first establish two properties of occupational indifference and equity-market clearing that determine the comparative statics. Define the user cost of capital inclusive of the occupational wedge as

$$\mathcal{U} \equiv R^{f*}(1+\Delta) + \delta - 1.$$

**Lemma 3.1.** *Holding financing capacity fixed, occupational indifference defines a unique differentiable cutoff $Z(\lambda)$, and its elasticity with respect to the risk premium satisfies*

$$\frac{\partial \ln Z}{\partial \ln \lambda} = \frac{\Gamma}{1+e_D} > 0,$$

*where*

$$\Gamma \equiv \frac{\partial \ln(\lambda \mathcal{U})}{\partial \ln \lambda} = 1 + \frac{R^{f*}}{\mathcal{U}} \frac{\partial \Delta}{\partial \ln \lambda} > 0.$$

The proof is in Appendix B.4. A higher premium lowers capital valuations and raises investors' optimized certainty equivalent, making production less attractive relative to financial investment. Occupational indifference therefore requires a higher productivity cutoff.

**Lemma 3.2.** *Holding $\omega$ and $\xi$ fixed, equity-market clearing defines a unique differentiable premium $\lambda^*(\bar{z};\omega,\xi)$ that clears the equity market at a given cutoff, and its elasticity with respect to the cutoff satisfies*

$$\frac{\partial \ln \lambda^*}{\partial \ln \bar{z}} < 0.$$

*At a given cutoff, the market-clearing premium also decreases with equity-market integration:*

$$\frac{\partial \ln \lambda^*}{\partial \omega} = -\frac{1}{(1-\omega)(\varepsilon_\vartheta + \varepsilon_{\bar{\psi}})} < 0.$$

Here $\varepsilon_\vartheta \equiv \mathrm{d}\ln\vartheta^*/\mathrm{d}\ln\lambda > 0$ and $\varepsilon_{\bar{\psi}} \equiv \mathrm{d}\ln\bar{\psi}(h)/\mathrm{d}\ln h \in (0,1)$, at $h = \nu(\bar{z})/\lambda$, are the elasticities of investors' desired exposure and of the index's exposure. The proof is in Appendix B.4. A higher cutoff leaves fewer producers issuing equity and more investors demanding equity assets. Although the remaining producers' equity has greater exposure to aggregate risk, which implies an increase in the risk premium, the demand effect for equity assets dominates under the Pareto distribution assumption and diminishing returns to aggregate capital. Greater foreign ownership reduces the exposure domestic investors must absorb at the original cutoff. A lower premium reduces investors' desired exposure and the payoff of investing, which makes production more attractive.

**Corollary 3.1** (Equilibrium feedback)**.** *The equilibrium feedback of occupational adjustment is positive and given by*

$$\varrho \equiv -\frac{\partial \ln Z}{\partial \ln \lambda}\frac{\partial \ln \lambda^*}{\partial \ln \bar{z}} = -\frac{\Gamma}{1+e_D}\frac{\partial \ln \lambda^*}{\partial \ln \bar{z}} > 0.$$

**Proposition 3.2** (Equity market liberalization increases entry and lowers average productivity)**.** *An increase in foreign investors' ownership cap lowers the productivity cutoff:*

$$\frac{\mathrm{d}\ln \bar{z}}{\mathrm{d}\omega} = \frac{\partial \ln Z}{\partial \ln \lambda}\frac{\partial \ln \lambda}{\partial \omega} = \underbrace{\frac{\Gamma}{1+e_D}}_{\text{selection response } >0} \times \underbrace{\frac{1}{1+\varrho}}_{\text{equilibrium feedback } >0} \times \underbrace{\frac{\partial \ln \lambda^*}{\partial \omega}}_{\text{foreign risk absorption } <0} < 0,$$

*where*

$$\frac{\partial \ln \lambda}{\partial \omega} = \frac{1}{1+\varrho}\frac{\partial \ln \lambda^*}{\partial \omega}.$$

The proof is in Appendix B.5. When foreign investors are allowed to hold a larger share of domestic equity, they diversify more of the domestic aggregate risk. Given the original equilibrium cutoff, the reduced risk premium is captured by $\partial \ln \lambda^*/\partial\omega$. A lower risk premium makes producing more attractive as the cost of equity goes down and attracts more capitalists to enter production. The positive coefficient $\Gamma/(1+e_D)$ measures the selection response, including the adjustment of the occupational wedge. More entry partly offsets the initial decline in the premium by increasing the equity that domestic investors must absorb relative to their wealth. The resulting feedback scales the cutoff response by $1/(1+\varrho) \in (0,1)$, which reduces its magnitude but does not reverse the sign. Liberalization therefore lowers the productivity cutoff and average producer productivity.

Next, we consider the effect of a change in domestic financing capacity on occupational selection and average productivity.

**Proposition 3.3** (Financing capacity raises the cutoff and average productivity when the risk premium is small relative to the user cost)**.** *A larger financing capacity affects both producers' scale and the equity that investors must absorb. Its effect on the cutoff is*

$$\frac{\mathrm{d}\ln \bar{z}}{\mathrm{d}\xi} = \frac{1}{(1+e_D)(1+\varrho)}\left[\underbrace{\frac{1-\alpha}{1+\xi}}_{\text{capital deepening } >0} + \underbrace{\frac{R^{f*}}{\mathcal{U}}\frac{\partial \Delta}{\partial \xi}}_{\text{wedge reduction at given prices } \leq 0} + \underbrace{\Gamma\frac{\partial \ln \lambda^*}{\partial \xi}}_{\text{risk-premium channel } >0}\right].$$

The proof is in Appendix B.5. Changes in financing capacity affect the occupational cutoff through three channels. First, the direct effect of capital deepening raises the cutoff by increased cost of labor and reduced profit per unit of capital. In other words, when producers can operate more capital per unit of own wealth, they hire more labor and, given the fixed labor supply, the wage increases and operating profits per unit of efficiency capital fall. This raises the productivity required to make production worthwhile, and crowds out low-productivity producers.

Second, an increase in financing capacity reduces the occupational wedge at a given risk premium. The intuition is that when producers can scale up more capital per unit of own wealth, given that the price of outside equity is higher than replacement cost, they require less compensation for

the exposure mismatch with investors. The occupational wedge is therefore lower, which makes production more attractive and encourages entry.

Lastly, a larger financing capacity raises the cutoff through the risk premium. On the one hand, given the original cutoff, when producers issue more equity, it increases the total equity issued and the exposure domestic investors must absorb, which pushes up the risk premium. On the other hand, capital deepening reduces the equity index's exposure to the aggregate risk factor. In equilibrium, the increase in issuance dominates and $\partial \ln \lambda^* / \partial \xi > 0$. The higher premium makes production less attractive relative to investing, raises the productivity required to produce, and discourages entry.

The effect of financing capacity on the productivity cutoff is therefore ambiguous in general, and the bound below signs it under a condition on the risk premium. The cutoff rises if and only if

$$-\frac{R^{f*}}{\mathcal{U}} \left.\frac{\partial \Delta}{\partial \xi}\right|_{\lambda} < \underbrace{\frac{1-\alpha}{1+\xi}}_{\text{capital deepening}} + \underbrace{\Gamma \left.\frac{\partial \ln \lambda^*}{\partial \xi}\right|_{\bar{z}}}_{\text{risk-premium channel}}.$$

The left-hand side measures the proportional reduction in the wedge-adjusted user cost at a fixed premium. This reduction makes production more attractive and encourages entry. The right-hand side captures two opposing forces: capital deepening lowers expected marginal productivity through diminishing returns, while greater domestic equity issuance raises the premium required to clear the equity market at a given cutoff. The productivity cutoff rises when these two forces together outweigh the entry incentive from reducing the occupational wedge.

With log utility, the risk-premium channel is positive and thus greater financing capacity raises the cutoff whenever capital deepening offsets the reduction in the occupational wedge. When $\psi(\bar{z}) \leq \vartheta$, we have $\mathcal{C}(\psi(\bar{z}); \lambda) \geq 1$ and $\mathcal{V}(\lambda) \leq 1 + \vartheta(\lambda - 1)$, where $\mathcal{V}(\lambda) \equiv \max_x \mathcal{C}(x; \lambda) = \mathcal{C}(\vartheta; \lambda)$ is investors' maximized certainty equivalent (see details in Appendix B.1), which implies

$$(1+\xi)\Delta = \frac{\mathcal{V}(\lambda)}{\mathcal{C}(\psi(\bar{z}); \lambda)} - 1 \leq \vartheta(\lambda - 1).$$

Proposition B.1 in the appendix therefore establishes that the cutoff rises if

$$\vartheta(\lambda - 1) \leq (1-\alpha)\big(\psi^c + \xi\psi(\bar{z})\big),$$

where $\psi^c \in (0,1)$ is the expected share of operating profits in the marginal producer's capital payoff (Appendix B.2). Under the same exposure hypothesis, a stronger sufficient condition for $\frac{\mathrm{d} \ln \bar{z}}{\mathrm{d}\xi} > 0$ is

$$R^{f*}(\lambda - 1) \leq (1-\alpha)(1+\xi)(R^{f*} + \delta - 1).$$

This condition compares the expected excess return on the claim to operating profits with the labor share times the user cost of capital, scaled by $1 + \xi$.

# 4 Discussion of the Model

In this section, we first illustrate how the equilibrium risk premium $\lambda$ and the productivity cutoff $\bar{z}$ respond to a rise in the foreign ownership cap $\omega$ and to a rise in financing capacity $\xi$. The second part discusses the pecuniary externalities of entry and how they affect long-run output.

## 4.1 The equilibrium in the $(\lambda, \bar{z})$ plane

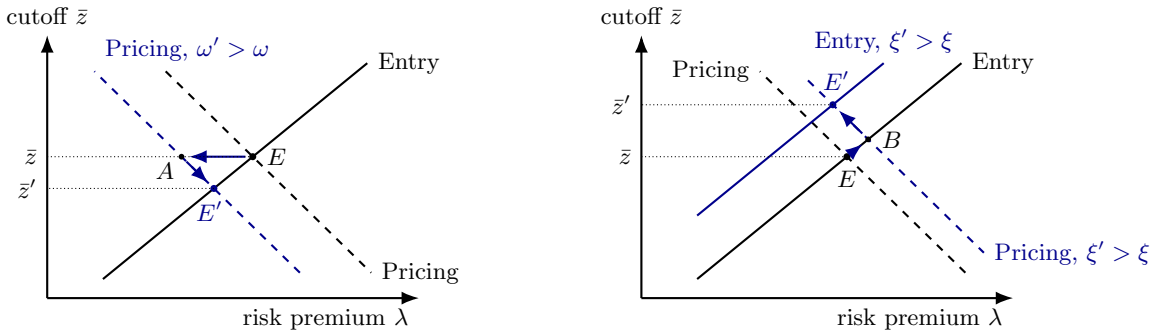
Under log utility and at given aggregate wealth $W$, the within-period equilibrium reduces to the Entry and Pricing equations of Section 3 with two unknowns: the risk premium $\lambda$ and the productivity cutoff $\bar{z}$. Figure 2 plots these conditions as curves whose intersection determines the equilibrium.

The Entry curve, $\bar{z} = Z(\lambda)$, characterizes the marginal capitalist indifferent between producing and investing. It is upward sloping (Lemma 3.1), since a higher premium lowers capital valuations and raises investors' certainty equivalent. This makes investing more attractive than producing and thus requires a more productive marginal producer. The Pricing curve, $\lambda = \lambda^*(\bar{z}; \omega, \xi)$, clears the equity market by equating investors' desired risk exposure with the exposure they must hold. It is downward sloping (Lemma 3.2), as a higher cutoff reduces equity issuance and expands the pool of investors, so a lower premium clears the market. The two curves have at most one intersection, consistent with Proposition 3.1. With log utility, we can write the analytical expressions of the two curves as

$$\text{(Entry)}: \quad \underbrace{\bar{z}\,\mathbb{E}[D]}_{\text{marginal producer's expected profit}} = \underbrace{\lambda}_{\text{risk premium}} \times \Big( \underbrace{R^{f*} + \delta - 1}_{\text{user cost}} + \underbrace{R^{f*}\Delta(\lambda, R^{f*}; \xi)}_{\text{wedge}} \Big),$$

$$\text{(Pricing)}: \quad \underbrace{\theta^*(\lambda)\, W^I}_{\text{equity demand, rises in } \lambda} = \underbrace{(1-\omega)\, \xi\, W^P}_{\text{equity supply net of foreign holdings, falls in } \bar{z}},$$

where $\theta^*(\lambda) = \vartheta^*(\lambda)/\bar{\psi}$ is the equity share that gives investors their desired exposure at the premium $\lambda$, $\mathbb{E}[D] = \kappa(\xi, W)\, \bar{z}^{e_D}$ rises with the cutoff, and $W^I = \Phi(\bar{z})W$ and $W^P = \bar{z}^{-\eta}W$.




(a) Equity market opening, $\omega \uparrow$

(b) Financing expansion, $\xi \uparrow$

Figure 2: The two equilibrium conditions and shifts under $\omega$ or $\xi$ changes

*Notes:* Solid lines show Entry conditions, where the marginal capitalist is indifferent between producing and investing; dashed lines show Pricing conditions, where the risk premium clears the equity market at each cutoff. Black lines and $E$ denote the initial equilibrium; blue lines and $E'$ denote the equilibrium after the policy change. In panel (a), a higher foreign ownership cap $\omega$ shifts Pricing left; $A$ gives the market clearing premium at the initial cutoff. In panel (b), greater financing capacity shifts Pricing right and Entry up when capital deepening dominates wedge reduction; $B$ shows the equilibrium after the Pricing shift. Panel (b) illustrates the case in which the equilibrium premium falls.

**Increase in $\omega$.** A higher $\omega$ reduces domestic investors' exposure at the initial cutoff, lowering the market-clearing premium by

$$\frac{\partial \ln \lambda^*}{\partial \omega} = -\frac{1}{(1-\omega)(\varepsilon_\vartheta + \varepsilon_{\bar{\psi}})}$$

as shown in Lemma 3.2. Foreign risk absorption moves the economy from $E$ to $A$ in Figure 2(a). The lower premium makes production more attractive than investing and thus attracts entry. The entrants' equity issuance raises investors' risk exposure and partially offsets the premium, moving the economy along the new Pricing curve to $E'$ on the original Entry curve. To first order, entry offsets the fraction $\varrho/(1+\varrho)$ of the initial premium decline, and results in an equilibrium decline of $1/(1+\varrho)$ times the foreign risk absorption (Proposition 3.2).

Since TFP $= (\eta\bar{z}/(\eta-1))^\alpha$, a larger risk premium decline implies a larger TFP decline and more entry of low-productivity capitalists at a given aggregate wealth level. Section 5 documents the corresponding pattern across the liberalization episodes where openings with larger equity repricings were followed by weaker TFP growth (Fact 3). The calibrated transition in Section 6.3 illustrates the effect of entry on subsequent growth, as TFP continues to fall while aggregate wealth adjusts along the transition path.

**Increase in $\xi$.** Higher financing capacity shifts both equilibrium conditions. Producers operate at a larger scale and issue more equity. Capital deepening lowers profit per unit of capital and hence the index's risky share. At the initial cutoff, however, increased issuance outweighs the lower exposure per unit of equity, raising the market-clearing premium:

$$\frac{\partial \ln \lambda^*}{\partial \xi} = \frac{a_\xi}{\varepsilon_\vartheta + \varepsilon_{\bar{\psi}}} > 0, \qquad a_\xi = \frac{1}{\xi} - \frac{(1-\alpha)\varepsilon_{\bar{\psi}}}{1+\xi} > 0$$

(see the proofs in Appendix B.4). The first term captures issuance growth while the second captures the lower exposure per claim as capital deepening reduces profit per unit of capital. In Figure 2(b), this rightward shift of Pricing moves the intersection from $E$ to $B$ along the original Entry curve.

At a given premium, the shift of Entry satisfies $\partial \ln Z/\partial \xi = \mathcal{B}/(1+e_D)$, where

$$\mathcal{B} = \underbrace{\frac{1-\alpha}{1+\xi}}_{\text{capital deepening}} - \underbrace{\frac{\Delta}{\mathcal{A}}}_{\text{wedge reduction}}, \qquad \mathcal{A} > 0$$

(see the proofs in Appendix B.4). Capital deepening raises wages, lowers profit per unit of capital, and raises the cutoff. The effect on the occupational wedge works in the opposite direction. Operating more capital per unit of net worth raises producers' returns at a given risk premium, reducing the occupational wedge needed to compensate them for being unable to choose their optimal risk exposure. When $\psi(\bar{z}) \leq \vartheta$ and the premium is small relative to the user cost, capital deepening dominates and the Entry curve shifts up (Proposition B.1). The intersection then moves from $B$ to $E'$ along the new Pricing curve as marginal producers exit, reducing issuance and the risk premium.

Both shifts therefore raise the cutoff and TFP when $\mathcal{B} \geq 0$, but their effects on the risk premium have opposite signs. Combining (69) and (70) gives

$$(1+\varrho)\frac{\mathrm{d}\ln\lambda}{\mathrm{d}\xi} = \underbrace{\frac{\partial \ln \lambda^*}{\partial \xi}}_{\text{Pricing shift } >0} + \underbrace{\frac{\partial \ln \lambda^*}{\partial \ln \bar{z}}\frac{\mathcal{B}}{1+e_D}}_{\text{Entry shift } \leq 0 \text{ if } \mathcal{B}\geq 0}.$$

The premium response is therefore ambiguous. Panel (b) illustrates the calibrated outcome, in which exit dominates and the premium falls below its initial level (Section 6.3).

**Beyond log utility.** With $\gamma > 1$, continuation weights $\Lambda$ enter the certainty equivalents in Entry and investors' demand for exposure in Pricing (see details in Appendix A.5). Because these weights respond to both $\omega$ and $\xi$, we evaluate the policy effects quantitatively. The full CRRA calibration in Section 6 preserves the selection responses described above.

### 4.2 Income distribution and long-run output

The long-run effect of an equity market opening depends on how the additional income affects capitalists' aggregate wealth accumulation. Output is divided among producers, workers, and domestic and foreign investors. The saving of producers and domestic investors replenishes the net worth that supports future production and outside financing. Income received by workers is consumed, while payments to foreign shareholders leave the country. Higher current output therefore may not imply greater productive wealth and higher long-run output.

Substituting the external positions (38) into capitalists' cash on hand (41) gives

$$H = R^{f*}W + \alpha Y - \mathcal{U}_0 K - F, \qquad \mathcal{U}_0 \equiv R^{f*} + \delta - 1, \qquad F \equiv \omega\xi W^P(R^e - R^{f*}).$$

Entry raises aggregate output and wages, reducing incumbents' profit per unit of capital. However, individual entrants take wages as given without internalizing the fact that entry raises wages and reduces incumbents' profits. This pecuniary externality transfers income from capitalists to workers, who are hand-to-mouth and consume their wage income every period. At fixed initial wealth, the expected change in cash on hand for all capitalists is

$$d\mathbb{E}[H] = \alpha\, d\mathbb{E}[Y] - \big(R^{f*} + \delta - 1\big) dK - d\mathbb{E}[F].$$

Expected cash on hand falls if the user cost of additional capital and the change in foreign excess payouts exceed capitalists' share of the output gain. Thus, entry induced by an opening can raise current output but also reduce capitalists' wealth accumulation, which leads to lower long-run output.

Under log utility, where $W' = \beta H$, lower expected cash on hand gives wealth a negative drift. Across conditional-mean steady states, $\beta\mathbb{E}[H] = W$, and (32), (33), and the indifference condition imply, for a marginal opening (i.e., an increase in $\omega$) at a fixed $\xi$

$$d\ln \mathbb{E}[Y_\infty] = \frac{\alpha}{1-\alpha}\left[d\ln \bar{z}_\infty - \Gamma\, d\ln \lambda_\infty\right].$$

Long-run output falls precisely when $d\ln \bar{z}_\infty < \Gamma\, d\ln \lambda_\infty$. Upon a change in $\omega$ with aggregate wealth fixed, Entry implies $d\ln \bar{z} = \Gamma\, d\ln \lambda/(1+e_D)$, meaning a decline in the risk premium but a boom in output due to more entry despite a higher aggregate wage. The subsequent wealth dynamics can overturn that initial output gain: lower wealth lowers the subsequent wage, which admits more low-productivity entrants, and thus the long-run cutoff can fall by more than the initial drop. Moreover, foreign payouts can further attenuate domestic wealth accumulation and eventually lead to a lower long-run output level compared to its level before the opening. This boom-bust pattern in output is also captured in the calibrated results in Section 6.3.

The distributional effects do not necessarily imply a welfare loss. In this model, as producers and investors face different rates of return such that their certainty equivalents are not equalized due to the scale constraint, a Pareto improvement can be made by transferring wealth from investors to all producers. But given the scale constraint, the laissez-faire equilibrium features a competitive wage, endogenous occupational choice and risk premium. Entry nevertheless changes prices and future wealth in ways individual capitalists do not internalize, creating potential distributive and dynamic pecuniary externalities (Greenwald and Stiglitz, 1986; Dávila and Korinek, 2018; Itskhoki and Moll, 2019). Their welfare implications require a planner's problem, which we leave, together with optimal policy, to future work.

# 5 Empirical Evidence

This section tests the model's predictions with the equity market liberalizations of emerging markets in the late 1980s and 1990s. Past decades have seen increased cross-border capital flows and financial

integration. The total amount of cross-border assets and liabilities increased from 44.3% of world GDP in 1970 to 395% of world GDP in 2023 (Lane and Milesi-Ferretti, 2007). On the legislative side, countries have been gradually opening their financial markets to foreign investors. A standard measure of this de jure openness is the index of Chinn and Ito (2006). The index is based on cross-border financial transaction restrictions reported in the IMF's *Annual Report on Exchange Arrangements and Exchange Restrictions*, ranging from zero where all restrictions are imposed to one where no restrictions are imposed. Averaged across countries, the index started from about 0.35 in the 1970s, rose sharply around the 1990s, and has stayed around 0.5 since. In particular, Figure 3 shows a concentrated opening of capital accounts for emerging market countries from 1980 to 2000. This wave of liberalization had a profound impact on emerging market countries' financial markets and the real economy. On the one hand, the liberalization contributed to local economic growth by improving the credit provision, reducing the cost of capital and increasing investment (Henry, 2000b, 2003; Bekaert et al., 2005). On the other hand, however, there were episodes of financial crises in emerging market countries after the liberalization, which led to the slowdown of economic growth (Kaminsky and Reinhart, 1999; Kaminsky and Schmukler, 2008). The evidence on the effect of financial integration on the real economy is therefore mixed (Kose et al., 2009; Rodrik and Subramanian, 2009).

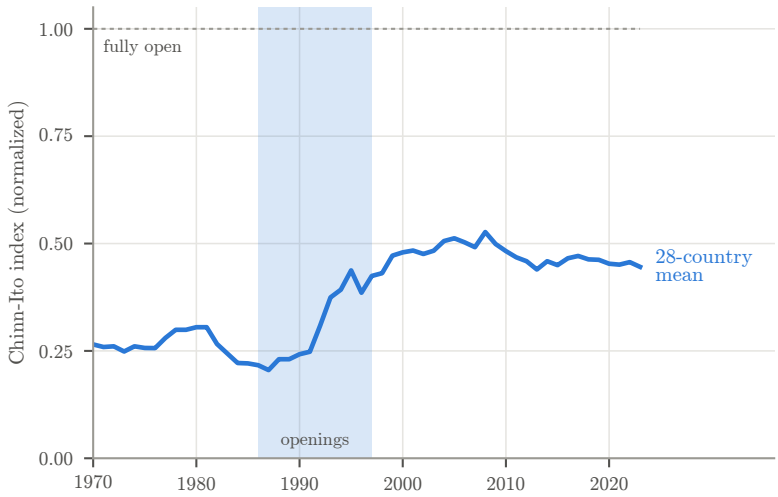



Figure 3: De jure capital account openness of emerging market countries

*Notes:* The figure plots the normalized index of capital account openness of Chinn and Ito (2006), averaged across the 28 opening countries in our sample (27 before 1984) over 1970–2023. The index equals zero for a country that imposes all of the restrictions categorized by the IMF, and one (dashed line) for a country that imposes none. The shaded area marks 1986–1997, the years of the equity market openings that we study.

In general, financial integration is a gradual process that is affected and accompanied by many macro-finance factors. It is difficult to detect the effect of financial integration on the real economy without being confounded by other economic conditions. We therefore focus on the liberalization episodes of emerging market countries between 1986 and 1997, when the liberalization was concentrated in a relatively short period of time and the legislative impact on the financial market was more direct. In this section, we first introduce the data and the empirical design, and then document

three stylized facts about the liberalization episodes that we relate to the model's predictions.

### 5.1 Data

**Liberalization dates.** We use the official liberalization dates documented in Bekaert and Harvey (2000) and Bekaert et al. (2005). In general, countries have different forms of legislation or practical implementation that increase their financial openness. For example, Henry (2000a) dates a country's first liberalization by the earliest date across official policy decrees, or the establishment of a traded country fund that allows foreigners to be exposed to emerging markets, or a jump in the investable share of the country shown in IFC's Investability Index. In this paper, we focus on the impact of direct foreign capital invested in the country and thus use the month of the first formal change in law or regulation after which foreign investors could buy equities directly on the domestic market as the equity market opening date. On top of allowing foreigners to invest in the domestic equity market, countries have different implementation details that vary in regulating foreign investment, such as imposing caps on the foreign ownership share. Our empirical strategy relies on the difference in outcome variables between pre- and post-liberalization dates, and we do not exploit the variation embedded in the intensive margin of the level of openness. For annual outcomes we date an opening by the year of its official policy implementation.

**Other data.** We merge the following country-level annual frequency macroeconomic variables: the external asset positions from the External Wealth of Nations database (Lane and Milesi-Ferretti, 2007); the number of listed domestic companies per million people from the Global Financial Development Database of the World Bank (Čihák et al., 2012); Solow residual TFP, nominal and real GDP from the Penn World Table (Feenstra et al., 2015). The monthly financial variables include the equity prices from the S&P/IFC Global price and total monthly dollar return indices of each market as distributed by Datastream. Appendix C.1 documents the detailed variable definitions and data sources. Our sample includes 28 opening episodes[1] that range from 1986 to 1997.

### 5.2 Empirical strategy

Since each country liberalizes at a different time, we adopt the staggered difference-in-differences approach of Callaway and Sant'Anna (2021), with countries that have not yet liberalized serving as the control group. The key identifying assumption is that, in the absence of liberalization, the outcome of interest, such as foreign capital inflows or changes in asset prices, would have evolved in the same way as in countries that remained unliberalized over the corresponding comparison period. As for the baseline analysis, we use the unconditional estimator of Callaway and Sant'Anna (2021) to estimate the average treatment effect on the treated (ATT). We also report the conditional

[1]The countries are: Argentina, Brazil, Chile, Colombia, Côte d'Ivoire, Greece, India, Indonesia, Israel, Jamaica, Jordan, Kenya, Korea, Malaysia, Mexico, Morocco, Nigeria, Pakistan, the Philippines, Portugal, South Africa, Sri Lanka, Thailand, Trinidad and Tobago, Tunisia, Turkey, Venezuela, and Zimbabwe.

estimator, which adjusts the change among the comparison countries for log real GDP per capita in the year before the opening, so that openers are compared with countries of similar income (Appendix Table C.5).

We construct the estimates in two steps. First, we compare each group of countries that liberalized in the same period with countries that had not yet liberalized. Second, we average these comparisons across groups benchmarked at the same post period relative to the liberalization date.

Let $y_{c,t}$ denote the outcome of interest for country $c$ at time $t$, measured at either annual or monthly frequency. Countries that liberalized at time $t_g$ form a cohort $g$. For each cohort, we measure the average change in the outcome of interest between a pre-liberalization base time, $b(g)$, and any time $t$. We then subtract the average change over the same period (between $b(g)$ and $t$) among the countries that have not yet liberalized, which we call the control group countries. The resulting difference-in-differences estimate is:

$$\widehat{ATT}(g,t) = \frac{1}{N_{g,t}} \sum_{c \in \mathcal{T}_{g,t}} \left(y_{c,t} - y_{c,b(g)}\right) - \frac{1}{M_{g,t}} \sum_{c \in \mathcal{C}_{g,t}} \left(y_{c,t} - y_{c,b(g)}\right). \tag{42}$$

Here, $\mathcal{T}_{g,t}$ contains countries in cohort $g$, and $\mathcal{C}_{g,t}$ contains the control group countries. Both groups include only countries observed in the base period and in period $t$; $N_{g,t}$ and $M_{g,t}$ denote their respective number of countries in the group.

The comparison group changes over time as countries liberalize. Let $G_c$ denote country $c$'s liberalization date. A country is eligible as a comparison if $G_c > \max(g,t) + a$, meaning that its own liberalization occurs more than $a$ periods after the later of the cohort's opening date and the observation period. For annual outcomes, we use the year before liberalization as the base time $b(g)$ and set $a = 0$. For monthly outcomes, we use month $-6$ as the base to take into account the anticipation of the implementation and set $a = 5$, allowing outcomes to respond during the five months before liberalization. Countries whose liberalization occurs sufficiently far beyond the sample period also supply comparisons. In our sample, countries that liberalized in 1986–1993 will be used as treated countries cohort by cohort. Countries that liberalized after 1993 are always included in the control group. Appendix C.1 reports the liberalization dates and the sample of will-be-treated and always-control countries.

We then construct the estimates in event time, $k = t - g$, by aggregating the cohort estimates. For example, $k = 1$ denotes one year or one month after liberalization. At each event time, we average the available cohort estimates, weighting each estimate by the number of contributing countries in the treated group:

$$\hat{\theta}_k = \sum_g w_{g,k} \, \widehat{ATT}(g, g+k), \qquad w_{g,k} = \frac{N_{g,g+k}}{\sum_{g'} N_{g',g'+k}}. \tag{43}$$

The sums include only cohorts for which the comparison can be constructed. Thus, $\hat{\theta}_k$ summarizes how outcomes changed around liberalization relative to contemporaneous changes in countries that remained closed. We report analytical leave-one-out jackknife standard errors following MacKinnon

et al. (2023) in the main text and show both 90% and 95% confidence bands in the DiD dynamic estimate plots.

We use these estimates to describe changes around liberalization relative to contemporaneous changes in countries that had not yet liberalized. A causal interpretation requires that, absent liberalization, the outcome variables of treated and control groups would have experienced the same trajectory over time. This requires that decisions of liberalization are unaffected by anticipated future outcomes, and that treated countries do not affect outcomes in the control group countries. For example, countries might decide to liberalize their equity market if they expect to have TFP growth in the future. Or countries that have just liberalized their equity market will affect the economic performance of the control group countries through trade. Because openings often coincided with other reforms and macroeconomic changes (Henry, 2000a, 2007), we interpret the estimates as descriptive evidence that verifies the model's predictions.

### 5.3 Three facts about the liberalization episodes

We document three stylized facts about the emerging country equity market liberalization episodes that correspond to the model's predictions for an opening. In the model, an opening is a rise in the share of domestic equity held by foreign investors (Fact 1). A lower risk premium reduces the cost of equity and draws new producers into production (Fact 2). The entrants are less productive than the incumbents, so a larger fall in the premium comes with a larger fall in aggregate TFP (Fact 3). Facts 1 and 2 adopt the DiD approach and Fact 3 is a cross-country analysis. In Section 6, we will calibrate the model to key features of those countries and characteristics of the liberalization episodes to illustrate the mechanism.

**Fact 1: Liberalization attracts more foreign portfolio equity inflows.** We first examine whether equity-market liberalization is followed by greater accumulation of foreign portfolio equity liabilities. An increase in foreign liability indicates an inflow of foreign capital into the country. Let $F_{c,t}$ denote the foreign liability position in portfolio equity as a percentage of contemporaneous GDP,

$$F_{c,t} = 100 \times \frac{L_{c,t}}{Y_{c,t}}, \tag{44}$$

where $L_{c,t}$ is the dollar value of foreign equity portfolio holdings at the end of year $t$ and $Y_{c,t}$ is dollar GDP. In the DiD estimation, $F_{c,t}$ is the outcome variable. Given a base year $b(g)$, the change in foreign holdings in portfolio equity relative to the year before the opening is $F_{c,t} - F_{c,b(g)}$.

We choose the base year $b(g)$ to be one year before the official implementation of the liberalization. Thus, the event-time effect $\hat{\theta}_k$ estimated in equation (43) is interpreted as the extra inflow that the openers received over years $-1$ to $k$ around the liberalization compared to the countries that have not yet opened in the same period.

Figure 4 shows the dynamic event-time estimates $\hat{\theta}_k$, which are the differential changes in foreign holdings of domestic portfolio equity, $F_{c,t}$, relative to the year before the opening between

the openers and the countries that have not yet opened. Before the opening, the openers' foreign holdings rose slightly faster than those of the countries that had not yet opened, which mostly reflects valuation gains on existing holdings, as we show below. But the difference is rather small and only accounts for 0.4 percentage points of GDP over the four years before the opening. Appendix C.2 reports the joint test for pre-trends, which covers the four pre-opening estimates for years −5 to −2. After the opening, the difference widens. By event year 4, the difference reaches 2.8 percentage points, significant at the 5 percent level.

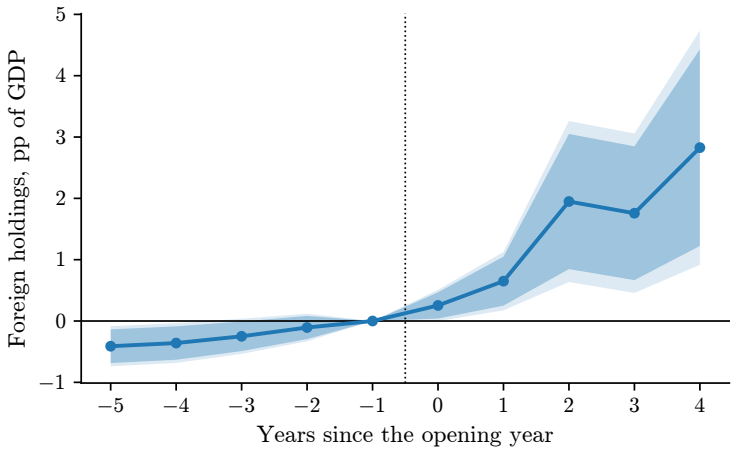



Figure 4: Foreign holdings of domestic equity around equity market openings

*Notes:* The figure reports event-time estimates $\hat{\theta}_k$ from equation (43) for foreign holdings of domestic portfolio equity, $F_{c,t}$, defined in equation (44). Each opening cohort is compared with countries that remain unopened through both the cohort's opening year and the observation year. Changes are measured relative to the year before opening ($k = -1$), and cohort estimates are weighted by the number of opening countries. Positive post-opening estimates indicate greater increases in holdings among openers. Negative pre-opening estimates indicate that holdings increased more among openers between the indicated year and the year before opening. Shaded bands show pointwise 90 and 95 percent confidence intervals based on leave-one-country-out jackknife standard errors. The sample contains 364 country-year observations across 26 countries during 1981–1994, including 19 countries that open during 1981–1993 and seven that open after 1994. The latter serve as control group countries throughout. Estimates at event years 2, 3, and 4 include 17, 14, and 9 openers, respectively. Data: External Wealth of Nations (Lane and Milesi-Ferretti, 2007).

Foreign portfolio equity holdings are recorded at market value, so changes in these positions reflect both net purchases and valuation gains or losses. Previous studies document increases in equity prices around liberalization (Henry, 2000a; Bekaert and Harvey, 2000), suggesting that revaluation may account for part of the observed increase in foreign holdings. To separate the valuation effect from the net purchase of portfolio equity, we use annual dollar price returns on each country's S&P/IFC Global index from the S&P Emerging Markets Data Base to proxy the valuation effect. These returns are available for 16 of the 26 countries in our sample.

We estimate annual revaluation as the previous observed end-of-year foreign holdings multiplied by the current year's index price return. For each opening cohort, we construct an adjusted position benchmarked at observed holdings in the base year, $b(g) = g - 1$, and then accumulate subsequent changes in dollar holdings net of estimated revaluation. Pre-opening positions are constructed

analogously by subtracting adjusted changes backward from the base year. We then divide the adjusted positions by contemporaneous GDP. The difference between observed and adjusted positions represents cumulative estimated revaluation relative to the base year, expressed as a percentage of current GDP. This procedure assumes that the index price return approximates the return on foreign investors' equity portfolios and does not account for valuation changes on purchases made within the year.

Figure 5 compares the observed and adjusted positions on the same restricted sample that includes 13 opening countries. At event year 4, the increase in observed holdings from the base year exceeds that in control group countries by 3.7 percentage points of GDP; adjusting for revaluation reduces this difference to 2.0 percentage points. Under this adjustment, revaluation accounts for approximately half of the estimated differential increase. The adjusted pre-opening estimates are close to zero, which suggests that valuation gains contributed to the pre-opening increase in foreign holdings within this subsample. After accounting for revaluation, the post-opening increase in foreign holdings remains economically and statistically significant for liberalized countries compared to not yet opened countries. This shows that liberalization attracted substantial foreign capital inflows in the equity market.

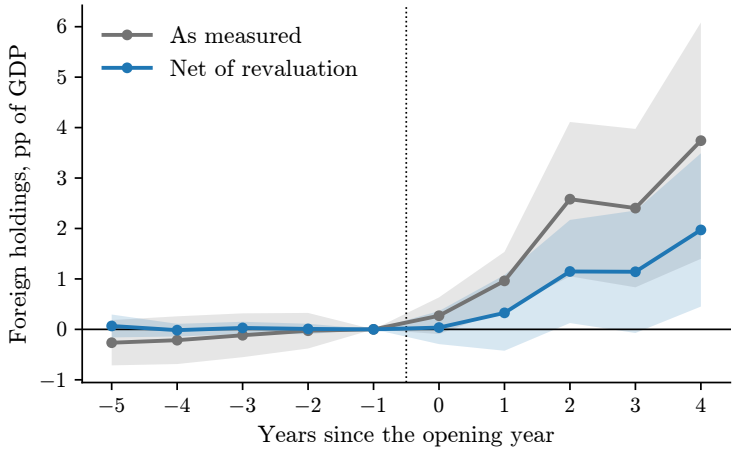



Figure 5: Foreign holdings of domestic equity, as measured and net of revaluation

*Notes:* The figure reports event-time estimates $\hat{\theta}_k$ from equation (43) for foreign holdings of domestic portfolio equity, $F_{c,t}$, defined in equation (44), as measured (gray) and adjusted for estimated valuation changes (blue). For each cohort, adjusted dollar holdings are compared to their observed value in the base year, $b(g) = g - 1$. For each year, we estimate valuation gains as the previous end-of-year's holdings multiplied by that year's dollar price return on the domestic S&P/IFC Global index. The adjusted holdings equal observed holdings minus the valuation gains accumulated since the base year. Before the base year, adjusted holdings equal observed holdings plus the valuation gains accumulated between that year and the base year. We express both positions as percentages of GDP in the observation year. Shaded bands show pointwise 95 percent confidence intervals based on leave-one-country-out jackknife standard errors. The sample contains 169 country-year observations with available index returns across 16 countries, including 13 that open during 1987–1992. The year-4 estimate includes six openers from 1987–1990. Data: External Wealth of Nations (Lane and Milesi-Ferretti, 2007); S&P Emerging Markets Data Base.

Appendix C.3 reports analogous estimates for other external positions and total net capital

inflows. We find no comparably clear increase in FDI liabilities or aggregate net capital inflows. There is some evidence of more accumulation of reserves, but the results are not statistically significant. These patterns are consistent with the targeted nature of the reforms, which specifically relaxed restrictions on foreign purchases of domestic equity. The evidence therefore points primarily to an expansion of foreign portfolio equity investment, without establishing a broader increase in total net capital inflows.

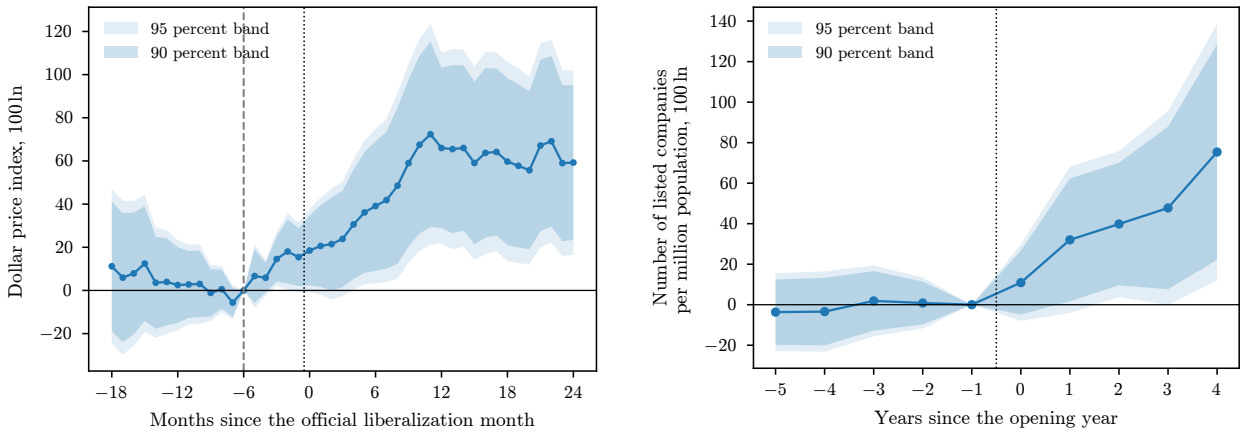
**Fact 2: Openings increased domestic equity prices and the number of listed companies.** Past studies have documented that equity prices rise around liberalizations. Henry (2000a) finds abnormal returns of 3.3 percent a month in real dollars over the eight months before a country's first liberalization, and Bekaert and Harvey (2000) find that dividend yields fall. Chari and Henry (2004) find that the stocks opened to foreign investors revalue by about 15 percent and attribute roughly two fifths of the revaluation to risk sharing. We measure the re-pricing of each market as a whole around its official month, against the markets not yet open. We test whether the equity market liberalization is followed by a re-pricing that is higher than the counterfactuals using the DiD approach.

We obtain monthly observations of the S&P/IFC Global price index in current US dollars from Datastream. Our estimation sample covers 17 countries over January 1980–December 1994. Since our event dates correspond to the official implementation month of the new regulation, there could be policy announcements or earlier indications that foreign investment restrictions will be relaxed. Therefore, equity prices may respond before the official implementation date. To accommodate the anticipation, we allow for five months of anticipation and measure changes relative to six months before the opening, setting $a = 5$ and $b(g) = g - 6$ in equation (42), following the anticipation convention of Callaway and Sant'Anna (2021). We also restrict control group countries to be the ones that have not yet opened during this anticipation window.

Figure 6a reports the event-time estimates $\hat{\theta}_k$ for the log price index. The figure shows that there is a significant increase in equity prices once liberalization occurs. The average event-time estimate over the next 24 months corresponds to a relative price increase of approximately 67 percent, significant at the 5% level. The corresponding estimate over the first twelve months implies an increase of 49 percent. The stock price differences between the openers and the control group are statistically insignificant before the anticipation window, but the difference becomes more prominent during the anticipation period. After the implementation of the new policy, the stock prices of the openers continue to rise rapidly relative to the control group in the first year and gradually stabilize at a level approximately 60 percent higher on average than the stock prices of the control group.

More capital inflow in the equity market also makes it more attractive for firms to raise outside equity. In the model, the capitalists who enter production run new firms that sell outside equity, and we take the number of listed companies as the empirical counterpart of the number of producers. We use the annual data of the number of listed domestic companies per million people, $\ell_{c,t}$, from

the Global Financial Development Database. We estimate equation (43) at annual frequency with $y_{c,t} = 100 \ln \ell_{c,t}$ and use the year before the opening as the base year, $b(g) = g-1$. Figure 6b reports the event-time estimates. Before the opening, the number of listed companies per capita among opener countries evolved similarly compared to the control group countries. After the opening, the number of listed companies grew faster in the openers. The average increase over years 0 to 4 post opening is approximately 51 percent more in liberalized countries, significant at the 5% level, and the difference reaches approximately 113 percent by year 4.




(a) Dollar price index

(b) Number of listed companies per million people

Figure 6: Equity prices and number of listed companies around equity market openings

*Notes:* Panel (a) reports event-time estimates $\hat{\theta}_k$ from equation (43) for 100 ln of the S&P/IFC Global price index in current US dollars. Event time is measured in months relative to the official liberalization month (dotted line), and the reference period is six months before opening (dashed line). The specification allows for five months of anticipation: $b(g) = g-6$, and comparison countries satisfy $G_c > \max(g,t)+5$. Cohort estimates are weighted by the number of opening countries. Shaded bands show pointwise 90 and 95 percent confidence intervals based on leave-one-country-out jackknife standard errors. The sample contains 2,531 country-month observations across 17 countries during January 1980–December 1994, including 15 countries that open between September 1987 and June 1993. Data: S&P/IFC Global indices from Datastream.
Panel (b) reports the corresponding annual estimates for 100 ln of the number of listed domestic companies per million people, $\ell_{c,t}$. Event time is measured in years relative to the opening year (dotted line), the reference period is the year before opening ($k = -1$), and comparison countries satisfy $G_c > \max(g,t)$. Cohort weights and confidence bands are constructed as in panel (a). The sample contains 210 country-year observations across 16 countries during 1980–1994. Estimates at event years 2, 3, and 4 include 14, 11, and 8 openers, respectively. Data: Global Financial Development Database (Čihák et al., 2012).

Appendix C.4 reports the corresponding estimates for dividend yields (Table C.2 and Figure C.7). As a robustness check, Figure C.8 presents annual estimates using December observations and grouping countries by their opening year, as in Fact 1. This pools the small monthly cohorts into larger annual cohorts, while the number of countries remains unchanged. The annual estimates also indicate an increase in equity prices around liberalization. Relative to the preceding December and the change in control group countries, the opening-year estimate corresponds to a 116 percent increase in the dollar price index.

**Fact 3: Larger equity re-pricings and larger increases in listed companies were followed by weaker TFP growth.** We measure each country's re-pricing around the liberalization event using the change in its dollar equity price index from six months before the official opening to twelve months afterward:

$$\Delta_c = 100 \left[ \ln I_c(m_c + 12) - \ln I_c(m_c - 6) \right], \tag{45}$$

where $m_c$ is the official liberalization month and $I_c$ is the S&P/IFC Global price index used in Fact 2. This eighteen-month window includes both the potential anticipatory response and the first year after the opening. We then measure subsequent aggregate TFP growth over the nine years following the opening year, using the Solow residual from the Penn World Table (Feenstra et al., 2015):

$$g_c = \frac{100 \ln \mathrm{TFP}_{c,Y_c+10} - 100 \ln \mathrm{TFP}_{c,Y_c+1}}{9}, \tag{46}$$

where $Y_c$ is the year when the country implements the equity market opening. $g_c$ measures annualized TFP growth over the nine years from year +1 to year +10 so that $g_c$ represents the average annual growth rate of TFP over the period that is not overlapping with the re-pricing window. We estimate a simple OLS regression of $g_c$ on $\Delta_c$:

$$g_c = a + b \frac{\Delta_c}{10} + \nu_c. \tag{47}$$

The coefficient $b$ measures the difference in annual TFP growth associated with a ten-log-point larger equity price increase. We report small sample heteroskedasticity-robust standard errors and permutation p-values based on 20,000 reassignments of the re-pricing across episodes. The sample consists of the 17 openers that opened between 1987 and 1996, for which both the price index and TFP data are available.[2]

Figure 7a shows a significant negative relationship between the magnitude of increased equity re-pricing and subsequent TFP growth. The estimated slope is −0.180, with an HC3 standard error of 0.043 and a permutation p-value of 0.003. A doubling of the index corresponds to approximately 1.2 percentage points lower annual TFP growth. In all eight countries where the dollar equity price index rose by more than 80 percent over the re-pricing window, aggregate TFP declined in the subsequent decade after liberalization. These declines indicate that those countries became less efficient in using their capital and labor to produce output, despite the increase in capital accumulation that accompanied the liberalization.

The result is robust across alternative growth measures, controls, and samples. It remains when we vary the growth horizon, measure growth relative to the world median, or exclude crisis and post-crisis years. It is also robust to adding controls one at a time, such as prior TFP growth, the opening date, high inflation, the capital account opening that followed, the financing expansion

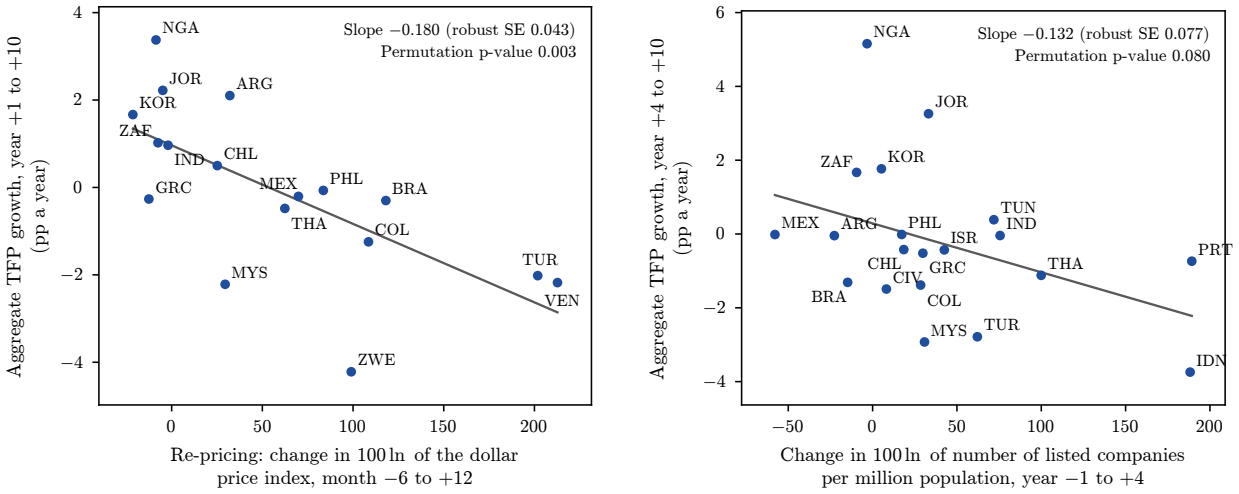
[2]The sample countries are: Argentina, Brazil, Chile, Colombia, Greece, India, Jordan, Korea, Malaysia, Mexico, Nigeria, the Philippines, South Africa, Thailand, Turkey, Venezuela, and Zimbabwe.

measured in Section 6.3, or stabilization programs; each also leaves the slope negative and similar in magnitude. Appendix Table C.3 reports these robustness checks.

Openers whose number of listed companies grew more also experienced weaker subsequent TFP growth. For each opener, we measure the change in listed companies per million people from the year before the opening to four years afterward, and annualized TFP growth from year +4 to year +10:

$$\Delta_c^{\ell} = 100\left[\ln \ell_{c,Y_c+4} - \ln \ell_{c,Y_c-1}\right], \qquad g_c^{\ell} = \frac{100 \ln \mathrm{TFP}_{c,Y_c+10} - 100 \ln \mathrm{TFP}_{c,Y_c+4}}{6}. \tag{48}$$

We estimate the OLS regression of equation (47) with $g_c^{\ell}$ and $\Delta_c^{\ell}$ in place of $g_c$ and $\Delta_c$ for the 20 openers with both measures. Figure 7b shows that a ten-log-point increase in the number of listed companies is associated with 0.13 percentage points lower annual TFP growth over the following six years. A doubling of the number of listed companies corresponds to approximately 0.9 percentage points lower annual TFP growth, and the correlation is −0.40. Appendix Table C.4 reports other timing windows and robustness with respect to adding controls.




(a) Equity re-pricing

(b) Change in number of listed companies

Figure 7: Equity re-pricing, changes in number of listed companies, and subsequent TFP growth across liberalization episodes

*Notes:* In panel (a), the horizontal axis measures $\Delta_c$, the change in 100 ln of the dollar equity price index from six months before the official opening to twelve months afterward. The vertical axis measures $g_c$, annualized log TFP growth from year +1 to year +10, in percent. The fitted line is the ordinary least squares estimate of equation (47). The slope is −0.180 percentage points of annual TFP growth per ten log points of re-pricing, with a small sample robust standard error of 0.043 and a permutation p-value of 0.003. In panel (b), the horizontal axis measures $\Delta_c^{\ell}$, the change in 100 ln of number of listed domestic companies per million people from the year before the opening to four years afterward. The vertical axis measures $g_c^{\ell}$, annualized log TFP growth from year +4 to year +10, in percent. Both are defined in equation (48). The fitted line is the ordinary least squares estimate of equation (47) with $\Delta_c^{\ell}$ and $g_c^{\ell}$ in place of $\Delta_c$ and $g_c$. The slope is −0.132 percentage points of annual TFP growth per ten log points of listings growth with a small sample robust standard error of 0.077 and a permutation p-value of 0.080, and the correlation is −0.40. Data: S&P/IFC Global indices from Datastream; aggregate TFP at constant national prices from the Penn World Table (Feenstra et al., 2015), Global Financial Development Database (Čihák et al., 2012).

The difference-in-differences estimates in Appendix C.6 provide a complementary perspective.

Relative to the year before liberalization, real GDP increased by approximately 6 percent more in the openers than in the control group countries, on average over the first five years. The estimated increase in aggregate TFP is smaller and sensitive to the reference period and the composition of the control group. Thus, the expansion in output was not accompanied by equally strong and robust productivity gains.

# 6 Quantitative Analysis

In this section, we calibrate a representative emerging-market economy to quantify how international risk sharing and the relaxation of financing constraints contribute to changes in aggregate TFP following equity market liberalization. The calibration uses data from the 14 economies examined in the previous section for which the data used in the calibration are available. We construct a benchmark economy by matching cross-country medians of several country characteristics before equity market liberalization. We first present the calibration and evaluate how closely the benchmark economy matches the characteristics of countries just before liberalization. We then use the calibrated model to examine the dynamic impact of (i) increased foreign holdings of domestic portfolio equities and (ii) the expansion of outside financing for productive capital.

## 6.1 Calibration strategy

A model period maps to one calendar year in the data. Unless otherwise indicated, each empirical variable for the benchmark economy is constructed by first averaging the same variable of each country over the five years preceding its opening, and then taking the median across countries. We use the outside-financing share to calibrate $\xi$ and the foreign portfolio ownership share of listed equity not held by insiders to calibrate $\omega$, and these are the only two variables whose time series values are used. Appendix D.1 lists the countries and documents the sources, construction, and coverage of each moment.

We assign values to the capital share, depreciation rate, and Pareto tail parameter. We calculate the world interest rate, financing capacity, foreign ownership share, and shock standard deviation from observed data using the formulas described below. We jointly calibrate risk aversion, the discount factor, and initial capitalist wealth to match the pre-opening risk premium and net foreign bond position and to make expected wealth next period equal initial wealth. The mean shock is normalized so that $\mathbb{E}[e^{\alpha\varepsilon}] = 1$. Aggregate TFP is not a calibration target. Table 1 reports each value and how it is obtained.

**Assigned values.** We set the capital share to $\alpha = 1/3$ following Gollin (2002), which estimates that the labor income share is around 0.65–0.8 for most countries using data in the 1980s. The depreciation rate is set to $\delta = 0.06$, which is the lower end of the range of 0.06–0.075 for the capital of listed firms based on Compustat data. We set $\eta = 3$ as the benchmark Pareto tail parameter,

as it gives the productivity distribution a finite variance.[3] Appendix D.4 examines the sensitivity of our results to alternative values of $\eta$. Since the capital-output ratio depends on both $\eta$ and $\delta$, we reserve the capital-output ratio as an untargeted moment to check on the model fit.

**Parameters calculated from data.** We calculate the world risk-free rate, financing capacity, foreign ownership share, and the productivity shock standard deviation from the data as follows. The model's risk-free rate is mapped to the real one-year Treasury bill yield as one plus the one-year US Treasury yield minus US consumer price inflation in 1990, which gives $R^{f*} = 1.0248$.

Next, since the financing capacity $\xi$ is the outside finance per unit of producers' own equity, this implies that outside investors finance the $\xi/(1+\xi)$ share of total productive capital. We proxy this share by summing the private credit, listed book equity not held by insiders[4], and foreign direct investment liabilities, and then dividing it by the private capital stock from the IMF's Investment and Capital Stock Dataset. For each country $c$, we calculate this share, $x_c$, in year $-1$ and then calculate $\xi_c = x_c/(1-x_c)$. The median of $\xi_c$ across countries gives $\xi = 0.290$, corresponding to an outside-financing share of 0.225. To proxy the foreign share of the outside financing $\omega$, we use foreign portfolio equity liabilities (Lane and Milesi-Ferretti, 2007) divided by the market capitalization of listed firms available to outside investors. The median value for $\omega$ in year $-1$ before liberalization across the 13 opening countries with a usable private capital stock is 0.094.

The proxies for $\xi$ and $\omega$ rely on two assumptions. First, we ignore the heterogeneity across firms in terms of how much outside finance they require. In reality, firms differ in terms of their financing decisions, and we use $\xi$ as a proxy for a nationwide outside financing share. Second, the foreign share observed for listed outside equity is used to proxy the overall foreign participation in the entire pool of outside finance due to data limitations in foreign participation in private credit or outsider share in direct investment. We use the sum of foreign portfolio equity liabilities and foreign direct investment liabilities as a percentage of GDP as an untargeted moment to assess the model's implied foreign holdings relative to output. We include direct investment because it also enters our measure of outside finance. Appendix D.1 discusses other assumptions for the data choices in detail.

We then use dividend data to discipline the volatility of the aggregate shock. In the model, log aggregate operating profits vary with $\alpha\varepsilon$ conditional on the capital stock. We use innovations to listed-firm dividends as a proxy for this operating-profit risk. Specifically, we compute annual log changes in real US-dollar dividends on the IFC Global index country by country, demean within each country to remove the country-specific mean, and pool the available pre-opening observations.

[3] The evidence from Chile's 1998 sudden stop is suggestive of a low tail parameter: entering firms were substantially more productive despite a comparatively modest decline in their number (Ateş and Saffie, 2021). Under the model's pure selection mechanism, a larger productivity gain relative to the decline in producer numbers corresponds to a lower $\eta$. However, a literal application of this relationship to the reported cohort differences yields a value outside the model's admissible range. Since these differences can also reflect changes in potential entry and productivity growth after entry, we regard the evidence as suggestive rather than as a quantitative calibration target.

[4] When calculating listed book equity not held by insiders, we use one minus the closely held share of listed equity from Dahlquist et al. (2003) and multiply it by the book value of each country's listed equity, its market capitalization divided by the price-to-book ratio of its IFC index.

We calculate the robust standard deviation of the dividend innovation and the value is $\alpha\sigma = 0.322$. Given $\alpha = 1/3$, the value of $\sigma$ is 0.965. For the mean of the shock, we normalize such that the expected TFP shock is 1, i.e. $\mathbb{E}[e^{\alpha\varepsilon}] = 1$, which implies $\mu = -0.155$.

We measure the implied cost of equity as the country's IFC Index dividend yield plus the past five-year average growth rate of real GDP (Fama and French, 2002). Subtracting the real one-year US Treasury yield and applying the pre-opening averaging convention gives a median premium of 6.75 percentage points. We map this premium to the required excess return on the model's claim to operating profits, $\mathbb{E}[R^D] - R^{f*} = R^{f*}(\lambda - 1)$, which sets $\lambda = 1.0659$. We use the implied equity premium of listed firms as the empirical counterpart for the model's risk premium on operating profits. This assumes that the dividend payouts are representative of the underlying operating profits. Appendix D.1 discusses this assumption, and Appendix D.4 reports an alternative calibration that adjusts the observed premium for firms' debt financing.

**Joint calibration.** Given the parameters that are either assigned predetermined values or calculated from data, we estimate the rest of the parameters from the following moment conditions

$$\text{Operating-profit premium:} \qquad \lambda(W;\gamma,\beta) = 1 + \frac{0.0675}{R^{f*}}, \tag{49a}$$

$$\text{Net foreign bond position:} \qquad \frac{W}{\mathbb{E}[Y]} - \frac{K}{\mathbb{E}[Y]}\,\frac{1+(1-\omega)\xi}{1+\xi} = -0.267, \tag{49b}$$

$$\text{Conditional-mean steady state:} \qquad \mathbb{E}\big[W_{t+1} \mid W_t = W\big] = W. \tag{49c}$$

The three conditions jointly determine risk aversion $\gamma$, the discount factor $\beta$, and benchmark wealth $W_{\text{pre}}$. Other untargeted model moments are evaluated at $W = W_{\text{pre}}$ in the pre-liberalization equilibrium. $\lambda(W;\gamma,\beta)$ is the equilibrium risk premium given wealth $W$, risk aversion $\gamma$ and the discount factor $\beta$, which is derived from investors' first-order condition (11) given that their exposure $\vartheta = \theta\bar{\psi}$ equals the value implied by equity-market clearing (36), which implies

$$\mathbb{E}\Big[\Lambda(W_{t+1})\left(R^{f*} + \vartheta(W)\left(\lambda R^{f*}X - R^{f*}\right)\right)^{-\gamma}\left(\lambda R^{f*}X - R^{f*}\right)\Big] = 0, \tag{50}$$

$$\Lambda(W') = \big(1 - s(W';\gamma,\beta)\big)^{-\gamma}, \quad W_{t+1} = G(W,\varepsilon). \tag{51}$$

Risk aversion affects investors' marginal utility, and the discount factor and wealth affect the continuation weights $\Lambda_t$ and the exposure $\vartheta$. At a given exposure, more risk-averse investors require a larger premium.

To discipline the capitalist wealth, we target the net foreign bond position, which is computed as external debt assets plus reserve assets excluding gold, less external debt liabilities, relative to GDP (Lane and Milesi-Ferretti, 2007). Its pre-opening median across opener countries is −26.7 percent. Since the model has no government, we assume the country's net foreign bond position to be held by domestic capitalist investors. This assumption connects investors' borrowing and the domestic risk they bear, which in turn affects the risk aversion needed to match the observed

foreign bond position. Appendix D.4 reports a private-sector alternative.

The final condition requires $W_{\text{pre}}$ to be a conditional-mean steady state, which means that the conditional expectation of the next period's wealth equals current wealth at the benchmark, and the realized wealth can fluctuate with aggregate shocks. The above three moments are solved jointly. The resulting values are $\gamma = 2.70$, $\beta = 0.905$, and $W_{\text{pre}}/\mathbb{E}[Y] = 2.04$. The saving rate, averaged across shocks at benchmark wealth, is 0.917; with $\gamma > 1$, it depends on continuation values and need not equal $\beta$.

Table 1 summarizes the calibrated values and the way they are calibrated.

Table 1: Parameter values

| Parameter | Symbol | Value | How the value is obtained |
|---|---|---|---|
| *Assigned values* | | | |
| Capital share | $\alpha$ | 1/3 | Gollin (2002) |
| Depreciation rate | $\delta$ | 0.060 | Chosen at the lower end of the 0.06–0.075 range for listed firms |
| Pareto tail parameter | $\eta$ | 3 | Chosen to allow a thick productivity tail with finite variance; capital-output ratio not targeted |
| *Calculated from data* | | | |
| World safe rate | $R^{f*}$ | 1.0248 | One plus the one-year US Treasury yield (constant maturity) minus US CPI inflation, both measured in 1990 |
| Financing capacity | $\xi$ | 0.290 | Median of $x_c/(1-x_c)$, where $x_c$ is private credit plus listed book equity not held by insiders plus FDI liabilities, divided by private capital, in year $-1$. After financing expansion: 0.381 |
| Foreign share of outside finance | $\omega$ | 0.094 | Median of foreign portfolio equity liabilities divided by the market value of listed equity not held by insiders, in year $-1$. After opening: 0.249 |
| Shock standard deviation | $\sigma$ | 0.965 | Robust standard deviation of pooled, country-demeaned annual log changes in real dividends before opening, divided by $\alpha$ ($\alpha\sigma = 0.322$) |
| *Jointly calibrated* | | | |
| Risk aversion | $\gamma$ | 2.70 | Joint conditions (49a)–(49c) |
| Discount factor | $\beta$ | 0.905 | Joint conditions (49a)–(49c) |
| Initial capitalist wealth | $W_{\text{pre}}$ | 3.893 | Joint conditions (49a)–(49c) |
| *Normalization* | | | |
| Mean of the shock | $\mu$ | $-0.155$ | $\mu = -\alpha\sigma^2/2$, so that $\mathbb{E}[e^{\alpha\varepsilon}] = 1$ |

*Notes:* The risk premium and net foreign bond position are cross-country medians of country averages over years $-5$ to $-1$ before liberalization. The ratios used to calculate $\xi$ and $\omega$ are measured in year $-1$ before liberalization for 13 countries. Dividend volatility is estimated from pooled, country-demeaned annual log changes in real US-dollar dividends on the IFC Global index over all available pre-opening years. Risk aversion, the discount factor, and initial wealth are estimated together to match an operating-profit premium of 0.0675 and a net foreign bond position of $-26.7$ percent of output, and to satisfy $\mathbb{E}[W_{t+1} \mid W_t = W_{\text{pre}}] = W_{\text{pre}}$, as in (49a)–(49c). None of these conditions identifies a parameter on its own. The calibrated wealth level equals 2.04 times expected output. The post-liberalization financing and ownership values reported in the table combine the year $-1$ medians with the median country changes from year $-1$ to year $+4$, using the transformations described in the experiments below. Appendix Table D.1 defines each moment.

**Numerical solution.** The state variable of the recursive equilibrium is the aggregate capitalist wealth. With $\gamma > 1$, current portfolio and production choices depend on future investment opportunities through the continuation function $\bar{\Lambda}(W)$. We solve the recursive equilibrium of Section 2.5 over a grid of wealth levels. Given a candidate continuation function, we solve the static equilibrium

at each grid point, including the allocation-dependent continuation weights. Holding these static allocations fixed, we update the continuation function and saving policy. We repeat until the static allocations and continuation values are mutually consistent.

We combine the recursive equilibrium solution with the joint calibration of $(\gamma, \beta, W_{\text{pre}})$. The cutoff condition and aggregation give the capital-output ratio,

$$\frac{K}{\mathbb{E}[Y]} = \frac{\alpha(\eta-1)}{\eta\,\lambda\,[R^{f*} - (1-\delta) + \Delta R^{f*}]},$$

the bond-position target (49b) gives $W/\mathbb{E}[Y]$, the capital stock $K = (1+\xi)\bar{z}^{-\eta}W$ gives the share of producers $\bar{z}^{-\eta}$, and $\mathbb{E}[Y] = \text{TFP}^{1/(1-\alpha)}(K/\mathbb{E}[Y])^{\alpha/(1-\alpha)}$, with $\text{TFP} = \big(\eta\bar{z}/(\eta-1)\big)^{\alpha}$, gives the level of output and hence $W_{\text{pre}}$. We then search over $(\gamma, \beta, \Delta)$. For each set of parameters, we solve the recursive equilibrium that requires: investors' first-order condition holds at the target premium, $W_{\text{pre}}$ is stationary, (49c), and the wedge of the equilibrium, $q(\bar{z}) - 1$, equals the given $\Delta$. Appendix D.2 gives the details.

## 6.2 The benchmark economy

Before examining how financial integration affects the aggregate TFP via foreign risk sharing and expansion of financing capacity, we briefly describe what our benchmark economy looks like. This benchmark economy can be considered as an emerging market before liberalization. Table 2 reports the model's fit to the untargeted moments and the values of key model variables at the benchmark. The calibration shows that, before the opening, 90 percent of capitalists produce while 10 percent of capitalists are investors. Producers operate 1.3 units of capital per unit of their own wealth, and investors hold domestic equity worth 2.26 times their wealth and borrow the difference abroad. The occupational wedge is small. The marginal producer's capital is worth 0.37 percent more than its replacement cost, and the average firm of the economy is worth 1.05 times its capital, meaning the model-implied price-to-book ratio is about 1.05.

The model matches well with the macroeconomic variables. The capital-output ratio is 2.35 in the model against 2.47 on average across opening countries based on data from the Penn World Table (Feenstra et al., 2015). The replacement investment[5] is 14.1% of output in the model against 15.3% in the data, and the depreciation rate implied by the data is 0.058, which is close to $\delta = 0.06$.

Two untargeted moments related to asset prices are not matched from the data, and both are due to the model's simplification on firms' characteristics. The excess return on the equity index

[5]Replacement investment is the investment that offsets depreciation. Given the capital stock dynamics $K_{t+1} = (1-\delta)K_t + I_t$ and along a path where capital grows with output at rate $g$, the investment share can be expressed as

$$\frac{I}{Y} = (\delta + g)\,\frac{K}{Y}.$$

We therefore measure replacement investment relative to output as $I/Y - g\,K/Y$, where $I/Y$ is the investment share of GDP at national prices, $g$ the growth rate of real GDP, and $K/Y$ the capital-output ratio, each averaged over the five years before the opening (Feenstra et al., 2015). Its median across the openers is 0.153, against $\delta K/\mathbb{E}[Y] = 0.141$ in the model, which has no trend growth. The depreciation rate implied by these data, $(I/Y - g\,K/Y)/(K/Y)$, has a median of 0.058, close to $\delta = 0.06$.

is 0.8 percent a year in the model against 11.9 percent in the data, and the volatility of the index return is 0.04 against 0.33 (Appendix Table D.2). The firm in the model lives one period, and its undepreciated capital is assumed to be resold at a safe price at the end of the period. This implies that a large part of the firm value comes from undepreciated capital, which is considered as risk free. The only risky part is the operating profit of the period that is subject to the country's aggregate risk. Therefore, the premium on the firm's equity return is rather small and the equity index is not as volatile. The same feature results in the model's calibrated response of firm values and entry to foreign risk sharing being small, as Appendix D.3 shows. In other words, the model's estimate of the risk-sharing channel is a conservative one. The volatility of annual changes in log dividends is also higher in the model than in the data, 0.446 against 0.322, because the model's shock is independent over time (Appendix D.2). The model's price-to-book ratio is 1.05, against an unlevered price-to-book ratio of 1.19 for the index, because a firm that lives one period is worth its capital plus one period's rent.

Table 2: The benchmark economy before the opening

| | Data | Model |
|---|---|---|
| *Untargeted moment* | | |
| Capital / output | 2.471 | 2.351 |
| Replacement investment / output, $I/Y - g\,K/Y$ | 0.153 | 0.141 |
| Implied depreciation rate (data) against the set $\delta$ (model) | 0.058 | 0.060 |
| Price-to-book, unlevered, $(E+D)/(B+D)$ | 1.193 | 1.047 |
| Foreign equity / output, portfolio and direct investment | 0.0566 | 0.0499 |
| Excess return on the equity index | 0.119 | 0.008 |
| Volatility of the index return | 0.334 | 0.044 |
| *Model objects* | | |
| Share of capitalists who produce | | 0.896 |
| Investors' equity share $\theta$ | | 2.256 |
| Investors' exposure $\vartheta$ | | 0.273 |
| Average risky share of producers' equity $\bar{\psi}$ | | 0.121 |
| Risky share of the marginal producer's equity $\psi(\bar{z})$ | | 0.086 |
| Occupational wedge $\Delta$ | | 0.0037 |
| Risk-premium term of the cutoff, $\ln\lambda$ | | 0.0638 |
| Wedge term of the cutoff | | 0.0440 |
| *Hedging and saving* | | |
| Tilt of the continuation weights, $c = \ln \mathbb{E}^{\Lambda}[X]$ | | 0.0144 |
| Correlation of $\ln\Lambda$ with the shock, across states | | 0.98 |
| Saving rate, mean of $S(H_t)$ at $W_{\text{pre}}$ | | 0.917 |
| Discount factor $\beta$ | | 0.905 |

*Notes:* Values computed from data are medians across the 14 opening countries where each value for a country is an average over years $-5$ to $-1$ before liberalization, except the foreign equity holdings, which are measured in year $-1$ before liberalization. Model values are the equilibrium given the pre-opening wealth level. The unlevered price-to-book ratio adds the debt of listed firms to their market and book values of equity; its model counterpart is $\bar{q}$. The wedge term of the cutoff is $\ln\left(1+\Delta R^{f*}/(R^{f*}+\delta-1)\right)$ of (35). Appendix Table D.1 defines every row, and Appendix Table D.2 reports the full set of moments.

### 6.3 The effects of foreign risk sharing and financing capacity

We use the calibrated economy to separate the effects of more foreign risk sharing from those of an expansion in financing capacity. On impact, both changes lower the risk premium and raise output in the benchmark, but they have opposite effects on aggregate TFP. Foreign risk sharing encourages less productive capitalists to enter production, whereas greater financing capacity expands the capital operated by each producer and induces the least productive producers to exit.

First, we estimate the changes in foreign risk sharing $\omega$ and financing capacity $\xi$ from the data. We measure the change in foreign risk sharing by the percentage-point change in $\omega$, which captures the additional share of outside equity held by foreign investors. We measure the change in financing expansion by $\Delta \ln(1+\xi)$. Since producers operate a total amount of capital equal to $(1+\xi)W^P$, this measure captures the proportional increase in capital given a wealth level. The pre-opening values for those two parameters are $\omega_{\text{pre}} = 0.094$ and $\xi_{\text{pre}} = 0.290$ as shown in Table 2. We then calculate the post-opening changes using the median change in the foreign share of listed outside equity between the year before liberalization and year +4 after the liberalization:

$$\omega_{\text{post}} = \omega_{\text{pre}} + \text{median}_c\big(\omega_{c,+4} - \omega_{c,-1}\big) = 0.249.$$

In other words, the post-opening value is the benchmark's initial share plus the median country change after five years of the liberalization.

Similarly, we measure $\xi_{+4}$ following the same construction in the calibration, take the median change in $\ln(1+\xi)$ and apply it to the benchmark:

$$\begin{aligned}\Delta \ln(1+\xi) &= \text{median}_c\left[\ln(1+\xi_{c,+4}) - \ln(1+\xi_{c,-1})\right] = 0.0683,\\ \xi_{\text{post}} &= (1+\xi_{\text{pre}})e^{\Delta \ln(1+\xi)} - 1 = 0.381.\end{aligned}$$

Appendix D.1 documents the construction and Appendix D.4 the alternative measures.

We first examine the impact of increased foreign risk sharing and financing capacity separately, holding all other parameters fixed at their pre-opening values, and then consider the joint effect of both changes. Table 3 shows the immediate effects, where the first two columns show the effects of increased foreign risk sharing and financing capacity separately, and the third column shows their combined effect.

Table 3: Foreign risk sharing and financing capacity: immediate effects

| | Risk sharing $\omega$ 0.094 $\to$ 0.249 | Financing $\xi$ 0.290 $\to$ 0.381 | Both changes $\omega$ and $\xi$ |
|---|---|---|---|
| Operating-profit premium (pp) | −0.42 | −0.50 | −1.03 |
| TFP (%) | −0.15 | 0.45 | 0.28 |
| Expected output (%) | 0.30 | 1.38 | 1.72 |
| Number of producers (%) | 1.34 | −4.01 | −2.48 |
| Capital per producer (%) | 0.00 | 6.83 | 6.83 |

*Notes:* Each experiment compares the new equilibrium at $W_{\text{pre}}$ with the pre-opening equilibrium, using the continuation values of its own recursive equilibrium. The premium is $\mathbb{E}[R^D] - R^{f*}$, and its change is in percentage points. Quantity changes are $100\,\Delta \ln$, described as percentage changes in the text. Capital per producer is $(1+\xi)W$.

With the financing capacity unchanged, the increased foreign risk sharing around the liberalization events lowers the premium by 0.42 percentage points. As foreign investors bear more domestic risk, the required return on risky capital falls, making producing more attractive than investing. Capitalists just below the initial productivity cutoff enter production, which leads to a rise in the number of producers by 1.34 percent. The entrants are less productive than the incumbent producers, so TFP falls by 0.15 percent. However, the total expected output still increases by 0.3 percent due to capital deepening.

With the foreign holding of domestic equity index share fixed, the increased financing capacity around liberalization events raises capital per producer by 6.83 percent. As producers are able to operate at a larger scale, diminishing returns reduce the return to capital, and thus it becomes more attractive for the least productive producers to become investors. The number of producers falls by 4.01 percent, TFP rises by 0.45 percent and the expected output rises by 1.38 percent. The risk premium also falls by 0.50 percentage points as the occupational adjustment changes domestic investors' exposure to risk. Higher financing capacity increases each producer's demand for outside finance, putting upward pressure on the risk premium. At the same time, some producers become investors, expanding the wealth available to supply as outside finance. In the calibrated economy, investors' aggregate wealth rises by 34 percent, while total outside equity demand rises by 26 percent. The expansion in financing supply therefore dominates, reducing investors' risk exposure per unit of wealth and lowering the premium by 0.50 percentage points.

Appendix D.5 decomposes the productivity responses into the risk-premium, occupational-wedge, and capital-deepening terms for each of the two changes. The calibrated example shows that a lower premium is not a sufficient statistic for the productivity response. When the lower premium is caused by the increased risk sharing, the aggregate TFP can fall as capital can be allocated to less productive producers. If the lower premium is driven by an increase in financing capacity, the productivity gain can be substantial.

**Dynamics and persistence.** Figure 8 shows the dynamic effect after the initial increase in foreign risk sharing and the financing capacity as described above. Since the economy is subject to aggregate shocks, we also simulate 20,000 histories to capture the variation in outcomes due to

random shocks. The shaded bands show the 2.5th–97.5th percentiles of the effects across 20,000 simulated histories. Upon a shock of foreign risk sharing, the initial output gain fades and the entry of less productive producers leaves a persistent productivity loss. TFP is 0.18 percent below its initial level in the long run, and the output becomes lower than its initial level after around 26 years. However, when the financing capacity improves, the productivity gain persists at 0.46 percent, and output is 1.46 percent higher in the long run.

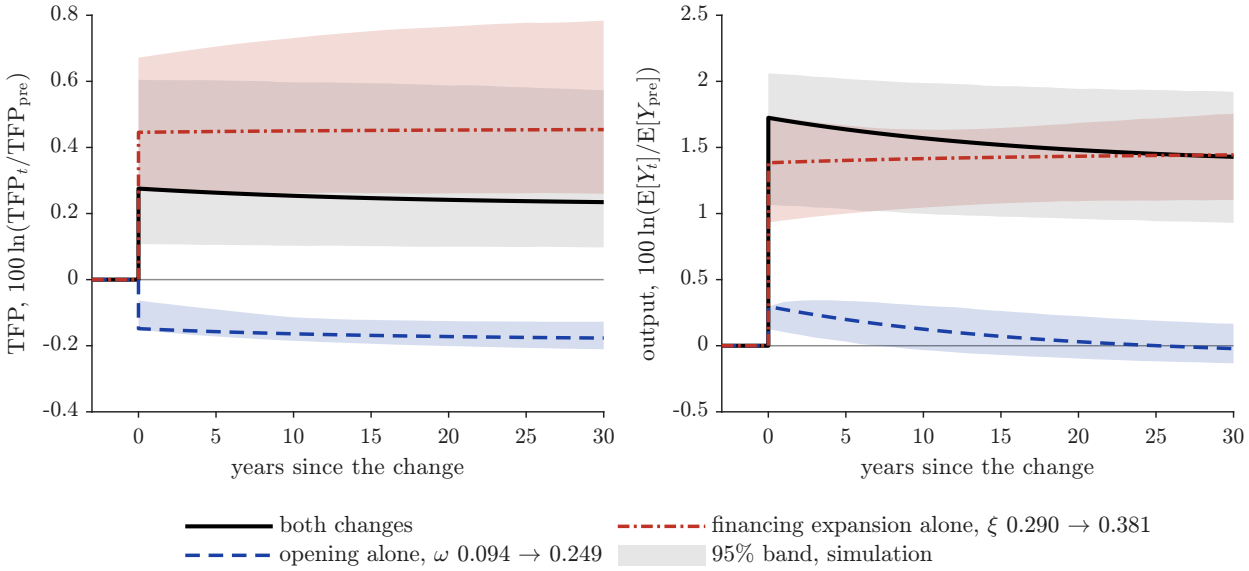



Figure 8: Productivity and output after the two changes

*Notes:* Effects in $100\,\Delta \ln$; the left panel is TFP changes. The permanent changes arrive in year 0: $\omega$ rises from 0.094 to 0.249 under foreign risk sharing (dashed), $\xi$ from 0.290 to 0.381 under the financing expansion (dash-dot), or both change (solid). These lines follow conditional-mean wealth paths from $W_{\text{pre}}$, using each regime's recursive equilibrium. Shaded bands show the 2.5th–97.5th percentiles of the effects across 20,000 simulated histories, comparing economies with and without the changes under identical shocks. The bands describe variation across shock histories at fixed parameters, rather than uncertainty about the estimated parameters. The lines need not lie at the center of the bands, because the responses are nonlinear in wealth. Appendix D.5 shows details of the simulation design.

The total effect of both foreign risk sharing and financing capacity turns out to be a 0.23 percent increase in TFP and a 1.36 percent increase in output in the long run. However, the foreign risk sharing attenuates the effect of the financing expansion by 51 percent. The initial output gain is larger than the sum of the two separate effects, as the occupational adjustment from the financing expansion amplifies the effect of foreign risk sharing. But in the long run, the joint effect on TFP is smaller than the sum of the two separate effects, as the entry of less productive producers due to foreign risk sharing offsets some of the productivity gain from the financing expansion.

For the benchmark economy, the joint effect on both TFP and output is positive, as the impact of financing capacity expansion dominates the effect of foreign risk sharing. However, if financial liberalization only increases foreign risk sharing without improving financing capacity, the joint effect on TFP can be negative.

# 7 Conclusion

This paper studies how global financial integration affects aggregate productivity through the pricing of risk and the selection of producers. We develop a small open economy model in which heterogeneous capitalists choose between producing and investing, and equilibrium equity prices determine who produces. Greater foreign participation in domestic equity markets shifts part of the country's aggregate risk abroad and lowers the domestic risk premium. This results in higher equity valuations, which lower the hurdle rate and draw less productive capitalists into production. This increases the level of capital and output, but lowers aggregate TFP. The entry of producers increases wages, so income shifts toward the workers in the economy. In addition, part of the equity return accrues to the foreign shareholders. These two effects of increasing wages and foreign ownership can weaken domestic wealth accumulation and reverse the initial output gain in the long run. An expansion of domestic financing capacity works in the opposite direction, where producers operate more capital, wages rise, and the least productive producers exit.

We show that the equity market liberalizations of emerging markets between 1986 and 1997 are consistent with the model predictions. After liberalization, foreign holdings of domestic equity rose, equity prices rose sharply, and more companies listed. Countries with larger increases in equity prices or listings subsequently experienced slower TFP growth.

In a benchmark economy calibrated using a subset of the liberalization episodes, greater foreign risk sharing and the financing expansion both lower the risk premium but have opposite effects on productivity. Greater foreign risk sharing alone raises output only temporarily and lowers TFP persistently. In contrast, the financing expansion observed in the data raises both output and TFP persistently. When the two occur together, the financing expansion dominates, and both output and TFP increase in the long run. A lower risk premium is therefore not a sufficient statistic for the productivity effects of integration. These effects depend on whether the fall reflects foreign investors absorbing domestic risk or productive firms gaining access to finance. As a result, opening to foreign investors raises productivity when it is accompanied by reforms that expand firms' access to finance.

## References


Acemoglu, D. and F. Zilibotti (1997): "Was Prometheus Unbound by Chance? Risk, Diversification, and Growth," *Journal of Political Economy*, 105, 709–751.

Acharya, V. V., M. Crosignani, T. Eisert, and C. Eufinger (2022): "Zombie Lending: Theoretical, International, and Historical Perspectives," *Annual Review of Financial Economics*, 14, 21–38.

Aoki, K., G. Benigno, and N. Kiyotaki (2010): "Adjusting to Capital Account Liberalization," CEPR Discussion Paper 8087, Centre for Economic Policy Research.

Asriyan, V., L. Laeven, A. Martin, A. Van der Ghote, and V. Vanasco (2025): "Falling Interest Rates and Credit Reallocation: Lessons from General Equilibrium," *Review of Economic Studies*, 92, 2197–2227.

Ateş, S. T. and F. E. Saffie (2021): "Fewer but Better: Sudden Stops, Firm Entry, and Financial Selection," *American Economic Journal: Macroeconomics*, 13, 304–356.

Banerjee, A. V. and B. Moll (2010): "Why Does Misallocation Persist?" *American Economic Journal: Macroeconomics*, 2, 189–206.

Bartelsman, E., J. Haltiwanger, and S. Scarpetta (2013): "Cross-Country Differences in Productivity: The Role of Allocation and Selection," *American Economic Review*, 103, 305–334.

Bau, N. and A. Matray (2023): "Misallocation and Capital Market Integration: Evidence from India," *Econometrica*, 91, 67–106.

Bekaert, G. and C. R. Harvey (2000): "Foreign Speculators and Emerging Equity Markets," *Journal of Finance*, 55, 565–613.

Bekaert, G., C. R. Harvey, and R. L. Lumsdaine (2002): "The Dynamics of Emerging Market Equity Flows," *Journal of International Money and Finance*, 21, 295–350.

Bekaert, G., C. R. Harvey, and C. Lundblad (2005): "Does Financial Liberalization Spur Growth?" *Journal of Financial Economics*, 77, 3–55.

——— (2011a): "Financial Openness and Productivity," *World Development*, 39, 1–19.

Bekaert, G., C. R. Harvey, C. T. Lundblad, and S. Siegel (2011b): "What Segments Equity Markets?" *Review of Financial Studies*, 24, 3841–3890.

Benigno, G., N. Converse, and L. Fornaro (2015): "Large Capital Inflows, Sectoral Allocation, and Economic Performance," *Journal of International Money and Finance*, 55, 60–87.

Bertrand, M., A. Schoar, and D. Thesmar (2007): "Banking Deregulation and Industry Structure: Evidence from the French Banking Reforms of 1985," *Journal of Finance*, 62, 597–628.

Bonfiglioli, A. (2008): "Financial Integration, Productivity and Capital Accumulation," *Journal of International Economics*, 76, 337–355.

Buera, F. J., J. P. Kaboski, and Y. Shin (2011): "Finance and Development: A Tale of Two Sectors," *American Economic Review*, 101, 1964–2002.

Buera, F. J. and B. Moll (2015): "Aggregate implications of a credit crunch: The importance of heterogeneity," *American Economic Journal: Macroeconomics*, 7, 1–42.

Buera, F. J. and Y. Shin (2017): "Productivity Growth and Capital Flows: The Dynamics of Reforms," *American Economic Journal: Macroeconomics*, 9, 147–185.

Caballero, R. J., E. Farhi, and P.-O. Gourinchas (2017): "Rents, Technical Change, and Risk Premia: Accounting for Secular Trends in Interest Rates, Returns on Capital, Earning Yields, and Factor Shares," *American Economic Review*, 107, 614–620.

Caballero, R. J., T. Hoshi, and A. K. Kashyap (2008): "Zombie Lending and Depressed Restructuring in Japan," *American Economic Review*, 98, 1943–1977.

Callaway, B. and P. H. C. Sant'Anna (2021): "Difference-in-Differences with Multiple Time Periods," *Journal of Econometrics*, 225, 200–230.

Chari, A. and P. B. Henry (2004): "Risk Sharing and Asset Prices: Evidence from a Natural Experiment," *Journal of Finance*, 59, 1295–1324.

——— (2008): "Firm-Specific Information and the Efficiency of Investment," *Journal of Financial Economics*, 87, 636–655.

Chinn, M. D. and H. Ito (2006): "What Matters for Financial Development? Capital Controls, Institutions, and Interactions," *Journal of Development Economics*, 81, 163–192.

Čihák, M., A. Demirgüç-Kunt, E. Feyen, and R. Levine (2012): "Benchmarking Financial Systems around the World," Policy Research Working Paper 6175, World Bank, Washington, DC.

Coeurdacier, N. and H. Rey (2013): "Home Bias in Open Economy Financial Macroeconomics," *Journal of Economic Literature*, 51, 63–115.

Dahlquist, M., L. Pinkowitz, R. M. Stulz, and R. Williamson (2003): "Corporate Governance and the Home Bias," *Journal of Financial and Quantitative Analysis*, 38, 87–110.

David, J. M., L. Schmid, and D. Zeke (2022): "Risk-Adjusted Capital Allocation and Misallocation," *Journal of Financial Economics*, 145, 684–705.

Dávila, E. and A. Korinek (2018): "Pecuniary Externalities in Economies with Financial Frictions," *Review of Economic Studies*, 85, 352–395.

Del Negro, M., D. Giannone, M. P. Giannoni, and A. Tambalotti (2019): "Global Trends in Interest Rates," *Journal of International Economics*, 118, 248–262.

Errunza, V. and E. Losq (1985): "International Asset Pricing under Mild Segmentation: Theory and Test," *Journal of Finance*, 40, 105–124.

Evans, D. S. and B. Jovanovic (1989): "An Estimated Model of Entrepreneurial Choice under Liquidity Constraints," *Journal of Political Economy*, 97, 808–827.

Fama, E. F. and K. R. French (2002): "The Equity Premium," *Journal of Finance*, 57, 637–659.

——— (2004): "New Lists: Fundamentals and Survival Rates," *Journal of Financial Economics*, 73, 229–269.

Farhi, E. and F. Gourio (2018): "Accounting for Macro-Finance Trends: Market Power, Intangibles, and Risk Premia," *Brookings Papers on Economic Activity*, 49, 147–223.

Feenstra, R. C., R. Inklaar, and M. P. Timmer (2015): "The Next Generation of the Penn World Table," *American Economic Review*, 105, 3150–3182.

Gollin, D. (2002): "Getting Income Shares Right," *Journal of Political Economy*, 110, 458–474.

Gopinath, G., S. Kalemli-Ozcan, L. Karabarbounis, and C. Villegas-Sanchez (2017): "Capital Allocation and Productivity in South Europe," *Quarterly Journal of Economics*, 132, 1915–1967.

Gourinchas, P.-O. and O. Jeanne (2006): "The Elusive Gains from International Financial Integration," *Review of Economic Studies*, 73, 715–741.

——— (2013): "Capital Flows to Developing Countries: The Allocation Puzzle," *Review of Economic Studies*, 80, 1484–1515.

Greenwald, B. C. and J. E. Stiglitz (1986): "Externalities in Economies with Imperfect Information and Incomplete Markets," *Quarterly Journal of Economics*, 101, 229–264.

Gupta, N. and K. Yuan (2009): "On the Growth Effect of Stock Market Liberalizations," *Review of Financial Studies*, 22, 4715–4752.

Henry, P. B. (2000a): "Stock Market Liberalization, Economic Reform, and Emerging Market Equity Prices," *Journal of Finance*, 55, 529–564.

——— (2000b): "Do Stock Market Liberalizations Cause Investment Booms?" *Journal of Financial Economics*, 58, 301–334.

——— (2002): "Is Disinflation Good for the Stock Market?" *Journal of Finance*, 57, 1617–1648.

——— (2003): "Capital-Account Liberalization, the Cost of Capital, and Economic Growth," *American Economic Review*, 93, 91–96.

——— (2007): "Capital Account Liberalization: Theory, Evidence, and Speculation," *Journal of Economic Literature*, 45, 887–935.

Hopenhayn, H. A. (1992): "Entry, Exit, and Firm Dynamics in Long Run Equilibrium," *Econometrica*, 60, 1127–1150.

Hsieh, C.-T. and P. J. Klenow (2009): "Misallocation and Manufacturing TFP in China and India," *Quarterly Journal of Economics*, 124, 1403–1448.

Itskhoki, O. and B. Moll (2019): "Optimal Development Policies with Financial Frictions," *Econometrica*, 87, 139–173.

Kaminsky, G. L. and C. M. Reinhart (1999): "The Twin Crises: The Causes of Banking and Balance-of-Payments Problems," *American Economic Review*, 89, 473–500.

Kaminsky, G. L. and S. L. Schmukler (2008): "Short-Run Pain, Long-Run Gain: Financial Liberalization and Stock Market Cycles," *Review of Finance*, 12, 253–292.

Kerr, W. R. and R. Nanda (2009): "Democratizing Entry: Banking Deregulations, Financing Constraints, and Entrepreneurship," *Journal of Financial Economics*, 94, 124–149.

Kose, M. A., E. Prasad, K. Rogoff, and S.-J. Wei (2009): "Financial Globalization: A Reappraisal," *IMF Staff Papers*, 56, 8–62.

Laeven, L. and F. Valencia (2020): "Systemic Banking Crises Database II," *IMF Economic Review*, 68, 307–361.

Lane, P. R. and G. M. Milesi-Ferretti (2007): "The External Wealth of Nations Mark II: Revised and Extended Estimates of Foreign Assets and Liabilities, 1970–2004," *Journal of*

*International Economics*, 73, 223–250.

Larrain, M. and S. Stumpner (2017): "Capital Account Liberalization and Aggregate Productivity: The Role of Firm Capital Allocation," *Journal of Finance*, 72, 1825–1858.

Levchenko, A. A., R. Rancière, and M. Thoenig (2009): "Growth and Risk at the Industry Level: The Real Effects of Financial Liberalization," *Journal of Development Economics*, 89, 210–222.

Liu, E., A. Mian, and A. Sufi (2022): "Low Interest Rates, Market Power, and Productivity Growth," *Econometrica*, 90, 193–221.

Lucas, Robert E., J. (1978): "On the Size Distribution of Business Firms," *Bell Journal of Economics*, 9, 508–523.

MacKinnon, J. G., M. Ø. Nielsen, and M. D. Webb (2023): "Cluster-Robust Inference: A Guide to Empirical Practice," *Journal of Econometrics*, 232, 272–299.

Maggiori, M., B. Neiman, and J. Schreger (2020): "International Currencies and Capital Allocation," *Journal of Political Economy*, 128, 2019–2066.

Melitz, M. J. (2003): "The Impact of Trade on Intra-Industry Reallocations and Aggregate Industry Productivity," *Econometrica*, 71, 1695–1725.

Mendoza, E. G., V. Quadrini, and J.-V. Ríos-Rull (2009): "Financial Integration, Financial Development, and Global Imbalances," *Journal of Political Economy*, 117, 371–416.

Midrigan, V. and D. Y. Xu (2014): "Finance and Misallocation: Evidence from Plant-Level Data," *American Economic Review*, 104, 422–458.

Moll, B. (2014): "Productivity Losses from Financial Frictions: Can Self-Financing Undo Capital Misallocation?" *American Economic Review*, 104, 3186–3221.

Obstfeld, M. (1994): "Risk-Taking, Global Diversification, and Growth," *American Economic Review*, 84, 1310–1329.

Prasad, E. S., R. G. Rajan, and A. Subramanian (2007): "Foreign Capital and Economic Growth," *Brookings Papers on Economic Activity*, 2007, 153–230.

Reis, R. (2013): "The Portuguese Slump and Crash and the Euro Crisis," *Brookings Papers on Economic Activity*, 2013, 143–193.

Restuccia, D. and R. Rogerson (2008): "Policy Distortions and Aggregate Productivity with Heterogeneous Establishments," *Review of Economic Dynamics*, 11, 707–720.

Rodrik, D. and A. Subramanian (2009): "Why Did Financial Globalization Disappoint?" *IMF Staff Papers*, 56, 112–138.

Saffie, F., L. Varela, and K.-M. Yi (2026): "The Micro and Macro Dynamics of Capital Inflows," *Review of Economic Studies*, nBER Working Paper 27371.

Schivardi, F., E. Sette, and G. Tabellini (2022): "Credit Misallocation During the European Financial Crisis," *Economic Journal*, 132, 391–423.

Song, Z., K. Storesletten, and F. Zilibotti (2011): "Growing Like China," *American Economic Review*, 101, 196–233.

Stulz, R. M. (1981): "On the Effects of Barriers to International Investment," *Journal of Finance*,

36, 923–934.

Varela, L. (2018): "Reallocation, Competition, and Productivity: Evidence from a Financial Liberalization Episode," *Review of Economic Studies*, 85, 1279–1313.

# A The Model: Proofs and Supporting Results

This appendix proves the results of Section 2. Appendices A.1 to A.4 concern the model of Section 2 for general CRRA utility, including the occupational choice rule (Proposition 2.2), aggregation and the law of motion of wealth, asset prices (Proposition 2.1), the productivity cutoff and the occupational wedge (Proposition 2.3). Appendix A.5 characterizes the equilibrium to the two conditions that Section 3 analyzes. Appendix B proves the log-utility results of Section 3.

## A.1 Dynamic occupational choices

We prove Proposition 2.2 which reduces the capitalist's dynamic problem to the weighted certainty-equivalent rule (15), and we bound the continuation coefficient and the weights by the value of holding only the world bond (Lemma A.1).

**The capitalist's problem.** A capitalist with net worth $a$ and productivity $z$ chooses an occupation $o \in \{P, I\}$, given the optimal production and portfolio choice decisions $x$ within the occupation, and chooses how much cash on hand to save at the end of the period. For a producer, $x$ is the equity issuance per unit of net worth, $x = E/a \in [0, \xi]$, which sets the capital stock $k = (1 + x)\, a$. Labor is hired after the shock and the producer's labor demand $l(\varepsilon)$ is contingent on the shock. Taking the wage as given, the labor demand that maximizes operating profit in every state, (3), also maximizes the producer's return in every state. We substitute this rule into $R^K$, and $x$ is the producer's only remaining choice. For an investor, $x$ is the equity share $\theta$ where its exposure is $\theta\bar{\psi} \in [0, 1]$ by Assumption A0 (Section 2.4). Since the productivity draws and the aggregate shocks are i.i.d. across time and individuals, the gross return $R^o(z; x)$ on net worth does not depend on the capitalist's net worth $a$. But it is affected by the realization of the shock and the aggregate wealth of the capitalists $W$ through labor wages. In other words, the aggregate wealth $W$ is the only state variable of the economy. The value function of capitalists at the beginning of the period is given by (12) of Section 2.3, which we write as

$$\begin{aligned} V(a, z; W) &= \max_{o \in \{P,I\}} V^o(a, z; W), \qquad V^o(a, z; W) = \max_{x}\ \mathbb{E}\left[U\left(R^o(z; x)\, a; G(W, \varepsilon)\right)\right], \\ U(m; W') &\equiv \max_{0 \le a' \le m} \left\{ u(m - a') + \beta\, \mathbb{E}_{z'}\left[V(a', z'; W')\right] \right\}, \qquad u(c) = \frac{c^{1-\gamma}}{1-\gamma}, \end{aligned} \tag{52}$$

with $u(c) = \ln c$ when $\gamma = 1$. Here $W' \equiv G(W, \varepsilon)$ is next period's aggregate wealth, and the law of motion $G$ follows (41) in equilibrium. The expectation $\mathbb{E}$ is over the shock $\varepsilon$, $\mathbb{E}_{z'}$ is over next period's draw, and $U(m; W')$ is the value of end-of-period cash on hand $m$.

*Proof of Proposition 2.2.*
The proof has three steps. Step 1 shows that the occupation chosen today has no option value. Step 2 solves for the value of each occupation by conjecturing the form of the value function and verifying it. Step 3 compares the two values.

*Step 1: the occupational choice has no option value.* The capitalist enters next period with the state $(a', z'; W')$. The draw $z'$ is i.i.d., and its distribution is the same whichever occupation the capitalist chooses today. Since the shock is i.i.d., aggregate wealth is the only aggregate state, and next period's aggregate wealth $W' = G(W, \varepsilon)$ depends only on current aggregate wealth and the shock. Hence, for every $a'$ and $W'$, the expected continuation value $\mathbb{E}_{z'}\big[V(a', z'; W')\big]$ is the same for a capitalist who produces today and for one who invests, and so is the end-of-period value $U(m; W')$ in (52). The two occupations differ only in the stochastic return and the resulting cash on hand $m = R^o(z; x)\, a$ that they deliver, and both returns are valued with the same utility function $U$. The occupational choice therefore has no option value.

*Step 2: the value of each occupation.* We conjecture the form of the value function

$$V(a, z; W) = v(z; W)\, \frac{a^{1-\gamma}}{1-\gamma}, \tag{53}$$

for a function $v$ of productivity and aggregate wealth, and we verify the conjecture below.

We first solve the end-of-period saving problem under (53). Because $z'$ is independent of $a'$ and $W'$,

$$\mathbb{E}_{z'}\big[V(a', z'; W')\big] = \bar{\Lambda}(W')\, \frac{a'^{1-\gamma}}{1-\gamma}, \qquad \bar{\Lambda}(W') \equiv \mathbb{E}_{z'}\big[v(z'; W')\big] > 0,$$

and $\bar{\Lambda}(W')$ is the same for all capitalists. Both terms of the saving problem have infinite marginal value at zero and are strictly concave in $a'$, so the first-order condition $(m - a')^{-\gamma} = \beta \bar{\Lambda}(W')\, a'^{-\gamma}$ characterizes the solution and gives the saving policy $a' = s(W')\, m$, with $s(W') = b/(1+b)$ and $b \equiv \big(\beta \bar{\Lambda}(W')\big)^{1/\gamma}$. Substituting, and using $\beta \bar{\Lambda}(W') = b^\gamma$,

$$U(m; W') = \frac{m^{1-\gamma}}{1-\gamma} \left[ (1+b)^{\gamma-1} + b^\gamma \left( \frac{b}{1+b} \right)^{1-\gamma} \right] = \Lambda(W')\, \frac{m^{1-\gamma}}{1-\gamma},$$

with $\Lambda(W') = (1+b)^\gamma = \big(1 - s(W')\big)^{-\gamma}$.

We next derive the value function of each occupation. The next period's wealth $W_{t+1} = G(W, \varepsilon)$ depends on the shock, and thus $\Lambda_t \equiv \Lambda(W_{t+1})$ is a random variable, common to all capitalists. Given the return $R^o(z; x)$ of occupation $o$ and the definitions of $\mathbb{E}^\Lambda$ and of $\mathrm{CE}^\Lambda$ in (13), the value function of occupation $o$ is

$$\mathbb{E}\Big[U\big(R^o(z; x)\, a; W_{t+1}\big)\Big] = \mathbb{E}\Big[\Lambda_t\, R^o(z; x)^{1-\gamma}\Big] \frac{a^{1-\gamma}}{1-\gamma} = \mathbb{E}[\Lambda_t]\, \frac{\mathrm{CE}^\Lambda\big(R^o(z; x)\big)^{1-\gamma}}{1-\gamma}\, a^{1-\gamma}.$$

And the value function of occupation $o$ is

$$V^o(a, z; W) = \max_x\ \mathbb{E}\left[U\big(R^o(z; x)\, a; G(W, \varepsilon)\big)\right] = \mathbb{E}[\Lambda_t]\, \frac{\max_x \mathrm{CE}^\Lambda\big(R^o(z; x)\big)^{1-\gamma}}{1-\gamma}\, a^{1-\gamma}.$$

For the producer's problem, the optimal scale choice is state-independent, and for the investor's problem, the optimal portfolio choice is consistent with optimizing $\mathrm{CE}^\Lambda\big(R^I(z; x\theta)\big)$. The optimal

financial choice within occupation $o$ therefore maximizes $\mathrm{CE}^{\Lambda}\big(R^o(z;x)\big)$ over $x$.

It remains to verify the conjectured form. By (14),

$$V(a,z;W)=\max_{o\in\{P,I\}}V^o(a,z;W)=v(z;W)\,\frac{a^{1-\gamma}}{1-\gamma},\qquad v(z;W)=\mathbb{E}[\Lambda_t]\,\mathrm{CE}^{\Lambda*}(z)^{1-\gamma}>0,$$

where $\mathrm{CE}^{\Lambda*}(z)\equiv\max_x\mathrm{CE}^{\Lambda}\big(R^o(z,x)\big)$, and the weights and the returns are those of the period with wealth $W$. The value has the form (53), which verifies the conjecture, and averaging $v$ over the draw gives the continuation coefficient $\bar\Lambda(W)=\mathbb{E}_z\big[v(z;W)\big]$ of the saving problem one period earlier. Lemma A.1 below bounds $\bar\Lambda$, $s$, and $\Lambda$ at every level of wealth and shows that the transversality condition holds along the plan these choices generate, so the solution of (52) constructed here is the capitalist's expected lifetime utility.

*Step 3: the occupational choice.* By (14) the values of the two occupations share the factor $\mathbb{E}[\Lambda_t]\,a^{1-\gamma}/(1-\gamma)$, and each is strictly increasing in the weighted certainty equivalent of its return. The capitalist therefore produces if and only if $\mathrm{CE}^{\Lambda}\big(R^P(z)\big)\geq\mathrm{CE}^{\Lambda}\big(R^I\big)$, which is (15).

*Log utility.* Let $\gamma=1$ and we conjecture the value function has the form

$$V(a,z;W)=\frac{\ln a}{1-\beta}+g(z;W)$$

for a function $g$ of productivity and aggregate wealth. Under the conjecture, $\mathbb{E}_{z'}\big[V(a',z';W')\big]=\ln a'/(1-\beta)+\bar g(W')$ with $\bar g(W')\equiv\mathbb{E}_{z'}\big[g(z';W')\big]$. The first-order condition of the saving problem, $1/(m-a')=\beta/\big[(1-\beta)a'\big]$, gives $a'=\beta m$, and

$$U(m;W')=\frac{\ln m}{1-\beta}+h(W'),\qquad h(W')=\ln(1-\beta)+\frac{\beta\ln\beta}{1-\beta}+\beta\,\bar g(W'),$$

For the return $R^o(z;x)$ of a financial choice $x$ in occupation $o$,

$$\mathbb{E}\Big[U\big(R^o(z;x)\,a;W_{t+1}\big)\Big]=\frac{\mathbb{E}\big[\ln R^o(z;x)\big]+\ln a}{1-\beta}+\mathbb{E}\big[h(W_{t+1})\big].$$

The last term is the same for every choice of the capitalist. The financial choice maximizes $\mathbb{E}\big[\ln R^o(z;x)\big]$, which is equivalent to maximizing the certainty equivalent $\mathrm{CE}(R)=\exp\mathbb{E}[\ln R]$, and

$$V(a,z;W)=\frac{\ln a}{1-\beta}+g(z;W),\qquad g(z;W)=\max_{o\in\{P,I\}}\frac{\mathbb{E}\big[\ln R^o(z)\big]}{1-\beta}+\mathbb{E}\big[h(W_{t+1})\big],$$

which verifies the conjecture. The capitalist produces if and only if $\mathbb{E}\big[\ln R^P(z)\big]\geq\mathbb{E}\big[\ln R^I\big]$. □

**Bounds from holding the bond.** As a capitalist, the lower bound of his lifetime utility is to invest in the world risk-free bond forever. Thus, the condition $\beta\big(R^{f*}\big)^{1-\gamma}<1$ of Proposition 2.2 bounds the continuation coefficient and the stochastic weights $\Lambda$ from below. It holds for every $\beta<1$ when $\gamma>1$ and $R^{f*}\geq1$.

**Lemma A.1** (Bond-only bounds and transversality)**.** *Let $\gamma > 1$ and $\beta\big(R^{f*}\big)^{1-\gamma} < 1$, so that $\bar{s} \equiv \big(\beta(R^{f*})^{1-\gamma}\big)^{1/\gamma} \in (0,1)$. For any prices and any law of motion of aggregate wealth that capitalists take as given:*

*(a) the lifetime utility of a capitalist is at least that of holding only the world bond and saving the fraction $\bar{s}$ of cash on hand in every period:*

$$V(a,z;W) \geq \big(R^{f*}\big)^{1-\gamma}(1-\bar{s})^{-\gamma}\,\frac{a^{1-\gamma}}{1-\gamma}, \qquad U(m;W') \geq (1-\bar{s})^{-\gamma}\,\frac{m^{1-\gamma}}{1-\gamma}$$

*for every $a$, $m$, $z$, $W$, and $W'$;*

*(b) the continuation coefficient, the saving rate, and the weight satisfy*

$$0 < \bar{\Lambda}(W') \leq \big(R^{f*}\big)^{1-\gamma}(1-\bar{s})^{-\gamma}, \qquad 0 < s(W') \leq \bar{s}, \qquad 1 < \Lambda(W') \leq (1-\bar{s})^{-\gamma},$$

*so that $S(H) \leq \bar{s}$ at any aggregate cash on hand $H$;*

*(c) along the optimal occupational and consumption paths,*

$$\Big|\beta^T\,\mathbb{E}_0\big[V(a_T,z_T;W_T)\big]\Big| \leq \bar{s}^{\,T}\,\Big|\mathbb{E}_0\big[V(a_0,z_0;W_0)\big]\Big| \qquad \text{for every } T \geq 0,$$

*where $\mathbb{E}_0$ is the expectation at the start of period 0, before the productivity draw. Hence the transversality condition $\lim_{T\to\infty}\beta^T\,\mathbb{E}_0\big[V(a_T,z_T;W_T)\big] = 0$ holds.*

*Proof.*
(a) Every capitalist can always invest all of the net worth into the risk-free bond which earns $R^{f*}$ regardless of the aggregate shock, and save the fraction $\bar{s}$ of cash on hand, in every period. From net worth $a$, this leads to a consumption schedule $c_t = (1-\bar{s})R^{f*}\big(\bar{s}R^{f*}\big)^t a$ in period $t$ in every state. Since $\beta\big(R^{f*}\big)^{1-\gamma} = \bar{s}^{\gamma}$, we have $\beta\big(\bar{s}R^{f*}\big)^{1-\gamma} = \bar{s} < 1$, and the plan's lifetime utility is

$$\sum_{t\geq 0}\beta^t\,\frac{\big[(1-\bar{s})R^{f*}\big]^{1-\gamma}\big(\bar{s}R^{f*}\big)^{t(1-\gamma)}a^{1-\gamma}}{1-\gamma} = \frac{\big[(1-\bar{s})R^{f*}\big]^{1-\gamma}}{1-\bar{s}}\,\frac{a^{1-\gamma}}{1-\gamma} = \big(R^{f*}\big)^{1-\gamma}(1-\bar{s})^{-\gamma}\,\frac{a^{1-\gamma}}{1-\gamma}.$$

Since the investor can optimize over all feasible portfolios, the value of the capitalist's problem in (52) is bounded by the value obtained by holding the bond forever. The same argument applies to the end-of-period saving problem, which gives the bound on $U(m;W')$.

(b) By the proof of Proposition 2.2, the value function takes the form

$$V(a,z;W) = v(z;W)\frac{a^{1-\gamma}}{1-\gamma}, \qquad v(z;W) > 0.$$

Since $1-\gamma < 0$, part (a) implies

$$v(z;W) \leq (R^{f*})^{1-\gamma}(1-\bar{s})^{-\gamma}.$$

Averaging over the draw therefore yields the same upper bound for $\bar{\Lambda}(W')$. Together with positivity, this gives

$$0 < b := \big(\beta\bar{\Lambda}(W')\big)^{1/\gamma} \le \frac{\big(\beta(R^{f*})^{1-\gamma}\big)^{1/\gamma}}{1-\bar{s}} = \frac{\bar{s}}{1-\bar{s}}.$$

Because $b \mapsto b/(1+b)$ is increasing, it follows that

$$0 < s(W') = \frac{b}{1+b} \le \bar{s}.$$

Similarly,

$$1 < \Lambda(W') = (1+b)^{\gamma} \le (1-\bar{s})^{-\gamma}.$$

Finally, applying the bound on $s$ to the identity $S(H) = s\big(S(H)H\big)$ gives $0 < S(H) \le \bar{s}$.

(c) Let $\mathcal{F}_T$ denote the information available after observing $z_T$ but before the period-$T$ aggregate shock, and let $\mathcal{G}_T$ denote the information available after that shock but before the next productivity draw $z_{T+1}$. Write

$$V_T := V(a_T, z_T; W_T), \qquad U_T := U(m_T; W_{T+1}), \qquad s_T := s(W_{T+1}).$$

Along the optimal choices, the Bellman equation gives

$$V_T = \mathbb{E}[U_T \mid \mathcal{F}_T].$$

Conditional on $\mathcal{G}_T$, the quantities $m_T$, $W_{T+1}$, and $a_{T+1} = s_T m_T$ are known, while $z_{T+1}$ is an independent new draw. Hence

$$\begin{aligned}\beta\,\mathbb{E}[V_{T+1} \mid \mathcal{G}_T] &= \beta\bar{\Lambda}(W_{T+1})\frac{(s_T m_T)^{1-\gamma}}{1-\gamma} \\ &= s_T\Lambda(W_{T+1})\frac{m_T^{1-\gamma}}{1-\gamma} = s_T U_T,\end{aligned}$$

where the second equality follows from

$$\beta\bar{\Lambda}(W_{T+1})s_T^{1-\gamma} = \left(\frac{s_T}{1-s_T}\right)^{\gamma} s_T^{1-\gamma} = s_T(1-s_T)^{-\gamma} = s_T\Lambda(W_{T+1}).$$

By (b), $0 < s_T \le \bar{s}$. Since $U_T < 0$, this implies

$$0 > s_T U_T \ge \bar{s}\, U_T.$$

Define

$$A_T := \beta^T \mathbb{E}_0[V_T],$$

where $\mathbb{E}_0$ is the conditional expectation on information before the initial productivity draw. Taking

expectations yields

$$A_{T+1} = \beta^{T+1}\mathbb{E}_0[V_{T+1}] = \beta^T\mathbb{E}_0[s_T U_T] \\ \geq \bar{s}\,\beta^T\mathbb{E}_0[U_T] = \bar{s}\,\beta^T\mathbb{E}_0[V_T] = \bar{s}\,A_T.$$

For fixed positive initial wealth, part (a) supplies a finite lower bound uniform in the initial productivity draw, so $-\infty < A_0 < 0$. The bound $0 > A_T \geq \bar{s}^T A_0$ holds at $T = 0$. If it holds at $T$, the recursive inequality gives

$$0 > A_{T+1} \geq \bar{s}\,A_T \geq \bar{s}^{T+1}A_0.$$

Thus, by induction, every $A_T$ is finite and

$$\left|\beta^T\mathbb{E}_0[V_T]\right| \leq \bar{s}^T\left|\mathbb{E}_0[V_0]\right|.$$

Since $0 < \bar{s} < 1$, the right-hand side converges to zero, proving the transversality condition. □

## A.2 Aggregation and wealth dynamics

We derive the aggregate production function and the law of motion of wealth (41), including the payoff $\omega\Pi^e$ that foreign shareholders take out of the country. The derivations use three facts: the producers are the capitalists with $z \geq \bar{z}$, each raises $E = \xi a$, and net worth is independent of the productivity draw. None of them involves the weights, so the results hold for every $\gamma \geq 1$.

**Proposition A.1** (Aggregate production function)**.** *With aggregate labor equal to one, output takes the Cobb–Douglas form* (32)*. Labor income is* $w = (1-\alpha)Y$*, aggregate operating profit is*

$$\Pi^A \equiv \int_{i\in\mathcal{P}} \pi\,\tilde{z}_i k_i\,di = D\tilde{K} = \alpha Y,$$

*and the wage and the aggregate factor are given by* (33)*.*

*Proof.*
By (3), producer $i$'s labor demand is $l_i = ((1-\alpha)/w)^{1/\alpha}\tilde{z}_i k_i$. Aggregating across producers and imposing labor-market clearing gives

$$1 = \int_{\mathcal{P}} l_i\,di = \left(\frac{1-\alpha}{w}\right)^{1/\alpha} e^{\varepsilon}\tilde{K},$$

so the state-contingent wage is

$$w(\varepsilon) = (1-\alpha)(e^{\varepsilon}\tilde{K})^{\alpha}.$$

Substituting this wage into each producer's output and aggregating yields

$$y_i = \tilde{z}_i k_i (e^{\varepsilon}\tilde{K})^{\alpha-1}, \qquad Y = (e^{\varepsilon}\tilde{K})^{\alpha-1}\int_{\mathcal{P}} \tilde{z}_i k_i\,di = (e^{\varepsilon}\tilde{K})^{\alpha}.$$

The aggregate labor income is therefore $wL = (1-\alpha)Y$, and aggregate operating profit is

$$\Pi^A = Y - wL = \alpha Y.$$

By (3), $\Pi^A = \pi e^{\varepsilon}\tilde{K} = D\tilde{K}$, which implies

$$D = \frac{\alpha Y}{\tilde{K}} = \alpha e^{\alpha\varepsilon}\tilde{K}^{\alpha-1}.$$

Finally, substituting $\tilde{K} = \text{TFP}^{1/\alpha} \times K$, established below in (30) and (31), into aggregate output gives the first expression in (32). □

**Dynamics of aggregate wealth.** By Proposition 2.3(b), each producer operates $k_i = (1+\xi)a_i$ and issues equity $E_i = \xi a_i$. Outside shareholders own the fraction $E_i/[q(z_i)k_i]$ of the firm, leaving the producer with

$$R^P(z_i)a_i = R^K(z_i)k_i - \frac{E_i R^K(z_i)}{q(z_i)}.$$

Using $R^K(z) = zD + 1 - \delta$, (30), and $D\tilde{K} = \alpha Y$, producers' aggregate cash on hand is

$$\int_{\mathcal{P}} R^P(z_i)a_i\, di = D\tilde{K} + (1-\delta)K - \Pi^e = \alpha Y + (1-\delta)K - \Pi^e,$$

where $\Pi^e$ is the total payoff on the equity index.

Investors choose a common equity share $\theta$, since their portfolio problem is independent of net worth and productivity and has a unique solution (Proposition 2.2 and Section 2.3). Their aggregate cash on hand is

$$\int_{i\notin\mathcal{P}} R^I a_i\, di = \left[(1-\theta)R^{f*} + \theta R^e\right]W^I = R^{f*}\text{NFB} + (1-\omega)\Pi^e.$$

The last equality uses $\text{NFB} = (1-\theta)W^I$, equity-market clearing $\theta W^I = (1-\omega)\xi W^P$, and $R^e = \Pi^e/(\xi W^P)$; see (37) and Appendix A.3.

Since all capitalists save the same fraction $s_t$, aggregate wealth follows (41):

$$\begin{aligned} W_{t+1} &= s_t H(W_t), \\ H(W_t) &\equiv \alpha Y(W_t) + (1-\delta)K(W_t) + R^{f*}\text{NFB}(W_t) - \omega\Pi^e(W_t), \end{aligned}$$

with dependence on the aggregate shock suppressed.

## A.3 Asset prices

We first prove Proposition 2.1. We then show that investors' stochastic discount factor prices every claim they trade (Proposition A.2). Next, we characterize investors' exposure to aggregate risks for given continuation weights and determine when foreign holdings reach the ownership cap. The

pricing result in Proposition A.2 is also used in the proofs of Propositions 2.3 and 3.1. Throughout, the continuation weights $\Lambda_t$ are a positive, bounded function of the shock, common to all capitalists and constant when $\gamma = 1$ (Proposition 2.2 and Lemma A.1), and $g_x \equiv 1 - x + x\lambda X$ is the return on a unit of wealth with exposure $x$ relative to the world rate, where $X \equiv e^{\alpha\varepsilon}/\mathbb{E}[e^{\alpha\varepsilon}]$ (Section 2.4).

*Proof of Proposition 2.1.*
With $E_i = \xi a_i$, the index pools all the equity issued by producers and the supply of the index is $Q^e$. The payoff of the index is the sum of all producers' payoffs to outside shareholders, $\Pi^e$, where we can derive

$$Q^e = \xi \int_{i\in\mathcal{P}} a_i \, di, \qquad \Pi^e = \int_{i\in\mathcal{P}} (1-\iota_i)[\Pi_i(\varepsilon) + (1-\delta)k_i]di = \xi \int_{i\in\mathcal{P}} a_i \frac{R^K(z_i)}{q(z_i)} \, di.$$

Since the productivity draws are i.i.d. and the producers are the capitalists whose productivity level $z$ is above a cutoff $\bar{z}$, $\mathcal{P} = \{i : z_i \geq \bar{z}\}$ in equilibrium (Proposition 2.3(a)), net worth therefore factors out of both integrals: for any integrable $f(z_i)$,

$$\int_{i\in\mathcal{P}} a_i \, f(z_i) \, di = W \int_{\bar{z}}^{\infty} f(z) \, d\Phi(z).$$

Dividing $\Pi^e$ by $Q^e$, the common factor $\xi W$ cancels and we get the return on the equity index as

$$R^e = \frac{\Pi^e}{Q^e} = \frac{\int_{\bar{z}}^{\infty} \left[R^K(z)/q(z)\right] d\Phi(z)}{1 - \Phi(\bar{z})} = \mathbb{E}_z \left[ \frac{R^K(z)}{q(z)} \,\middle|\, z \geq \bar{z} \right].$$

□

**Proposition A.2** (Stochastic discount factor)**.** *Let $M^\Lambda$ be the SDF defined in (16) from the portfolio return $R^I = R^{f*} + \theta(R^e - R^{f*})$ and continuation weights $\Lambda_t$. Given that the investor's FOC (11) holds, the SDF prices the world bond, the equity index, and capital of every productivity type as follows:*

$$R^{f*} = \frac{1}{\mathbb{E}[M^\Lambda]}, \qquad \mathbb{E}[M^\Lambda R^e] = 1, \qquad q(z) = \mathbb{E}[M^\Lambda R^K(z)] \quad \textit{for every } z.$$

*Proof.*
The cost of equity of a producer with productivity $z$ is given by (20). Thus, averaging it over the same distribution gives $R^e = (1-\bar{\psi})R^{f*} + \bar{\psi}R^D$ with $\bar{\psi} = \mathbb{E}_z[\psi(z) \mid z \geq \bar{z}]$ which is (21). Finally, as $q(z) = \mathbb{E}[M^\Lambda R^K(z)]$ for every $z$ and the independence of productivity draws and shocks, Tonelli's theorem gives

$$\mathbb{E}[M^\Lambda R^e] = \mathbb{E}_z \left[ \frac{\mathbb{E}[M^\Lambda R^K(z)]}{q(z)} \,\middle|\, z \geq \bar{z} \right] = 1.$$

□

**Investors' exposure to aggregate risk.** Assumption A0 restricts the equilibrium portfolio share invested in the risky index, which is equivalent to restricting the investor's exposure to aggregate risk. Define:

$$F^{\Lambda}(x,\lambda) \equiv \mathbb{E}^{\Lambda}\left[g_x^{-\gamma}\left(\lambda X - 1\right)\right], \qquad x \in [0,1],$$

where $\mathbb{E}^{\Lambda}[Y] \equiv \mathbb{E}[\Lambda_t Y]/\mathbb{E}[\Lambda_t]$ is the weighted expectation of Proposition 2.2 and $g_x = 1 - x + x\lambda X$ is the gross return relative to the risk-free rate on a position with risk exposure $x$ (Section 2.4). Since $\partial_x \mathbb{E}\left[\Lambda_t\, g_x^{1-\gamma}\right]/(1-\gamma) = \mathbb{E}[\Lambda_t]\, F^{\Lambda}(x,\lambda)$, $F^{\Lambda}$ is proportional to the marginal value of exposure.

**Lemma A.2** (Interior exposure given $\Lambda$)**.** *Let $\gamma \geq 1$, let the weights $\Lambda_t$ be a positive, bounded function of the shock, constant if $\gamma = 1$, and let $\lambda > 0$.*

*(i) Investors' optimal exposure $x^{*\Lambda}(\lambda)$ is unique and continuous in $\lambda$. It is zero if and only if $\lambda \leq 1/\mathbb{E}^{\Lambda}[X]$, one if and only if $\lambda \geq \lambda_1^{\Lambda} \equiv \mathbb{E}^{\Lambda}[X^{-\gamma}]/\mathbb{E}^{\Lambda}[X^{1-\gamma}]$, and interior if and only if $\lambda \in \left(1/\mathbb{E}^{\Lambda}[X], \lambda_1^{\Lambda}\right)$, where it is the unique solution of $F^{\Lambda}(x,\lambda) = 0$.*

*(ii) The interval is nonempty: $\lambda_1^{\Lambda}\, \mathbb{E}^{\Lambda}[X] > 1$.*

*Proof.*

(i) For $\gamma > 1$, maximizing $\mathcal{C}^{\Lambda}(x;\lambda)$ is equivalent to maximizing $\mathbb{E}^{\Lambda}[g_x^{1-\gamma}]/(1-\gamma)$; for $\gamma = 1$, it is equivalent to maximizing $\mathbb{E}^{\Lambda}[\ln g_x]$. In both cases, the derivative with respect to exposure is $F^{\Lambda}(x,\lambda)$, and

$$\frac{\partial F^{\Lambda}}{\partial x} = -\gamma\, \mathbb{E}^{\Lambda}\left[g_x^{-\gamma-1}(\lambda X - 1)^2\right] < 0.$$

Thus the objective is strictly concave. Its unique maximizer is zero if and only if $F^{\Lambda}(0,\lambda) \leq 0$, one if and only if $F^{\Lambda}(1,\lambda) \geq 0$, and otherwise the unique interior solution of $F^{\Lambda}(x,\lambda) = 0$.

The endpoint derivatives are

$$F^{\Lambda}(0,\lambda) = \lambda\, \mathbb{E}^{\Lambda}[X] - 1,$$
$$F^{\Lambda}(1,\lambda) = \lambda^{-\gamma}\mathbb{E}^{\Lambda}[X^{1-\gamma}](\lambda - \lambda_1^{\Lambda}).$$

These expressions give the stated three cases.

Note that $g_x$ lies between 1 and $\lambda X$. The bounded weights and the finite positive and negative moments of $X$ therefore provide integrable bounds for the objective and its first two derivatives, uniformly over $x \in [0,1]$ and $\lambda$ in any compact subset of $(0,\infty)$. Dominated convergence justifies differentiation under the expectation, with one-sided derivatives at the endpoints, and establishes joint continuity of the objective. Compactness of $[0,1]$ ensures existence of a maximizer; uniqueness and the maximum theorem imply continuity of $x^{*\Lambda}(\lambda)$.

(ii) Since $F^{\Lambda}$ is strictly decreasing in $x$, $\lambda_1^{\Lambda}\, \mathbb{E}^{\Lambda}[X] - 1 = F^{\Lambda}(0,\lambda_1^{\Lambda}) > F^{\Lambda}(1,\lambda_1^{\Lambda}) = 0$. □

**Foreign investors and the holding cap.** Foreign investors hold diversified world portfolios in which the country's aggregate shock is diversifiable (Section 2.2). Therefore, their valuation of the domestic risky equity index is at the risk-free rate. In other words, without the investment cap $\omega$, foreigners would like to hold all the equity index supply, as the index is priced by domestic investors, as long as $\lambda \geq 1$. In this paper, we focus on the case where $\lambda \geq 1$, so the cap binds in all equilibria. With log utility and $\xi > 0$, $\lambda > 1$ holds in equilibrium (Lemma B.1(i)). With $\gamma > 1$ we verify it numerically: in every equilibrium computed for Section 6, $\lambda \geq 1$ at every level of wealth on the grid.

## A.4 The cutoff and the wedge under CRRA

*Proof of Proposition 2.3.*
(a) By (8), optimal financing gives

$$R^P(z) = \left[1 + \xi \max\left\{0, 1 - \frac{1}{q(z)}\right\}\right] R^K(z).$$

since the issuance increases the producer's return in every state when $q(z) > 1$ and decreases it when $q(z) < 1$.

Since $D > 0$, the return $R^K(z) = zD + 1 - \delta$ increases strictly in $z$ in every state. Its weighted certainty equivalent is therefore strictly increasing. It is also continuous by dominated convergence. For $\gamma > 1$, $R^K(z) \geq 1 - \delta > 0$ implies

$$0 < (R^K(z))^{1-\gamma} \leq (1-\delta)^{1-\gamma}.$$

For $\gamma = 1$,

$$|\ln R^K(z)| \leq |\ln(1-\delta)| + zD + 1 - \delta.$$

Since $\mathbb{E}[D] < \infty$, the latter bound has finite expectation uniformly over any bounded productivity interval. Together with bounded continuation weights, these bounds justify dominated convergence in general CRRA utility form. Moreover, monotonicity and homogeneity imply

$$\mathrm{CE}^\Lambda(R^K(z)) \geq \mathrm{CE}^\Lambda(zD) = z\,\mathrm{CE}^\Lambda(D) \longrightarrow \infty,$$

since $\mathrm{CE}^\Lambda(D) > 0$.

Homogeneity gives

$$\mathrm{CE}^\Lambda(R^P(z)) = \left[1 + \xi \max\left\{0, 1 - \frac{1}{q(z)}\right\}\right] \mathrm{CE}^\Lambda(R^K(z)).$$

By (17), the expression in brackets is continuous, nondecreasing, and at least one. Since $R^K(z)$ is positive, continuous, strictly increasing in $z$, and unbounded, their product has these same properties.

The investor's certainty equivalent is independent of productivity. If $\mathrm{CE}^{\Lambda}(R^P(1)) \geq \mathrm{CE}^{\Lambda}(R^I)$, strict monotonicity would imply that every type $z > 1$ strictly prefers producing. Because the productivity distribution is continuous, this would leave zero mass of investors, contradicting the hypothesis. Therefore,

$$\mathrm{CE}^{\Lambda}(R^P(1)) < \mathrm{CE}^{\Lambda}(R^I).$$

Continuity and unboundedness now imply that the two certainty equivalents intersect at some $\bar{z} > 1$; strict monotonicity makes this intersection unique. Thus (25) holds, and (15) gives the cutoff rule, separating producers from investors.

(b) We first compare firm equity with the investor's portfolio. Since $q(z) = \mathbb{E}[M^{\Lambda} R^K(z)]$ and $\mathbb{E}[M^{\Lambda} R^I] = 1$, concavity of utility and (16) imply

$$\mathbb{E}\left[\Lambda_t \left(u\left(\frac{R^K(z)}{q(z)}\right) - u(R^I)\right)\right] \leq \mathbb{E}\left[\Lambda_t (R^I)^{-\gamma} \left(\frac{R^K(z)}{q(z)} - R^I\right)\right] = 0.$$

Consequently,

$$\mathrm{CE}^{\Lambda}\left(\frac{R^K(z)}{q(z)}\right) \leq \mathrm{CE}^{\Lambda}(R^I),$$

with equality if and only if $R^K(z)/q(z) = R^I$ almost surely.

If $q(\bar{z}) < 1$, the marginal producer does not issue any equity, so

$$\begin{aligned}\mathrm{CE}^{\Lambda}(R^P(\bar{z})) &= q(\bar{z})\,\mathrm{CE}^{\Lambda}\left(\frac{R^K(\bar{z})}{q(\bar{z})}\right) \\ &\leq q(\bar{z})\,\mathrm{CE}^{\Lambda}(R^I) < \mathrm{CE}^{\Lambda}(R^I),\end{aligned}$$

contradicting (25). Thus in equilibrium, $q(\bar{z}) \geq 1$. Since $q$ is increasing, every producer will have $q(z) \geq 1$, and issuing until the scale constraint is binding is optimal. With full issuance, (20), (23), and homogeneity turn the occupational indifference condition into

$$\big[(1+\xi)q(\bar{z}) - \xi\big]\,\mathcal{C}^{\Lambda}\big(\psi(\bar{z})\big) = \mathcal{C}^{\Lambda}(\vartheta).$$

Substituting $\Delta = q(\bar{z}) - 1$ gives (26). Strict concavity implies

$$\Delta = 0 \quad\Longleftrightarrow\quad \frac{R^K(\bar{z})}{q(\bar{z})} = R^I \text{ almost surely} \quad\Longleftrightarrow\quad \psi(\bar{z}) = \vartheta,$$

where the last equivalence follows from (20), (23), and the nondegeneracy of $D$.

(c) Finally, substituting $q(\bar{z}) = 1 + \Delta$ into (19) and (17) gives

$$\psi(\bar{z}) = 1 - \frac{1-\delta}{R^{f*}(1+\Delta)}.$$

Using $p = \mathbb{E}[D]/(\lambda R^{f*})$ also gives

$$\bar{z}\,\mathbb{E}[D] = \lambda\big[R^{f*}(1+\Delta) - (1-\delta)\big].$$

□

### A.5 The two equilibrium conditions

Given the aggregate wealth $W$, a within-period equilibrium solves a system of two equations with two unknowns, namely the cutoff $\bar{z}$ and the risk premium $\lambda$. With general CRRA utility, there is no closed-form solution as the stochastic continuation weight $\Lambda$ affects the certainty equivalent value of occupational choices as both occupations' returns and the continuation weight loaded on the aggregate rate, which require computing the covariance between the returns and the continuation weights, and thus the saving rate depends on the aggregate wealth level. With log utility, however, the saving rate is the intertemporal discount factor $\beta$ and it is constant across all states. We first show how the continuation weights affect the tractability of the solution, and then derive the equilibrium conditions as a system of two equations with two unknowns, and then explain why log utility makes the solution tractable.

**Stochastic continuation weights.** At given aggregate wealth $W$, the current shock determines $W_{t+1} = G(W, \varepsilon)$ and hence the continuation weight $\Lambda_t = \Lambda(W_{t+1})$. These weights are positive and bounded by Lemma A.1(b).

By (20) and (23), firm equity and investors' portfolios both earn returns following the form $R^{f*}g_x$, where $g_x = 1 - x + x\lambda X$ and $x = \psi(z)$ is the exposure of firm equity to the aggregate risk factor $D(\varepsilon)$ and $x = \vartheta$ is the exposure of the investor's portfolio. By homogeneity, comparing these returns amounts to comparing

$$\mathcal{C}^{\Lambda}(x;\lambda) = \mathrm{CE}^{\Lambda}(g_x) = \left(\frac{\mathbb{E}[\Lambda_t g_x^{1-\gamma}]}{\mathbb{E}[\Lambda_t]}\right)^{1/(1-\gamma)}, \tag{54}$$

for $\gamma > 1$. When $\gamma = 1$, the continuation weights are constant and $\mathcal{C}^{\Lambda}(x;\lambda) = \exp(\mathbb{E}[\ln g_x])$.

For given continuation weights, investors choose their exposure optimally over $[0,1]$:

$$x^{*\Lambda}(\lambda) = \operatorname*{argmax}_{x\in[0,1]} \mathcal{C}^{\Lambda}(x;\lambda),$$

which is unique by Lemma A.2(i). The wedge in (26) compares the same certainty equivalent at investors' optimal exposure and at the marginal producer's exposure $\psi(\bar{z})$.

The factor price $p = \mathbb{E}[M^{\Lambda}D]$ is also endogenous. We define the risk premium $\lambda = \mathbb{E}[D]/(pR^{f*})$ as the price variable and solve for it jointly with the cutoff $\bar{z}$. The factor price is then calculated from $p = \mathbb{E}[D]/(\lambda R^{f*})$.

**Lemma A.3** (Returns in the cutoff and the risk premium)**.** *Let $\gamma \geq 1$, fix aggregate wealth $W$, a cutoff $\bar{z}$, a world rate $R^{f*}$, and a risk premium $\lambda > 0$, and let $p = \mathbb{E}[D]/(\lambda R^{f*})$. Write $X \equiv D/\mathbb{E}[D]$, and $g_x \equiv 1 - x + x\lambda X$ for $x \in [0,1]$. Then:*

*(i)* $\mathbb{E}[D] = \kappa(\xi, W)\,\bar{z}^{e_D}$ *with* $e_D = (\eta - 1)(1 - \alpha)$*, and the marginal producer's expected operating payoff relative to its undepreciated capital*

$$\nu(\bar{z}) \equiv \frac{\bar{z}\,\mathbb{E}[D]}{1-\delta} = \frac{\kappa(\xi, W)}{1-\delta}\,\bar{z}^{\,1+e_D},$$

*is continuous and strictly increasing in $\bar{z}$, with $\mathrm{d}\ln\nu(\bar{z})/\mathrm{d}\ln\bar{z} = 1 + e_D$;*

*(ii) the unit capital run by a producer with productivity $z$ is priced at $q(z) = (1-\delta)\big(1+\nu(z)/\lambda\big)/R^{f*}$, generates per unit return $R^K(z) = (1-\delta)\big(1+\nu(z)X\big)$, and has cost of equity $R^K(z)/q(z) = R^{f*}g_{\psi(z)}$, where $\nu(z) = \nu(\bar{z})\,z/\bar{z}$ and $\psi(z) = \nu(z)/\big(\lambda + \nu(z)\big)$ is the risky share of the capital (19);*

*(iii) when every producer sells the amount of equity in proportion to net worth, i.e. $E_i = \xi a_i$, the equity index pays $R^e = R^{f*}g_{\bar{\psi}}$, where the exposure $\bar{\psi}$ is*

$$\bar{\psi} = \bar{\psi}\big(\nu(\bar{z})/\lambda\big), \qquad \bar{\psi}(h) \equiv \int_1^\infty \frac{h\rho}{1+h\rho}\,\eta\rho^{-\eta-1}\,\mathrm{d}\rho,$$

*with $\rho \equiv z/\bar{z}$, and investors with equity share $\theta$ earn $R^I = R^{f*}g_\vartheta$ with exposure $\vartheta = \theta\bar{\psi}$;*

*(iv) $\bar{\psi}(h) \in (0,1)$ is continuously differentiable and strictly increasing in $h > 0$, with elasticity*

$$\bar{\psi}_h \equiv \frac{\mathrm{d}\ln\bar{\psi}}{\mathrm{d}\ln h} = \frac{1}{\bar{\psi}(h)}\int_1^\infty \frac{h\rho}{(1+h\rho)^2}\,\eta\rho^{-\eta-1}\,\mathrm{d}\rho \in (0,1).$$

*Proof.*
(i) By (30), (31), and (33) (Proposition A.1), $\mathbb{E}[D] = \kappa(\xi, W)\,\bar{z}^{e_D}$, which gives the expression for $\nu(\bar{z})$; it is strictly increasing because $1 + e_D > 0$.

(ii) By (17) and $pR^{f*} = \mathbb{E}[D]/\lambda$, $q(z) = \big[z\,\mathbb{E}[D]/\lambda + 1 - \delta\big]/R^{f*} = (1-\delta)\big(1+\nu(z)/\lambda\big)/R^{f*}$, with $\nu(z) = z\,\mathbb{E}[D]/(1-\delta) = \nu(\bar{z})\,z/\bar{z}$. By (4) and $D = \mathbb{E}[D]\,X$, $R^K(z) = z\,\mathbb{E}[D]\,X + 1 - \delta = (1-\delta)\big(1+\nu(z)X\big)$. Dividing,

$$\frac{R^K(z)}{q(z)} = R^{f*}\,\frac{1+\nu(z)X}{1+\nu(z)/\lambda} = R^{f*}\big(1 - \psi(z) + \psi(z)\,\lambda X\big) = R^{f*}g_{\psi(z)}.$$

(iii) With each producer selling the amount of equity in proportion to his net worth, the index return is the simple average of producers' costs of equity (proof of Proposition 2.1), and by (ii) the index return is $R^{f*}g_{\bar{\psi}}$ with $\bar{\psi}$ the average of $\psi(z)$ over producers. Since producers' net worth is independent of the productivity draw and given the Pareto productivity distribution, the distribution of $\rho = z/\bar{z}$ is also Pareto distributed on $[1,\infty)$ with tail parameter $\eta$ regardless of the cutoff, and

we can write $\psi(z) = h\rho/(1+h\rho)$ with $h = \nu(\bar{z})/\lambda$, and the risky index's exposure to the aggregate risk factor is $\bar{\psi}(h)$. Finally, $R^I = R^{f*} + \theta(R^e - R^{f*}) = R^{f*}\big(1 + \theta\bar{\psi}(\lambda X - 1)\big) = R^{f*} g_{\theta\bar{\psi}}$.

(iv) For $h > 0$, the Pareto density $\eta\rho^{-\eta-1}$ is positive on $[1,\infty)$ and integrates to one. Since $h\rho/(1+h\rho)$ is strictly between zero and one throughout this interval,

$$0 < \bar{\psi}(h) = \int_1^\infty \frac{h\rho}{1+h\rho}\,\eta\rho^{-\eta-1}\,d\rho < \int_1^\infty \eta\rho^{-\eta-1}\,d\rho = 1.$$

The derivative of $h\rho/(1+h\rho)$ with respect to $h$ is $\rho/(1+h\rho)^2 \in \big(0, 1/(4h)\big]$, which implies that the differentiated integrand is bounded by the integrable function $\eta\rho^{-\eta-1}/(4h)$, which allows differentiation under the integral. In other words, we can derive

$$\bar{\psi}'(h) = \mathbb{E}_\rho\left[\frac{\rho}{(1+h\rho)^2}\right] > 0.$$

Moreover,

$$0 < h\bar{\psi}'(h) = \mathbb{E}_\rho\left[\frac{h\rho}{(1+h\rho)^2}\right] < \mathbb{E}_\rho\left[\frac{h\rho}{1+h\rho}\right] = \bar{\psi}(h).$$

Dividing by $\bar{\psi}(h) > 0$ concludes that the elasticity of $\bar{\psi}$ with respect to $h$ satisfies $h\bar{\psi}'(h)/\bar{\psi}(h) \in (0,1)$. □

**Lemma A.4** (The two equilibrium conditions)**.** *Let $\gamma \geq 1$, $\xi > 0$, $\omega < 1$, and $R^{f*} > 1 - \delta$, and given an aggregate wealth $W$. In an equilibrium with interior exposure (Assumption A0) in which foreign investors hold the share $\omega$ of the index, the cutoff $\bar{z}$ and the risk premium $\lambda$ satisfy the system of two equations*

$$\kappa(\xi, W)\,\bar{z}^{\,1+e_D} = \lambda\left[R^{f*}(1+\Delta) + \delta - 1\right], \tag{55}$$

$$x^{*\Lambda}(\lambda) = \frac{(1-\omega)\,\xi}{\bar{z}^\eta - 1}\,\bar{\psi}\left(\frac{\nu(\bar{z})}{\lambda}\right), \tag{56}$$

*where $e_D = (\eta-1)(1-\alpha)$, $\nu(\bar{z}) \equiv \kappa(\xi, W)\,\bar{z}^{\,1+e_D}/(1-\delta)$, $\bar{\psi}$ is the index's risky share of Section 3, and the wedge $\Delta \geq 0$ satisfies*

$$(1+\xi)\,\Delta = \frac{\mathcal{C}^\Lambda\big(x^{*\Lambda}(\lambda);\lambda\big)}{\mathcal{C}^\Lambda\big(\psi(\bar{z});\lambda\big)} - 1, \qquad \psi(\bar{z}) = 1 - \frac{1-\delta}{R^{f*}(1+\Delta)}. \tag{57}$$

*Equation (55) is (Entry) and (56) is (Pricing) of Section 3, with $\mathcal{C}^\Lambda$ and $x^{*\Lambda}$ in place of $\mathcal{C}$ and $x^*$.*

*Proof.*
*Entry.* By (30), (31), and (33), $\mathbb{E}[D] = \kappa(\xi, W)\bar{z}^{e_D}$. The cutoff identity (28) can therefore be written as

$$\kappa(\xi, W)\bar{z}^{1+e_D} = \bar{z}\,\mathbb{E}[D] = \lambda\big[R^{f*}(1+\Delta) + \delta - 1\big],$$

which is (55). Investors choose optimal exposure $\vartheta = x^{*\Lambda}(\lambda)$. Substituting this into (26) and (27)

gives

$$(1+\xi)\Delta = \frac{\mathcal{C}^{\Lambda}(x^{*\Lambda}(\lambda);\lambda)}{\mathcal{C}^{\Lambda}(\psi(\bar{z});\lambda)} - 1,$$
$$\psi(\bar{z}) = 1 - \frac{1-\delta}{R^{f*}(1+\Delta)}.$$

These are (57), with $\Delta \geq 0$ by Proposition 2.3(b).

*Pricing.* Equity-market clearing must equate investors' demand for equity with the equity supplied by producers. This is equivalent to equating the investor's desired exposure level to the aggregate risk factor to the exposure level supplied by the equity index. Using (17), (18), and the definition of $\nu(\bar{z})$, each producer's risky share of its capital is

$$\psi(z) = \frac{z\,\mathbb{E}[D]}{z\,\mathbb{E}[D] + \lambda(1-\delta)} = \frac{(\nu(\bar{z})/\lambda)(z/\bar{z})}{1 + (\nu(\bar{z})/\lambda)(z/\bar{z})}.$$

Since issuance is proportional to net worth and net worth is independent of productivity, the risky share of the index's return is $\mathbb{E}_z[\psi(z) \mid z \geq \bar{z}]$. The conditional Pareto distribution of $z/\bar{z}$ and the definition of $\bar{\psi}(\cdot)$ therefore imply

$$\bar{\psi} = \bar{\psi}\left(\frac{\nu(\bar{z})}{\lambda}\right).$$

Equity-market clearing (36) now gives

$$x^{*\Lambda}(\lambda) = \vartheta = \theta\bar{\psi} = \frac{(1-\omega)\xi W^P}{W^I}\,\bar{\psi}\left(\frac{\nu(\bar{z})}{\lambda}\right)$$
$$= \frac{(1-\omega)\xi}{\bar{z}^{\eta}-1}\,\bar{\psi}\left(\frac{\nu(\bar{z})}{\lambda}\right),$$

which is (56).

At given $W$, preferences and continuation weights enter these conditions through $\mathcal{C}^{\Lambda}$ and its maximizer $x^{*\Lambda}$. □

**Tractability under log utility.** Log utility provides two simplifications: the saving rate is $s_t = \beta$, and the continuation weight $\Lambda_t$ is constant (Proposition 2.2(c)). Saving therefore does not depend on wealth or the realized shock, and the constant continuation weight cancels from the certainty equivalent:

$$\mathcal{C}^{\Lambda}(x;\lambda) = \exp\big(\mathbb{E}[\ln g_x]\big).$$

Investors' exposure demand and the occupational wedge can thus be derived without the covariance between the returns and the continuation weight. At given $W$, the two equilibrium conditions can be expressed in a tractable system of equations in $(\bar{z},\lambda)$, with wealth entering through $\kappa(\xi, W)$. Once the period by period allocation is determined, the wealth transition follows from (41) using the fixed saving rate $\beta$.

Appendix B establishes existence, uniqueness, and the stated comparative statics under log utility.

# B Theoretical Results under Log Utility

## B.1 Preliminaries

**Log utility.** With $\gamma = 1$, Proposition 2.2(c) shows that the continuation weight is a constant and the saving rate is $\beta$, meaning $W' = \beta H$. Given aggregate capitalists' wealth level $W$, a period by period equilibrium consists of the occupational and financial choices, prices, and allocation satisfying conditions (i)–(iii) of the recursive equilibrium in Section 2.5. Since $\mathbb{E}^\Lambda = \mathbb{E}$, we omit the superscript $\Lambda$ from the objects of Section 2 and Lemma A.2; Section 3 denotes $x^*$ by $\vartheta^*$.

Capitalists value returns by $\mathrm{CE}(R) = \exp(\mathbb{E}[\ln R])$ and $\mathcal{C}(x;\lambda) = \mathrm{CE}(g_x)$ is the certainty equivalent relative to the world rate. Investors choose optimal portfolio exposure $x^*(\lambda)$ and obtain $\mathcal{V}(\lambda)$

$$x^*(\lambda) \equiv \operatorname*{argmax}_{x\in[0,1]} \mathcal{C}(x;\lambda), \qquad \mathcal{V}(\lambda) \equiv \mathcal{C}(x^*(\lambda);\lambda). \tag{58}$$

The SDF (16) becomes $M = 1/R^I$, while the full exposure threshold of Lemma A.2 becomes $\lambda_1 = \mathbb{E}[X^{-1}] = e^{\alpha^2\sigma^2}$ (Lemma B.1(i)).

Finally, producers issue a positive amount of equity, and foreign investors hold only the share $\omega < 1$. Domestic investors therefore have positive exposure, which requires $\lambda > 1$ (Lemma B.1(i)) since $\mathbb{E}[X] = 1$. Consequently,

$$p = \frac{\mathbb{E}[D]}{\lambda R^{f*}} < \frac{\mathbb{E}[D]}{R^{f*}},$$

This shows that the valuation of diversified foreign investors on the aggregate risk factor is higher than the price at which the equity index is sold (Section 2.2), so foreign investors will hold $\omega$ share as they are allowed.

**The marginal producer's certainty equivalent.** Given a cutoff $\bar{z}$ and a risk premium $\lambda > 0$, we can write $\bar{\nu} = \nu(\bar{z})$ as in Lemma A.3. For $\bar{\nu} > 0$, let $\mathcal{K}(\bar{\nu}) \equiv \mathrm{CE}(1+\bar{\nu}X) = \exp\mathbb{E}[\ln(1+\bar{\nu}X)]$ denote the certainty-equivalent return on capital whose expected operating payoff per unit of undepreciated capital is $\bar{\nu}$; the marginal producer's is $\mathcal{K}(\nu(\bar{z}))$. $\mathcal{K}$ is continuous and strictly increasing in $\bar{\nu}$, since $1+\bar{\nu}X$ is strictly increasing in $\bar{\nu}$ in every state, so $\mathcal{K}(\nu(\bar{z}))$ is continuous and strictly increasing in $\bar{z}$ (Lemma A.3(i)). By Lemma A.3(ii), $R^K(\bar{z}) = (1-\delta)(1+\bar{\nu}X)$, and the marginal producer's risky share of capital $\psi(\bar{z}) = \bar{\nu}/(\lambda+\bar{\nu})$ gives $g_{\psi(\bar{z})} = (1+\bar{\nu}X)/(1+\bar{\nu}/\lambda)$. Since CE is homogeneous of degree one,

$$\mathrm{CE}\big(R^K(\bar{z})\big) = (1-\delta)\,\mathcal{K}(\bar{\nu}), \qquad \mathcal{C}\big(\psi(\bar{z});\lambda\big) = \frac{\mathcal{K}(\bar{\nu})}{1+\bar{\nu}/\lambda}, \qquad \bar{\nu} = \nu(\bar{z}). \tag{59}$$

**Investors' demand for risk exposure.** The next lemma summarizes the properties of the investors' demand for exposure $x^*$ and their certainty equivalent $\mathcal{V}$ given by (58). By Lemma A.3(iii), investors who hold their optimal exposure earn $\mathrm{CE}(R^I) = R^{f*}\mathcal{V}(\lambda)$.

**Lemma B.1** (Investors' risk exposure)**.** *Under a positive risk premium $\lambda > 0$.*

*(i) The optimal exposure $x^*(\lambda)$ is unique and continuous in $\lambda$. It is zero for $\lambda \leq 1$, one for $\lambda \geq \lambda_1$, and interior for $\lambda \in (1, \lambda_1)$, where it is the unique solution of $\mathbb{E}\left[(\lambda X - 1)/g_x\right] = 0$. The threshold $\lambda_1 = \mathbb{E}[X^{-1}] = e^{\alpha^2\sigma^2}$ exceeds one.*

*(ii) The maximized certainty equivalent $\mathcal{V}$ is continuous on $\lambda \in (0, \infty)$ and strictly increasing on $(1, \infty)$, with $\mathcal{V}(\lambda) \geq 1$, and $\mathcal{V}(\lambda) = 1$ for $\lambda \in (0, 1]$.*

*(iii) The optimal exposure $x^*$ is continuously differentiable and strictly increasing on $(1, \lambda_1)$; its elasticity $x^*_\lambda = \mathrm{d} \ln x^* / \mathrm{d} \ln \lambda$ is positive.*

*Proof.*
Since $X$ is lognormally distributed, $\mathbb{E}[X^n] = e^{n(n-1)\tau^2/2}$ for every real $n$, with $\tau \equiv \alpha\sigma$, we have $\lambda_1 = \mathbb{E}[X^{-1}] = e^{\tau^2} = e^{\alpha^2\sigma^2}$.

(i) Maximizing $\mathcal{C}(x; \lambda)$ is equivalent to maximizing $\ln \mathcal{C}(x; \lambda) = \mathbb{E}[\ln g_x]$. This objective function is strictly concave in $x$, since $X$ is nondegenerate. Its derivative with respect to $x$ is

$$\frac{d\mathbb{E}[\ln g_x]}{dx} \equiv F(x, \lambda) = \mathbb{E}\left[\frac{\lambda X - 1}{g_x}\right]$$

which is therefore strictly decreasing in $x$, and the value of $F$ at the boundary points is

$$F(0, \lambda) = \lambda - 1, \qquad F(1, \lambda) = 1 - \frac{\mathbb{E}[X^{-1}]}{\lambda}.$$

Thus, the unique maximizer $x^*$ is zero when $\lambda \leq 1$ and is one when $\lambda \geq \lambda_1 = \mathbb{E}[X^{-1}]$. For $\lambda \in (1, \lambda_1)$, there is a unique interior solution of $F(x^*, \lambda) = 0$, where Jensen's inequality gives $\lambda_1 > 1/\mathbb{E}[X] = 1$.

Since ln is a strictly concave function, $g_x$ is affine in $x$, and $X$ is nondegenerate, we have that $\mathbb{E}[\ln g_x] = \ln \mathcal{C}(x; \lambda)$ is strictly concave in $x$. Given $X > 0$ and $g_x$ lies between 1 and $\lambda X$, we can bound the integrand and its derivative by

$$\max\left\{|\ln g_x|, \left|\frac{\lambda X - 1}{g_x}\right|\right\} \leq \lambda X + \frac{1}{\lambda X}.$$

The upper bound on the right-hand side of the inequality is integrable as $\mathbb{E}[X] = 1$ and $\mathbb{E}[X^{-1}] < \infty$. Moreover, $\lambda$ and $\lambda^{-1}$ are bounded on every compact interval in $(0, \infty)$. Thus $\mathbb{E}[|\ln g_x|] < \infty$, and dominated convergence implies that $\mathbb{E}[\ln g_x]$ is jointly continuous in $(x, \lambda)$ and differentiable in $x$:

$$\frac{\partial}{\partial x}\mathbb{E}[\ln g_x] = \mathbb{E}\left[\frac{\lambda X - 1}{g_x}\right] = F(x, \lambda),$$

given $x \in (0, 1)$.

For each fixed $\lambda > 0$, continuity in $x$ ensures that $\mathbb{E}[\ln g_x]$ attains its maximum on $[0, 1]$, and strict concavity makes the maximizing exposure $x^*(\lambda)$ unique. Joint continuity of $\mathbb{E}[\ln g_x]$ with the fixed compact choice set $[0, 1]$ implies continuity of $x^*(\lambda)$ by the maximum theorem.

(ii) Continuity on $(0,\infty)$ follows from the maximum theorem. For $x \in (0,1]$, $g_x$ is strictly increasing in $\lambda$ in every state, so $\mathcal{C}(x;\lambda)$ is strictly increasing in $\lambda$, and $\mathcal{C}(0;\lambda) = 1$. For any $1 < \lambda < \lambda'$, the maximum theorem implies $\mathcal{V}(\lambda) = \mathcal{C}(x^*(\lambda);\lambda) < \mathcal{C}(x^*(\lambda);\lambda') \leq \mathcal{V}(\lambda')$. Finally, $\mathcal{V} \geq \mathcal{C}(0;\lambda) = 1$, and for $\lambda \leq 1$, the optimal $x^*(\lambda) = 0$ by (i), so $\mathcal{V}(\lambda) = \mathcal{C}(0;\lambda) = 1$.

(iii) Since $g_x - x(\lambda X - 1) = 1$, we can compute the partial derivative of $F(x,\lambda)$ with respect to $\lambda$ and $x$ as:

$$F_\lambda(x,\lambda) = \mathbb{E}\left[\frac{X}{g_x^2}\right] > 0 \qquad \text{for } x \in [0,1),$$
$$F_x(x,\lambda) = -\mathbb{E}\left[\frac{(\lambda X - 1)^2}{g_x^2}\right] < 0 \quad \text{for } x \in (0,1).$$

For $0 < x < 1$, the inequalities $g_x \geq 1 - x$ and $g_x \geq x\lambda X$ imply

$$\left|\frac{\lambda X - 1}{g_x}\right| \leq \frac{1}{x} + \frac{1}{1-x}, \qquad \frac{X}{g_x^2} \leq \frac{X}{(1-x)^2}.$$

Given $\mathbb{E}[X] = 1$, the finite bounds ensure differentiation under the expectation and imply joint continuity of $F$, $F_x$, and $F_\lambda$ on $(0,1) \times (0,\infty)$. At $x = 0$, $F_\lambda(0,\lambda) = 1$ follows directly from $F(0,\lambda) = \lambda - 1$.

For $\lambda \in (1,\lambda_1)$, the unique optimal exposure is interior and satisfies $F(x^*(\lambda),\lambda) = 0$. Based on the implicit function theorem, $x^*$ is continuously differentiable, with

$$\frac{\mathrm{d}x^*}{\mathrm{d}\lambda} = -\frac{F_\lambda}{F_x} > 0, \qquad x^*_\lambda \equiv \frac{\mathrm{d}\ln x^*}{\mathrm{d}\ln\lambda} = -\frac{\lambda F_\lambda}{x^* F_x} > 0,$$

where the partial derivatives are evaluated at $(x^*(\lambda),\lambda)$, which concludes that $x^*$ is strictly increasing on $(1,\lambda_1)$. □

## B.2 The occupational wedge as a function of the prices

We derive the wedge equation and show that it is a continuously differentiable function $\Delta(\lambda, R^{f*};\xi)$ with respect to the risk premium, risk-free rate and the financing capacity. We also derive the properties of the wedge that Appendices B.4 and B.5 use. We adopt the notation of Lemma A.3 and Appendix B.1. Same as in Section 2.4, $g_x = 1 - x + x\lambda X$ represents the return on unit wealth relative to the world rate with share of $x \in [0,1]$ exposure to the aggregate risk factor, and we refer to $\psi(\bar{z}) - \vartheta$ as the exposure gap to the aggregate risk factor between the marginal producer and the investor.

**The wedge equation.** Consider an equilibrium with interior exposure (Assumption A0), $\xi > 0$, and $\omega < 1$. By Lemma A.4, investors hold a portfolio share that has the exposure $\vartheta = x^*(\lambda)$ to the aggregate risk factor per unit of net wealth. $\mathcal{C}(\vartheta;\lambda) = \mathcal{V}(\lambda)$. Equity-market clearing (36) imposes $\theta > 0$, so $\vartheta \in (0,1)$ by Assumption A0, and $\lambda \in (1,\lambda_1)$ by Lemma B.1(i). Thus, we can write the

wedge equation as

$$\Delta = \frac{1}{1+\xi}\left[\frac{\mathcal{V}(\lambda)}{\mathcal{C}\left(1-\frac{1-\delta}{R^{f*}(1+\Delta)};\lambda\right)}-1\right]. \tag{60}$$

The functions $\mathcal{V}$ and $\mathcal{C}$ depend on $X = e^{\alpha\varepsilon}/\mathbb{E}[e^{\alpha\varepsilon}]$, which is a lognormal variable with mean one and log variance $(\alpha\sigma)^2$.

**Lemma B.2** (The wedge magnitude)**.** *Let $\lambda \in [1, \lambda_1)$ and $\vartheta = x^*(\lambda)$. The function $\mathcal{C}(\cdot\,;\lambda)$ is twice continuously differentiable in $x$ and strictly concave on $x \in [0,1]$, with*

$$-\frac{\mathcal{C}''(x;\lambda)}{\mathcal{C}(x;\lambda)} = \mathrm{Var}\left[\frac{\lambda X - 1}{g_x}\right] > 0, \tag{61}$$

*where the variance is under the distribution of $\varepsilon$. For every $x \in [0,1]$ there exists an $\tilde{x}$ between $x$ and $\vartheta$ such that*

$$\frac{\mathcal{C}(\vartheta;\lambda)}{\mathcal{C}(x;\lambda)} - 1 = \frac{(x-\vartheta)^2}{2}\,\frac{-\mathcal{C}''(\tilde{x};\lambda)}{\mathcal{C}(x;\lambda)}. \tag{62}$$

*In particular, in equilibrium the wedge of (26) is*

$$\Delta = \frac{\big(\psi(\bar{z})-\vartheta\big)^2}{2(1+\xi)}\,\frac{-\mathcal{C}''(\tilde{x};\lambda)}{\mathcal{C}(\psi(\bar{z});\lambda)} \qquad \text{for some } \tilde{x} \text{ between } \psi(\bar{z}) \text{ and } \vartheta, \tag{63}$$

*so that $\Delta \geq 0$ with equality if and only if $\psi(\bar{z}) = \vartheta$.*

*Proof.*
Define $\Upsilon \equiv \lambda X - 1$ and $f(x) \equiv \ln \mathcal{C}(x;\lambda) = \mathbb{E}[\ln g_x]$. By the proof of Lemma B.1, $f'(x) = F(x,\lambda) = \mathbb{E}[\Upsilon/g_x]$ and $|\Upsilon|/g_x \leq \lambda X + 1/(\lambda X)$ for every $x \in [0,1]$. Thus, dominated convergence implies that $f$ is twice continuously differentiable on $[0,1]$ and $f''(x) = -\mathbb{E}\big[(\Upsilon/g_x)^2\big]$. Given $\mathcal{C} = e^f$, we can derive

$$\frac{\mathcal{C}''}{\mathcal{C}} = f'' + (f')^2 = -\,\mathrm{Var}\left[\frac{\Upsilon}{g_x}\right] < 0,$$

which is (61); the variance is positive because $\Upsilon/g_x$ is strictly increasing in $X$ (its derivative is $\lambda/g_x^2$) and $X$ is nondegenerate.

The derivative of $\mathcal{C}(\cdot\,;\lambda)$ equals 0 at $\vartheta$, i.e. $F(\vartheta,\lambda) = 0$ for $\lambda \in (1,\lambda_1)$ by Lemma B.1(i), and $F(0,1) = 0$. Applying Taylor's theorem taking a $\tilde{x}$ around $\vartheta$ gives (62). In equilibrium, investors hold $\vartheta = x^*(\lambda)$ with $\lambda \in (1,\lambda_1)$. Comparing (26) with (62) at $x = \psi(\bar{z})$ gives (63); $\Delta \geq 0$, with equality only if $\psi(\bar{z}) = \vartheta$, since $\mathcal{C}'' < 0$. □

Equation (60) solves a fixed point in the wedge, as the wedge moves the marginal producer's risky share. The next lemma shows that it has exactly one solution, so that the wedge can be expressed as a continuously differentiable function $\Delta(\lambda, R^{f*}; \xi)$. The next lemma also states the identities that are used in Appendices B.4 and B.5.

**Lemma B.3** (The uniqueness of the wedge value)**.** *Let $\lambda \in [1, \lambda_1)$ and $\vartheta = x^*(\lambda)$, so that $\vartheta = 0$ at $\lambda = 1$, and write $\varphi(x) \equiv \partial \ln \mathcal{C}(x;\lambda)/\partial\lambda$ for $x \in [0,1]$. For any $\bar{\nu} > 0$, consider a hypothetical marginal producer whose expected operating payoff per unit of undepreciated capital is $\bar{\nu}$. Define $\psi_{\bar{\nu}} \equiv \bar{\nu}/(\lambda + \bar{\nu})$ for the risky share of its capital, and let*

$$\ell_{\bar{\nu}} \equiv -\left.\frac{\partial \ln \mathcal{C}(x;\lambda)}{\partial x}\right|_{x=\psi_{\bar{\nu}}}, \qquad \mathcal{Q}_{\bar{\nu}} \equiv \frac{\mathcal{V}(\lambda)}{\mathcal{C}(\psi_{\bar{\nu}};\lambda)}, \qquad \Delta_{\bar{\nu}} \equiv \frac{\mathcal{Q}_{\bar{\nu}} - 1}{1+\xi},$$

*so that $\Delta_{\bar{\nu}}$ is the wedge (26) at a cutoff with $\nu(\bar{z}) = \bar{\nu}$ when investors hold the exposure $\vartheta$, and let*

$$\mathcal{J}_{\bar{\nu}} \equiv (1+\xi) - \frac{\mathcal{Q}_{\bar{\nu}}\, \ell_{\bar{\nu}}\, (1-\psi_{\bar{\nu}})}{1+\Delta_{\bar{\nu}}} \tag{64}$$

*denote the slope of the wedge equation. Let*

$$\psi^c_{\bar{\nu}} \equiv \mathbb{E}\left[\frac{\bar{\nu} X}{1+\bar{\nu} X}\right] = \frac{\mathrm{d}\ln \mathcal{K}(\bar{\nu})}{\mathrm{d}\ln \bar{\nu}} \in (0,1) \tag{65}$$

*denote the expected share of operating profits in the payoff of such a producer's capital, with $\mathcal{K}$ the certainty-equivalent return of (59). At $\bar{\nu} = \nu(\bar{z})$, $\psi_{\bar{\nu}} = \psi(\bar{z})$ is the marginal producer's risky share, $\psi^c$ its expected share of operating profits, and $\Delta$ the wedge (26). Then the following identities hold:*

*(i)* $\lambda\, \varphi(\vartheta) = \vartheta$, *and hence* $\mathrm{d}\ln \mathcal{V}/\mathrm{d}\ln\lambda = x^*(\lambda)$;

*(ii)* $\lambda\, \varphi(\psi_{\bar{\nu}}) = \psi^c_{\bar{\nu}}$;

*(iii)* $\psi^c_{\bar{\nu}} = \psi_{\bar{\nu}}\,[1 - \ell_{\bar{\nu}}\,(1-\psi_{\bar{\nu}})]$ *and* $\mathcal{J}_{\bar{\nu}}\,(1+\Delta_{\bar{\nu}}) = \mathcal{Q}_{\bar{\nu}}\,\psi^c_{\bar{\nu}}/\psi_{\bar{\nu}} + \xi > 0$, *for every* $\bar{\nu} > 0$.

*As a result, for every $\lambda \in [1, \lambda_1)$ and every $R^{f*} > 1 - \delta$, equation (60) has exactly one fixed point $\Delta(\lambda, R^{f*}; \xi) \geq 0$ which is continuously differentiable in $(\lambda, R^{f*}, \xi)$, right differentiable at $\lambda = 1$ and $\mathcal{J}_{\bar{\nu}} > 0$ at the $\bar{\nu}$ for which $\Delta_{\bar{\nu}}$ is the fixed point.*

*Proof.*
Similar to the argument in the proof of Lemma B.2, the partial derivative of $\ln \mathcal{C}(x;\lambda)$ with respect to $\lambda$ given $\partial g_x/\partial\lambda = xX$ and $0 \leq xX/g_x \leq X + 1/\lambda$ is

$$\varphi(x) = x\,\mathbb{E}\left[\frac{X}{g_x}\right] \geq 0, \tag{66}$$

with $\varphi(0) = 0$.

(i) For $\lambda \in (1, \lambda_1)$, the optimal exposure $\vartheta = x^*(\lambda)$ satisfies the FOC $\mathbb{E}[(\lambda X - 1)/g_\vartheta] = 0$. Since $g_\vartheta = 1 + \vartheta(\lambda X - 1)$, we can derive the following identities:

$$\mathbb{E}\left[\frac{1}{g_\vartheta}\right] = 1 - \vartheta\,\mathbb{E}\left[\frac{\lambda X - 1}{g_\vartheta}\right] = 1, \qquad \lambda\,\mathbb{E}\left[\frac{X}{g_\vartheta}\right] = \mathbb{E}\left[\frac{1}{g_\vartheta}\right] = 1.$$

By (66) and the envelope theorem,

$$\frac{\mathrm{d}\ln\mathcal{V}}{\mathrm{d}\lambda} = \varphi(\vartheta) = \vartheta\,\mathbb{E}\left[\frac{X}{g_\vartheta}\right] = \frac{\vartheta}{\lambda}, \qquad \frac{\mathrm{d}\ln\mathcal{V}}{\mathrm{d}\ln\lambda} = \vartheta = x^*(\lambda).$$

which gives (i).

At $\lambda = 1$, $\vartheta = 0$ and $\varphi(0) = 0$, so $\lambda\,\varphi(\vartheta) = \vartheta$ holds there as well. For $0 < \lambda \leq 1$, $\mathcal{V}(\lambda) = 1$ and $x^*(\lambda) = 0$, so the elasticity formula holds for $\lambda < 1$. As $\lambda \downarrow 1$, continuity of $x^*$ implies $x^*(\lambda)/\lambda \to 0$. By the continuity of $\ln\mathcal{V}$ and the mean value theorem, we get that the right derivative at $\lambda = 1$ is zero, which matches the zero left derivative. Thus $\mathcal{V}$ is continuously differentiable on $(0, \lambda_1)$, and the elasticity formula holds.

(ii) For any $\bar{\nu} > 0$, substituting $g_{\psi_{\bar{\nu}}} = (1 + \bar{\nu}X)/(1 + \bar{\nu}/\lambda)$ and $\lambda\psi_{\bar{\nu}} = \bar{\nu}/(1 + \bar{\nu}/\lambda)$ into (66) gives

$$\begin{aligned}\lambda\,\varphi(\psi_{\bar{\nu}}) = \lambda\psi_{\bar{\nu}}\,\mathbb{E}\left[\frac{X}{g_{\psi_{\bar{\nu}}}}\right] &= \lambda\psi_{\bar{\nu}}\,(1 + \bar{\nu}/\lambda)\,\mathbb{E}\left[\frac{X}{1 + \bar{\nu}X}\right] \\ &= \mathbb{E}\left[\frac{\bar{\nu}X}{1 + \bar{\nu}X}\right] = \psi^c_{\bar{\nu}},\end{aligned}$$

which gives (ii).

(iii) For any $\bar{\nu} > 0$, $g_{\psi_{\bar{\nu}}} = (1 + \bar{\nu}X)/(1 + \bar{\nu}/\lambda)$ implies $\ln\mathcal{C}(\psi_{\bar{\nu}}; \lambda) = \ln\mathcal{K}(\bar{\nu}) - \ln(1 + \bar{\nu}/\lambda)$. Differentiate both sides with respect to $\ln\bar{\nu}$ at a fixed $\lambda$. On the left-hand side, we have $\partial\ln\mathcal{C}/\partial x = -\ell_{\bar{\nu}}$ at $\psi_{\bar{\nu}}$ and $\mathrm{d}\psi_{\bar{\nu}}/\mathrm{d}\ln\bar{\nu} = \psi_{\bar{\nu}}(1 - \psi_{\bar{\nu}})$. On the right-hand side, the derivative is $\psi^c_{\bar{\nu}} - \psi_{\bar{\nu}}$. Hence $\psi^c_{\bar{\nu}} - \psi_{\bar{\nu}} = -\ell_{\bar{\nu}}\,\psi_{\bar{\nu}}(1 - \psi_{\bar{\nu}})$. By the definition of $\Delta_{\bar{\nu}}$, we have $\mathcal{Q}_{\bar{\nu}} = 1 + (1 + \xi)\Delta_{\bar{\nu}}$, so $(1 + \xi)(1 + \Delta_{\bar{\nu}}) = \mathcal{Q}_{\bar{\nu}} + \xi$ and thus we can derive

$$\mathcal{J}_{\bar{\nu}}\,(1 + \Delta_{\bar{\nu}}) = (1 + \xi)(1 + \Delta_{\bar{\nu}}) - \mathcal{Q}_{\bar{\nu}}\,\ell_{\bar{\nu}}\,(1 - \psi_{\bar{\nu}}) = \mathcal{Q}_{\bar{\nu}}\,[1 - \ell_{\bar{\nu}}\,(1 - \psi_{\bar{\nu}})] + \xi = \mathcal{Q}_{\bar{\nu}}\,\frac{\psi^c_{\bar{\nu}}}{\psi_{\bar{\nu}}} + \xi,$$

which is positive as $\psi^c_{\bar{\nu}} > 0$.

*Uniqueness and differentiability of the fixed point* $\Delta$. By homogeneity of CE, the fixed-point equation (60) is equivalent to

$$\left(1 + \frac{\xi\Delta}{1 + \Delta}\right)\mathrm{CE}\left(\frac{1 - \delta}{R^{f*}} + \lambda X\left[1 + \Delta - \frac{1 - \delta}{R^{f*}}\right]\right) = \mathcal{V}(\lambda).$$

For $\Delta \geq 0$, $\left(1 + \frac{\xi\Delta}{1+\Delta}\right)$ is positive and nondecreasing, while the payoff inside the certainty equivalent increases strictly in $\Delta$ in every state. The left-hand side is therefore continuous and strictly increasing in $\Delta$. When $\Delta = 0$, the left-hand side value is

$$\mathcal{C}\left(1 - \frac{1 - \delta}{R^{f*}}; \lambda\right) \leq \mathcal{V}(\lambda).$$

by optimality. Moreover, the left-hand side is at least $\lambda[1 + \Delta - (1 - \delta)/R^{f*}]\mathrm{CE}(X)$, which goes to infinity as $\Delta \to \infty$. By the intermediate value theorem, we conclude that there exists exactly one

nonnegative solution of $\Delta_{\bar{\nu}}$.

The wedge equation is continuously differentiable in $(\Delta, \lambda, R^{f*}, \xi)$, by the established differentiability of $\mathcal{C}$ and $\mathcal{V}$. At the fixed point $\Delta_{\bar{\nu}}$, part (iii) implies that the slope $\mathcal{J}_{\bar{\nu}}$ with respect to $\Delta$ is strictly positive. The implicit function theorem therefore implies that $\Delta(\lambda, R^{f*}; \xi)$ is continuously differentiable in $(\lambda, R^{f*}, \xi)$. □

## B.3 Equilibrium characterization

A period by period equilibrium with interior exposure (Section 2.5 and Assumption A0) is a pair $(\bar{z}, \lambda)$ that solves the system of equations, Entry and Pricing, of Section 3. We prove Proposition 3.1 by substitution: derive the risk premium as a function of the cutoff and the degree of integration from the Pricing equation, and then substitute it into the Entry equation. This results in one equilibrium equation with the cutoff being the unknown.

**The threshold for interior exposure.** At given aggregate capitalists' wealth level $W$, equity-market clearing supplies domestic investors with an amount of equity that has the following exposure to the aggregate risk factor:

$$\vartheta^s(\lambda; \bar{z}, \omega) = \theta(\bar{z}, \omega)\, \bar{\psi}\big(\nu(\bar{z})/\lambda\big).$$

Optimal portfolio choice characterizes the demand of investors' exposure to the aggregate risk factor $x^*(\lambda)$. Thus, the Pricing condition in Section 3 shows that the equilibrium risk premium should be such that the equity market clears, and the domestic investors' optimal portfolio choice is consistent with the equity index's exposure to aggregate risk, i.e. $x^*(\lambda) = \vartheta^s(\lambda; \bar{z}, \omega)$.

As $\lambda$ increases from 1 to $\lambda_1$, investors' desired exposure rises from zero to one, while the exposure required by market clearing is positive and decreasing. The two exposures therefore intersect below one if and only if $\vartheta^s(\lambda_1; \bar{z}, \omega) < 1$. Since $\vartheta^s$ also decreases in the value of the cutoff, define the critical cutoff by

$$\vartheta^s\big(\lambda_1; \bar{z}_{\text{crit}}(\omega; W), \omega\big) = 1. \tag{67}$$

The critical cutoff ensures that the equity market clearing condition can be satisfied with an interior exposure if and only if $\bar{z} > \bar{z}_{\text{crit}}(\omega; W)$. Lemma B.4 establishes that this cutoff exists and is unique, and that market clearing satisfies Assumption A0 precisely when $\bar{z} > \bar{z}_{\text{crit}}(\omega; W)$.

**Lemma B.4** (The risk premium)**.** *Let $\xi > 0$ and fix $\omega \in [0, 1)$.*

*(i) The threshold $\bar{z}_{\text{crit}}(\omega; W)$ of (67) exists and is unique. The portfolio condition has a solution that satisfies Assumption A0 if and only if $\bar{z} > \bar{z}_{\text{crit}}(\omega; W)$. The solution $\lambda(\bar{z}, \omega) \in (1, \lambda_1)$ is unique, continuous in $(\bar{z}, \omega)$, and converges to $\lambda_1$ as $\bar{z}$ decreases to $\bar{z}_{\text{crit}}(\omega; W)$; investors' exposure to the aggregate risk factor is then $\vartheta = x^*\big(\lambda(\bar{z}, \omega)\big)$.*

*(ii) The risk premium $\lambda(\bar{z}, \omega)$ and investors' exposure are strictly decreasing in $\bar{z}$ and in $\omega$.*

*Proof.*
With log utility, Lemma A.3(iii) gives

$$\mathbb{E}[M(R^e - R^{f*})] = \bar{\psi}\, F(\vartheta, \lambda).$$

Since $\bar{\psi} > 0$, the portfolio optimality condition is equivalent to $F(\vartheta, \lambda) = 0$. The domestic exposure supplied by the equity index is positive given $\xi > 0$ and $\omega < 1$. Lemma B.1(i) therefore implies that the portfolio condition satisfies Assumption A0 if and only if

$$1 < \lambda < \lambda_1, \qquad x^*(\lambda) = \vartheta^s(\lambda; \bar{z}, \omega).$$

The investors' equity demand implies the desired exposure $x^*(\lambda)$ increases continuously from zero to one on $\lambda \in [1, \lambda_1]$. The supplied exposure $\vartheta^s$ is strictly decreasing in $\lambda$, and proportional to $1 - \omega$. It also decreases strictly with the cutoff, since

$$\frac{\partial \ln \vartheta^s}{\partial \ln \bar{z}} = -\frac{\eta \bar{z}^\eta}{\bar{z}^\eta - 1} + (1 + e_D)\bar{\psi}_h < -\eta + (1 + e_D) = -\alpha(\eta - 1) < 0,$$

where $\bar{\psi}_h < 1$ by Lemma A.3(iv).

At $\lambda = 1$, investors' optimal exposure is zero and supplied exposure is positive, meaning the equity market will never clear at $\lambda = 1$. Thus continuity and the opposite monotonicity of the two curves give a unique intersection in $\lambda \in (1, \lambda_1)$ precisely when

$$\vartheta^s(\lambda_1; \bar{z}, \omega) < 1.$$

If this inequality is not satisfied, supplied exposure exceeds one on the interval $\lambda \in (1, \lambda_1)$, while desired exposure remains below one, and the model does not have an equilibrium.

When $\lambda = \lambda_1$, supplied exposure is

$$\vartheta^s(\lambda_1; \bar{z}, \omega) = \frac{(1 - \omega)\xi}{\bar{z}^\eta - 1}\, \bar{\psi}\left(\frac{\nu(\bar{z})}{\lambda_1}\right).$$

As $\bar{z} \downarrow 1$, the denominator $\bar{z}^\eta - 1$ approaches zero while the index's risky share converges to a positive value, so supplied exposure tends to infinity. As $\bar{z} \to \infty$, $\frac{(1-\omega)\xi}{\bar{z}^\eta - 1}$ approaches zero and $\bar{\psi} < 1$, so supplied exposure approaches zero. Since supplied exposure is continuous and strictly decreasing in $\bar{z}$, it equals one at exactly one cutoff, $\bar{z}_{\text{crit}}(\omega; W)$, defined by (67). Thus an interior clearing premium exists uniquely when $\bar{z} > \bar{z}_{\text{crit}}(\omega; W)$, with investors' exposure equal to $x^*(\lambda(\bar{z}, \omega))$. Continuity of $\lambda(\bar{z}, \omega)$ follows from compactness and uniqueness of the equilibrium risk premium.

Finally, given an interior clearing premium, the desired exposure is $x^*(\lambda) = \vartheta^s(\lambda; \bar{z}, \omega)$. Since

$\mathrm{d}x^*/\mathrm{d}\lambda > 0$ and $\partial\vartheta^s/\partial\lambda < 0$, the implicit function theorem gives

$$\frac{\partial\lambda}{\partial\bar{z}} = \frac{\partial\vartheta^s/\partial\bar{z}}{\mathrm{d}x^*/\mathrm{d}\lambda - \partial\vartheta^s/\partial\lambda} < 0,$$
$$\frac{\partial\lambda}{\partial\omega} = \frac{\partial\vartheta^s/\partial\omega}{\mathrm{d}x^*/\mathrm{d}\lambda - \partial\vartheta^s/\partial\lambda} < 0.$$

Since $x^*(\lambda)$ is strictly increasing in $\lambda$ on $\lambda \in (1, \lambda_1)$, investors' exposure $x^*(\lambda(\bar{z}, \omega))$ also decreases strictly in $\bar{z}$ and $\omega$. □

**The risk-free rate that supports the equilibrium cutoff.** Let $\xi > 0$ and fix $\omega \in [0, 1)$. For a cutoff $\bar{z} > \bar{z}_{\text{crit}}(\omega; W)$, let $\lambda = \lambda(\bar{z}, \omega)$ be the risk premium of Lemma B.4, $\bar{\nu} = \nu(\bar{z})$, and $\psi(\bar{z}) = \bar{\nu}/(\lambda + \bar{\nu})$, and define

$$\mathcal{R}_\omega(\bar{z}) \equiv (1-\delta)\,\frac{1+\bar{\nu}/\lambda}{1+\Delta_\omega(\bar{z})} = \frac{(1+\xi)(1-\delta)}{\mathcal{V}(\lambda)/\mathcal{K}(\bar{\nu}) + \xi/(1+\bar{\nu}/\lambda)}, \quad \Delta_\omega(\bar{z}) \equiv \frac{1}{1+\xi}\left[\frac{\mathcal{V}(\lambda)}{\mathcal{C}\big(\psi(\bar{z});\lambda\big)} - 1\right], \tag{68}$$

where the second expression for $\mathcal{R}_\omega$ follows from (59). The proof below shows that $\mathcal{R}_\omega(\bar{z})$ is the world rate at which the capitalist with productivity $\bar{z}$ is indifferent between producing and investing, the equity market clears at $\bar{z}$, and that $\Delta_\omega(\bar{z})$ is then the occupational wedge. Define

$$R_{\text{crit}}(\omega; W) \equiv \lim_{\bar{z}\downarrow\bar{z}_{\text{crit}}(\omega;W)} \mathcal{R}_\omega(\bar{z})$$

as the lower bound of the world risk-free rate that supports an interior equilibrium.

*Proof of Proposition 3.1.*
Fix $\bar{z} > \bar{z}_{\text{crit}}(\omega; W)$ and compute the index return under the candidate issuance rule $E_i = \xi a_i$. With $\theta = \theta(\bar{z}, \omega)$, Lemma B.4(i) proves the unique premium $\lambda = \lambda(\bar{z}, \omega) \in (1, \lambda_1)$ consistent with investor optimality and equity-market clearing. Let $p = \mathbb{E}[D]/(\lambda R^{f*})$. Investors then hold $\vartheta = x^*(\lambda)$ and obtain $\mathrm{CE}(R^I) = R^{f*}\mathcal{V}(\lambda)$.

Occupational indifference requires $q(\bar{z}) \geq 1$. If $q(\bar{z}) < 1$, the marginal producer will issue no equity and obtain

$$\mathrm{CE}(R^K(\bar{z})) = q(\bar{z})R^{f*}\mathcal{C}(\psi(\bar{z});\lambda) \leq q(\bar{z})R^{f*}\mathcal{V}(\lambda) < \mathrm{CE}(R^I),$$

contradicting the indifference condition. Full issuance is therefore optimal at the cutoff, and we can write the occupational indifference condition as

$$R^{f*}\mathcal{V}(\lambda) = (1+\xi)(1-\delta)\mathcal{K}(\bar{\nu}) - \xi R^{f*}\mathcal{C}(\psi(\bar{z});\lambda).$$

Using $\mathcal{C}(\psi(\bar{z});\lambda) = \mathcal{K}(\bar{\nu})/(1+\bar{\nu}/\lambda)$ from (59), we can derive $R^{f*} = \mathcal{R}_\omega(\bar{z})$. Conversely, $R^{f*} = \mathcal{R}_\omega(\bar{z})$ implies that $q(\bar{z}) = 1 + \Delta_\omega(\bar{z}) \geq 1$, since $\mathcal{V}(\lambda) \geq \mathcal{C}(\psi(\bar{z});\lambda)$. Full issuance is then the

producers' optimal scaling decision and establishes occupational indifference for marginal producers with productivity level $\bar{z}$.

The function $\mathcal{R}_\omega$ is continuous and strictly increasing in $\bar{z}$. As $\bar{z}$ increases, $\bar{\nu}$ and $\mathcal{K}(\bar{\nu})$ increase strictly, while $\lambda(\bar{z},\omega)$ decreases strictly by Lemma B.4(ii). Since $\mathcal{V}$ is strictly increasing in $\lambda$, both terms in the denominator of equation (68) decrease strictly. As $\bar{z}\to\infty$, $\bar{\nu}\to\infty$, we have

$$\mathcal{K}(\bar{\nu}) \geq \bar{\nu}\,\mathrm{CE}(X) \to \infty, \qquad \mathcal{V}(\lambda) \leq \mathcal{V}(\lambda_1) < \infty, \qquad \frac{\bar{\nu}}{\lambda} \geq \frac{\bar{\nu}}{\lambda_1} \to \infty.$$

Thus $\lim_{\bar{z}\to\infty} \mathcal{R}_\omega(\bar{z}) \to \infty$. As $\bar{z} \downarrow \bar{z}_{\mathrm{crit}}(\omega; W)$, Lemma B.4(i) shows $\lambda \to \lambda_1$, while $\bar{\nu}$ approaches a finite positive value. The denominator in (68) therefore has a finite positive limit, meaning $0 < R_{\mathrm{crit}}(\omega; W) < \infty$. As a result, for every $R^{f*} > R_{\mathrm{crit}}(\omega; W)$, the equation $R^{f*} = \mathcal{R}_\omega(\bar{z})$ has a unique solution with $\bar{z} > \bar{z}_{\mathrm{crit}}(\omega; W)$.

Next, we verify that the cutoff $\bar{z}$ that is the solution to the equation $R^{f*} = \mathcal{R}_\omega(\bar{z})$ satisfies the other equilibrium conditions. Suppose $R^{f*} > R_{\mathrm{crit}}(\omega; W)$ and let $\bar{z}$ be the unique solution of $R^{f*} = \mathcal{R}_\omega(\bar{z})$. Set $\lambda = \lambda(\bar{z},\omega)$ as determined by Lemma B.4(i), $p = \mathbb{E}[D]/(\lambda R^{f*})$, and $\theta = (1-\omega)\xi/(\bar{z}^\eta - 1)$, and set the wage according to (33). Occupational indifference holds at $\bar{z}$. The optimized production certainty equivalent is strictly increasing in $z$ (proof of Proposition 2.3(a)), so capitalists below the cutoff optimally invest and those above produce. Since $q(z) \geq q(\bar{z}) \geq 1$ for every producer, full issuance is optimal by (8).

Investor optimality and equity-market clearing follow from Lemma B.4(i) and (36); labor demand and labor-market clearing follow from Proposition A.1. Foreign investors fill the ownership cap because $\lambda > 1$ (Appendix A.3), and saving is optimal at $S(H) = \beta$ by Proposition 2.2(c).

To verify domestic asset pricing conditions, let $M = 1/R^I$. The portfolio optimality condition and $\mathbb{E}[MR^I] = 1$ imply

$$1 = R^{f*}\mathbb{E}[M] + \theta\,\mathbb{E}[M(R^e - R^{f*})] = R^{f*}\mathbb{E}[M], \qquad \mathbb{E}[MR^e] = 1.$$

Using $R^e = (1-\bar{\psi})R^{f*} + \bar{\psi}D/p$ with $\bar{\psi} > 0$ gives $\mathbb{E}[MD] = p$. Therefore, for every $z$,

$$\mathbb{E}[MR^K(z)] = z\,\mathbb{E}[MD] + (1-\delta)\mathbb{E}[M] = zp + \frac{1-\delta}{R^{f*}} = q(z).$$

These expectations are finite because $R^I \geq R^{f*}(1-\vartheta) > 0$ and $\mathbb{E}[D] < \infty$. Thus all pricing, optimality, and market-clearing conditions hold at wealth $W$.

Conversely, any equilibrium with interior investor exposure has a productivity cutoff and full issuance by Proposition 2.3. Positive investor exposure (Assumption A0) implies $\lambda > 1$, so foreign investors fill the cap and equity-market clearing gives $\theta = \theta(\bar{z},\omega)$. Lemma B.4(i) then implies $\bar{z} > \bar{z}_{\mathrm{crit}}(\omega; W)$ and $\lambda = \lambda(\bar{z},\omega)$. Occupational indifference requires $R^{f*} = \mathcal{R}_\omega(\bar{z})$, so $R^{f*} > R_{\mathrm{crit}}(\omega; W)$ and the cutoff equals the unique one constructed above. The associated prices and aggregate allocations are therefore unique. If $R^{f*} \leq R_{\mathrm{crit}}(\omega; W)$, no equilibrium with interior exposure exists.

Finally, at the constructed equilibrium, $q(\bar{z}) = 1 + \Delta_\omega(\bar{z})$ and (27) imply that $\Delta_\omega(\bar{z})$ satisfies

(60). Since $R^{f*} > 1-\delta$, uniqueness in Lemma B.3 gives $\Delta_\omega(\bar{z}) = \Delta(\lambda, R^{f*}; \xi)$, and the cutoff identity is given by the Entry equation. The investor optimality and equity-market clearing are captured by the Pricing equation. Conversely, any pair $\bar{z} > 1$ and $\lambda \in (1, \lambda_1)$ satisfying the Entry and Pricing equations has $\lambda = \lambda(\bar{z}, \omega)$, where the Entry equation implies $q(\bar{z}) = 1 + \Delta$. Equation (60) then gives

$$\big[(1+\xi)q(\bar{z}) - \xi\big]\mathcal{C}(\psi(\bar{z}); \lambda) = \mathcal{V}(\lambda),$$

which is occupational indifference. Hence the pair satisfies $R^{f*} = \mathcal{R}_\omega(\bar{z})$ and generates the unique equilibrium constructed above. □

## B.4 Occupational indifference, equity-market clearing, and equilibrium feedback

In this section, we prove Lemmas 3.1 and 3.2 and Corollary 3.1 of Section 3.2 by differentiating the Entry and Pricing equations. Appendix B.5 uses the same differentials for the comparative statics. We use $\psi = \psi(\bar{z}) \in (0,1)$ for the marginal producer's risky share of capital, $\varepsilon_\vartheta \equiv \mathrm{d}\ln\vartheta^*/\mathrm{d}\ln\lambda > 0$ for the elasticity of investors' demand for risk exposure, which is $x^*_\lambda$ of Lemma B.1(iii), and $\varepsilon_{\bar{\psi}} \equiv \bar{\psi}_h \in (0,1)$ for the elasticity of the index's risk exposure $\bar{\psi}$ (Lemma A.3(iv)). Differentiation under the expectation and under the Pareto integral is justified as in the proofs of Lemmas B.1(iii) and A.3(iv).

*Proof of Lemma 3.1.*
By equation (60), we define

$$\mathcal{Q} \equiv 1 + (1+\xi)\Delta = \frac{\mathcal{V}(\lambda)}{\mathcal{C}(\psi;\lambda)}, \qquad \psi = 1 - \frac{1-\delta}{R^{f*}(1+\Delta)},$$

and $\Delta \geq 0$, with equality if and only if $\psi = \vartheta$ (Lemma B.2). By Lemma B.3(ii), $\lambda\varphi(\psi) = \psi^c$, the marginal producer's expected profit share (65). Parts (i), (ii), and (iii) of Lemma B.3 show that

$$\frac{\mathrm{d}\ln\mathcal{V}}{\mathrm{d}\ln\lambda} = \vartheta, \qquad \frac{\partial\ln\mathcal{C}(\psi;\lambda)}{\partial\ln\lambda} = \psi^c, \qquad \frac{\partial\ln\mathcal{C}(x;\lambda)}{\partial x}\bigg|_{x=\psi} = \frac{\psi^c - \psi}{\psi(1-\psi)}.$$

Holding $R^{f*}$ and the remaining parameters fixed, take total differentials with respect to $\lambda$ and $\xi$, allowing $\Delta = \Delta(\lambda, R^{f*}; \xi)$ to adjust. Since $\psi = 1 - (1-\delta)/[R^{f*}(1+\Delta)]$, we have $\mathrm{d}\psi = (1-\psi)\,\mathrm{d}\Delta/(1+\Delta)$. The total differential of $\ln\mathcal{Q} = \ln\mathcal{V}(\lambda) - \ln\mathcal{C}(\psi;\lambda)$ is therefore

$$\frac{(1+\xi)\,\mathrm{d}\Delta + \Delta\,\mathrm{d}\xi}{\mathcal{Q}} = (\vartheta - \psi^c)\,\mathrm{d}\ln\lambda - \frac{\psi^c - \psi}{\psi(1+\Delta)}\,\mathrm{d}\Delta.$$

and collecting terms,

$$\mathrm{d}\Delta = \frac{(1+\Delta)\psi}{\mathcal{A}}\Big[\mathcal{Q}(\vartheta - \psi^c)\,\mathrm{d}\ln\lambda - \Delta\,\mathrm{d}\xi\Big], \qquad \mathcal{A} \equiv \mathcal{Q}\psi^c + \xi\psi > 0.$$

In particular $\partial\Delta/\partial\xi = -(1+\Delta)\psi\Delta/\mathcal{A} \leq 0$, strictly when $\Delta > 0$. Since $\mathcal{U} \equiv R^{f*}(1+\Delta) + \delta - 1 = R^{f*}(1+\Delta)\psi$, we can derive

$$\frac{R^{f*}}{\mathcal{U}}\frac{\partial\Delta}{\partial\xi} = -\frac{\Delta}{\mathcal{A}} \leq 0, \qquad \Gamma \equiv 1 + \frac{R^{f*}}{\mathcal{U}}\frac{\partial\Delta}{\partial\ln\lambda} = 1 + \frac{\mathcal{Q}(\vartheta - \psi^c)}{\mathcal{A}} = \frac{\mathcal{Q}\vartheta + \xi\psi}{\mathcal{Q}\psi^c + \xi\psi} > 0.$$

With investors holding their optimal risk exposure $\vartheta^*(\lambda)$, the occupational-indifference calculation in the proof of Proposition 3.1 gives the Entry equation. Since $\mathcal{U}(\lambda) = R^{f*}[1 + \Delta(\lambda, R^{f*}; \xi)] + \delta - 1 > 0$, the unique solution of the Entry equation is

$$Z(\lambda) = \left[\frac{\lambda\mathcal{U}(\lambda)}{\kappa(\xi, W)}\right]^{1/(1+e_D)}.$$

The function $Z$ is continuously differentiable by Lemma B.3. Near an interior equilibrium, continuity ensures that $Z(\lambda) > 1$, so this solution defines an admissible productivity cutoff.

Since $\mathrm{d}\ln\mathcal{U} = (R^{f*}/\mathcal{U})\,\mathrm{d}\Delta$ and $\partial\ln\kappa/\partial\xi = -(1-\alpha)/(1+\xi)$, taking total differentials of the logarithm of the Entry equation gives

$$(1+e_D)\,\mathrm{d}\ln\bar{z} = \Gamma\,\mathrm{d}\ln\lambda + \mathcal{B}\,\mathrm{d}\xi, \qquad \mathcal{B} \equiv \frac{1-\alpha}{1+\xi} + \frac{R^{f*}}{\mathcal{U}}\frac{\partial\Delta}{\partial\xi} = \frac{1-\alpha}{1+\xi} - \frac{\Delta}{\mathcal{A}}. \tag{69}$$

Holding $\xi$ fixed and using $\Gamma > 0$ therefore yields

$$\frac{\partial\ln Z}{\partial\ln\lambda} = \frac{\Gamma}{1+e_D} > 0,$$

which proves Lemma 3.1.

□

*Proof of Lemma 3.2.*

Let $h \equiv \nu(\bar{z})/\lambda$ and let $\vartheta^s(\lambda; \bar{z}, \omega, \xi) = (1-\omega)\,\xi\,\bar{\psi}(h)/(\bar{z}^\eta - 1)$ be the risk exposure that the equity market supplies to investors, so that the Pricing equation can be written as $\vartheta^*(\lambda) = \vartheta^s(\lambda; \bar{z}, \omega, \xi)$. At fixed wealth $\nu(\bar{z}) \propto (1+\xi)^{\alpha-1}\bar{z}^{1+e_D}$, differentiating the log of the Pricing equation gives

$$(\varepsilon_\vartheta + \varepsilon_{\bar{\psi}})\,\mathrm{d}\ln\lambda = \vartheta^s_z\,\mathrm{d}\ln\bar{z} - \frac{\mathrm{d}\omega}{1-\omega} + a_\xi\,\mathrm{d}\xi, \qquad a_\xi \equiv \frac{1}{\xi} - \frac{(1-\alpha)\,\varepsilon_{\bar{\psi}}}{1+\xi} > \frac{1}{\xi} - \frac{1}{1+\xi} > 0, \tag{70}$$

where $\vartheta^s_z = -\eta\bar{z}^\eta/(\bar{z}^\eta - 1) + (1+e_D)\varepsilon_{\bar{\psi}} < -\alpha(\eta-1) < 0$ is the elasticity bounded in the proof of Lemma B.4. Since $\varepsilon_\vartheta + \varepsilon_{\bar{\psi}} > 0$, the implicit function theorem implies that the market-clearing premium is a continuously differentiable function $\lambda^*(\bar{z}; \omega, \xi)$ around the equilibrium cutoff $\bar{z}$. It is also unique: at every cutoff above $\bar{z}_{\mathrm{crit}}(\omega; W)$, the risk premium $\lambda \in (1, \lambda_1)$ that clears the equity market exists and is unique, i.e. $\lambda^*(\bar{z}; \omega, \xi) = \lambda(\bar{z}, \omega)$ by Lemma B.4(i). By (70), we can compute

$$\frac{\partial\ln\lambda^*}{\partial\ln\bar{z}} = \frac{\vartheta^s_z}{\varepsilon_\vartheta + \varepsilon_{\bar{\psi}}} < 0, \qquad \frac{\partial\ln\lambda^*}{\partial\omega} = -\frac{1}{(1-\omega)(\varepsilon_\vartheta + \varepsilon_{\bar{\psi}})} < 0, \qquad \frac{\partial\ln\lambda^*}{\partial\xi} = \frac{a_\xi}{\varepsilon_\vartheta + \varepsilon_{\bar{\psi}}} > 0.$$

The first two are Lemma 3.2; the third is the risk-premium channel of Proposition 3.3. □

*Proof of Corollary 3.1.*
Lemmas 3.1 and 3.2 imply

$$\varrho = -\frac{\Gamma}{1+e_D}\frac{\partial \ln \lambda^*}{\partial \ln \bar{z}} = -\frac{\Gamma \vartheta_z^s}{(1+e_D)(\varepsilon_\vartheta + \varepsilon_{\bar{\psi}})} > 0.$$

By (69) and (70), the Jacobian determinant of the logarithmic Entry and Pricing conditions with respect to $(\ln \bar{z}, \ln \lambda)$ is $(1+e_D)(\varepsilon_\vartheta + \varepsilon_{\bar{\psi}})(1+\varrho) > 0$. The implicit function theorem therefore gives a continuously differentiable local interior solution with respect to $(\omega, \xi)$.

Using (69) and (70) to eliminate $\mathrm{d}\ln\lambda$ yields

$$(1+e_D)(\varepsilon_\vartheta + \varepsilon_{\bar{\psi}})(1+\varrho)\,\mathrm{d}\ln\bar{z} = -\frac{\Gamma}{1-\omega}\,\mathrm{d}\omega + \left[(\varepsilon_\vartheta + \varepsilon_{\bar{\psi}})\mathcal{B} + \Gamma a_\xi\right]\mathrm{d}\xi. \tag{71}$$

Using the partial derivatives from Lemma 3.2, the above equation becomes

$$\mathrm{d}\ln\bar{z} = \frac{1}{1+\varrho}\left\{\frac{\Gamma}{1+e_D}\frac{\partial \ln\lambda^*}{\partial\omega}\,\mathrm{d}\omega + \left[\frac{\mathcal{B}}{1+e_D} + \frac{\Gamma}{1+e_D}\frac{\partial\ln\lambda^*}{\partial\xi}\right]\mathrm{d}\xi\right\}.$$

Thus feedback multiplies each policy response by $1/(1+\varrho) \in (0,1)$, reducing the magnitude of the impact of each policy change. □

### B.5 Comparative statics under log utility

**Equity-market liberalization.** As we only adjust $\omega$ given a fixed $\xi$, setting $\mathrm{d}\xi = 0$ in (71) gives

$$\frac{\mathrm{d}\ln\bar{z}}{\mathrm{d}\omega} = -\frac{\Gamma}{(1+e_D)(1+\varrho)(1-\omega)(\varepsilon_\vartheta + \varepsilon_{\bar{\psi}})} = \frac{\Gamma}{1+e_D}\frac{1}{1+\varrho}\frac{\partial\ln\lambda^*}{\partial\omega} < 0.$$

From occupational indifference, we have

$$\frac{\mathrm{d}\ln\lambda}{\mathrm{d}\omega} = \frac{1}{1+\varrho}\frac{\partial\ln\lambda^*}{\partial\omega} < 0.$$

Thus entry offsets part of the fall in the premium at the original cutoff, while preserving the direction of the effect.

Because TFP $= \left(\frac{\eta}{\eta-1}\bar{z}\right)^\alpha$ and $K = (1+\xi)W\bar{z}^{-\eta}$, TFP falls with the cutoff, by $\alpha$ times its proportional fall, while the capital stock and the mass of producers rise.

**Domestic financing capacity.** Setting $\mathrm{d}\omega = 0$ in (71) yields

$$\frac{\mathrm{d}\ln\bar{z}}{\mathrm{d}\xi} = \frac{1}{(1+e_D)(1+\varrho)}\left[\frac{1-\alpha}{1+\xi} + \frac{R^{f*}}{\mathcal{U}}\frac{\partial\Delta}{\partial\xi} + \Gamma\frac{\partial\ln\lambda^*}{\partial\xi}\right] = \frac{\mathcal{B} + \Gamma\,\partial\ln\lambda^*/\partial\xi}{(1+e_D)(1+\varrho)}. \tag{72}$$

The first and third terms are strictly positive. The second is nonpositive and is zero when $\psi = \vartheta$.

The cutoff rises if and only if

$$\frac{\Delta}{\mathcal{A}} < \frac{1-\alpha}{1+\xi} + \Gamma \frac{a_\xi}{\varepsilon_\vartheta + \varepsilon_{\bar{\psi}}}.$$

A sufficient condition is

$$\frac{\Delta}{\mathcal{A}} \leq \frac{1-\alpha}{1+\xi},$$

or equivalently:

$$-\frac{R^{f*}}{\mathcal{U}} \frac{\partial \Delta}{\partial \xi} \leq \frac{1-\alpha}{1+\xi}.$$

Under this condition, capital deepening alone offsets the wedge reduction, and the additional issuance channel makes the total response strictly positive.

At an interior equilibrium with $\psi = \vartheta$, the wedge is zero and the sufficient condition holds strictly. Continuity therefore ensures that it also holds in a neighborhood of any such equilibrium: $\Delta/\mathcal{A}$ depends on the equilibrium only through $(\lambda, R^{f*}, \xi)$, in which it is continuous (Lemma B.3). By Lemma B.2, the wedge is second order in the exposure gap $\psi - \vartheta$, while $\mathcal{A} \geq \xi\psi > 0$, so the condition holds whenever the gap is small enough.

**Financing capacity raises the cutoff when the premium is small.** The argument above shows that the cutoff rises whenever $\mathcal{B} \geq 0$ and that this holds in a neighborhood of any equilibrium with $\psi = \vartheta$. The next proposition sharpens Proposition 3.3 by replacing the neighborhood with an explicit condition. The idea is that the wedge is bounded by the premium investors earn on their exposure, so that $\mathcal{B} \geq 0$ reduces to a comparison between that premium and the labor share.

**Proposition B.1** (Financing capacity raises the cutoff when the premium is small relative to the user cost)**.** *Consider an interior equilibrium with log utility in which the marginal producer is at most as exposed as investors, $\psi(\bar{z}) \leq \vartheta$. Then*

$$(1+\xi)\,\Delta \;\leq\; \vartheta\,(\lambda - 1), \tag{73}$$

*and if*

$$\vartheta\,(\lambda - 1) \;\leq\; (1-\alpha)\big(\psi^c + \xi\,\psi(\bar{z})\big), \tag{74}$$

*then $\mathcal{B} \geq 0$ and*

$$\frac{\mathrm{d}\ln \bar{z}}{\mathrm{d}\xi} > 0, \qquad \frac{\mathrm{d}\ln \mathrm{TFP}}{\mathrm{d}\xi} > 0.$$

*Condition (74) is implied by*

$$R^{f*}(\lambda - 1) \;\leq\; (1-\alpha)(1+\xi)\big(R^{f*} + \delta - 1\big), \tag{75}$$

*which compares the risk premium on operating profits with the labor share times the user cost of capital, scaled by $1+\xi$.*

*Proof.*
Let $\psi = \psi(\bar{z})$. Since $\mathcal{C}(\cdot;\lambda)$ is concave and maximized at $\vartheta \geq \psi$, it is nondecreasing on $[0,\vartheta]$, so $\mathcal{C}(\psi;\lambda) \geq \mathcal{C}(0;\lambda) = 1$. By Jensen's inequality and $\mathbb{E}[X] = 1$, we can sign $\mathcal{V}(\lambda) \leq \mathbb{E}[g_\vartheta] = 1+\vartheta(\lambda-1)$. The wedge equation therefore implies

$$(1+\xi)\Delta = \frac{\mathcal{V}(\lambda)}{\mathcal{C}(\psi;\lambda)} - 1 \leq \mathcal{V}(\lambda) - 1 \leq \vartheta(\lambda-1),$$

which proves equation (73).

Since $\mathcal{Q} = 1 + (1+\xi)\Delta \geq 1$, we have $\mathcal{A} = \mathcal{Q}\psi^c + \xi\psi \geq \psi^c + \xi\psi$. Under (74), we can derive

$$(1+\xi)\Delta \leq \vartheta(\lambda-1) \leq (1-\alpha)(\psi^c + \xi\psi) \leq (1-\alpha)\mathcal{A}.$$

Hence $\mathcal{B} = (1-\alpha)/(1+\xi) - \Delta/\mathcal{A} \geq 0$. Because $\Gamma\,\partial \ln \lambda^*/\partial\xi > 0$, equation (72) gives $\mathrm{d}\ln\bar{z}/\mathrm{d}\xi > 0$. The same sign holds for TFP because $\mathrm{TFP} = \big(\eta\bar{z}/(\eta-1)\big)^\alpha$.

Finally, the derivative of $\mathcal{C}(\cdot;\lambda)$ at $\psi$ is nonnegative, so Lemma B.3(iii) gives $\psi^c \geq \psi$. Moreover, $\Delta \geq 0$ and (27) imply $\psi \geq (R^{f*} + \delta - 1)/R^{f*}$. Since $\lambda > 1$ and $\vartheta < 1$, condition (75) therefore implies

$$\vartheta(\lambda-1) \leq \lambda - 1 \leq (1-\alpha)(1+\xi)\frac{R^{f*}+\delta-1}{R^{f*}} \leq (1-\alpha)(1+\xi)\psi \leq (1-\alpha)(\psi^c + \xi\psi),$$

which gives equation (74). □

# C Stylized Facts: Data and Additional Results

## C.1 Openings, samples, and data

**Dates.** We date each opening by its official liberalization month in Bekaert and Harvey (2000) and Bekaert et al. (2005), which is the month of the first formal change in law or regulation after which foreign investors could buy equities directly on the domestic market. Monthly outcomes use this month, and annual outcomes use its calendar year. Table C.1 lists the dates for the 28 openers of the panel and marks the countries used in monthly and annual analyses. For Côte d'Ivoire and South Africa the sources give the year only, which suffices for the annual outcomes; for the re-pricing of Fact 3, South Africa's opening is dated as January 1996.

Table C.1: Equity market openings: dates and samples

| Country | Official opening | In monthly analyses | In annual analyses |
|---|---|---|---|
| Argentina | Nov 1989 | ✓ | ✓ |
| Brazil | May 1991 | ✓ | ✓ |
| Chile | Jan 1992 | ✓ | ✓ |
| Colombia | Feb 1991 | ✓ | ✓ |
| Côte d'Ivoire | 1995, month unknown | | ✓ |
| Greece | Dec 1987 | ✓ | ✓ |
| India | Nov 1992 | ✓ | ✓ |
| Indonesia | Sep 1989 | | ✓ |
| Israel | Nov 1993 | | ✓ |
| Jamaica | Sep 1991 | | ✓ |
| Jordan | Dec 1995 | ✓ | ✓ |
| Kenya | Jan 1995 | | ✓ |
| Korea | Jan 1992 | ✓ | ✓ |
| Malaysia | Dec 1988 | ✓ | ✓ |
| Mexico | May 1989 | ✓ | ✓ |
| Morocco | Jun 1988 | | ✓ |
| Nigeria | Aug 1995 | ✓ | ✓ |
| Pakistan | Feb 1991 | ✓ | ✓ |
| Philippines | Jun 1991 | ✓ | ✓ |
| Portugal | Jul 1986 | | ✓ |
| South Africa | 1996, month unknown | ✓ | ✓ |
| Sri Lanka | Oct 1990 | | ✓ |
| Thailand | Sep 1987 | ✓ | ✓ |
| Trinidad and Tobago | Apr 1997 | | ✓ |
| Tunisia | Jun 1995 | | ✓ |
| Turkey | Aug 1989 | ✓ | ✓ |
| Venezuela | Jan 1990 | ✓ | ✓ |
| Zimbabwe | Jun 1993 | ✓ | ✓ |

*Notes:* Official opening: the official liberalization month of Bekaert and Harvey (2000), extended by Bekaert et al. (2005); Turkey's is August 1989, the month of Decree 32. In monthly analyses: a check mark indicates that the country enters at least one analysis of monthly data, namely the estimates for equity prices and dividend yields (Figure 6a and Table C.2) or the re-pricing of Fact 3 (Figure 7a). In annual analyses: a check mark indicates that the country enters at least one analysis of annual data, namely the estimates for foreign holdings (Figure 4), listed companies (Figure 6b), aggregate TFP and real GDP (Appendix C.6), or the TFP growth of Fact 3 (Figure 7).

**External positions.** Gross portfolio equity liabilities are the market value of foreign holdings of domestic portfolio equity, and gross portfolio equity assets are residents' holdings of foreign equity, both end-of-year stocks divided by that year's GDP. The foreign portfolio equity inflow is the change in liabilities, and the net portfolio equity inflow subtracts from it the change in assets; the same inflows net of revaluation subtract the previous year's liabilities times the dollar price return on the domestic index. The total net capital inflow is minus the current account; the net reserve inflow is minus the change in reserves, so reserve accumulation is a negative inflow; and the net inflow through debt and reserves adds to it the change in debt liabilities less debt assets. Each flow is winsorized at its own 1st and 99th percentiles, so the components of a total need not add to it exactly; the positions and the total net capital inflow are not winsorized. The database records zero portfolio equity positions in every year from 1980 to 1994 for three comparison countries (Jordan, Nigeria, and Trinidad and Tobago) and three openers (Jamaica, Sri Lanka, and Zimbabwe).

**Valuation series.** The S&P/IFC Global indices are monthly from December 1975 for the nine markets the IFC covered first and from December 1984 for six more; seven further indices begin between January 1978 and January 1997, and the other six markets have no index in the Datastream series. Datastream does not distribute the dividend yield of these indices, so we derive it from the price index $P$ and the total-return index $R$: the dividend paid in month $t$ per unit of the price index is $d_{c,t} = P_{c,t-1}\,(R_{c,t}/R_{c,t-1} - P_{c,t}/P_{c,t-1})$, and the dividend yield is the sum of $d_{c,t}$ over the past twelve months divided by $P_{c,t}$, in percent. This is the yield of Bekaert and Harvey (2000). In the years when both exist, it matches the year-end yield printed in the S&P/IFC Emerging Stock Markets Factbooks with a correlation of 0.90 and a median gap of 0.2 percentage points, and the Datastream and Factbook indices agree within 2 percent.

**Inflation and stabilizations.** Consumer price inflation data primarily come from the World Development Indicators. We supplement missing observations for Argentina, Venezuela, Zimbabwe, and Jordan, using the historical inflation series from the World Economic Outlook for 1988 onward and the country profiles in the 1995 Factbook for 1985–1987. We classify an opening as occurring in a high-inflation environment if annual inflation exceeds 40 percent in the calendar year before, during, or after the official opening date. Among the openers, Argentina, Brazil, Nigeria, Mexico, Turkey, Venezuela, and Zimbabwe meet this criterion. We separately identify openings that occurred close to a stabilization program launched during a high-inflation crisis. Argentina's opening coincided with the Bonex plan in November 1989 and preceded the Convertibility plan of April 1991. Brazil's opening occurred within 24 months of both the Collor plan of April 1990 and the stand-by agreement of January 1992. Mexico's opening occurred within 24 months of the Pacto of December 1987. Venezuela's IMF extended arrangement of June 1989 does not qualify because Henry (2002) does not classify it as a high-inflation stabilization program. Turkey's stand-by agreement of July 1994 falls outside the window, occurring five years after its opening. We conduct robustness checks to ensure that our results are not driven by these high-inflation episodes or stabilization programs.

## C.2 The difference-in-differences design

**The two base months.** Consider an opener whose official month is January 1991, so that $g$ is January 1991, and the effect in March 1991, event time +2. With the base month $-6$ of the main text, $b(g) = g - 6$ and $a = 5$: the cell of equation (42) compares the change in the index from July 1990 to March 1991 among the countries of this cohort with the same change among the countries whose opening comes after August 1991, five months beyond the later of the cohort month and the current month. With the base month $-1$, $b(g) = g - 1$ and $a = 0$: the cell compares the change from December 1990 to March 1991 with the same change among the countries whose opening comes after March 1991. For an effect before the opening, say October 1990 at event time $-3$, the first version compares the change from July 1990 to October 1990 with the countries whose opening comes after June 1991, and the second compares the change from December 1990 back to October 1990 with the countries whose opening comes after January 1991. In both versions event

time counts from the official month; what moves is the month from which changes are measured and the countries eligible for comparison. Setting $b(g) = g - 6$ with $a = 5$ is the anticipation option of Callaway and Sant'Anna (2021) with five periods of anticipation, whose base period is the period before the anticipation window begins.

**Inference.** Standard errors come from the influence function of $\hat{\theta}_k$, which accounts for the estimated weights $w_{g,k}$ of equation (43), and are clustered by country: a country's contributions to every cell it enters are summed before squaring, so any serial correlation within a country is allowed. Clustering does not allow for dependence across countries hit by the same regional shock. With 17 countries and comparison groups of two or three countries in the late monthly cells, we also report leave-one-country-out jackknife standard errors, which re-estimate every path without one country at a time; for the monthly two-year averages they are 11 to 21 percent larger than the clustered ones, and the figures of the main text show jackknife bands.

**Joint test of pre-trends.** For foreign holdings in Figure 4, the joint test that the four pre-opening estimates, $\hat{\theta}_{-5}$ to $\hat{\theta}_{-2}$, are all zero gives $\chi^2(4) = 7.91$ (p-value 0.095), with the leave-one-country-out jackknife covariance. This rise in holdings before liberalization is mostly due to the valuation effect. Net of revaluation the path before the opening is flat (Figure 5).

### C.3 External positions and flows

This subsection gives the details behind Figures 4 and 5 and the estimates for the other external positions and the total net capital inflow.

**The stock and the cumulated flow.** By the fourth year after the opening, foreign holdings $F_{c,t}$ are 2.8 percentage points of GDP higher, and the cumulated foreign portfolio equity inflow, which sums each year's change in holdings divided by that year's GDP, is 3.2 percent of GDP higher (jackknife standard error 1.3). The two measure the same holdings, because the inflow is the change in the stock. With each country's GDP fixed at its 1980–1994 average in both, so that only the numerator moves, they are identical, at 4.5 percent of GDP, apart from the winsorizing of the yearly inflow, which that version omits. The stock measured against current GDP is lower because the openers' dollar GDP is 32 percent higher relative to the comparison countries on average over years 0 to 4. Residents' cumulated purchases of foreign equity are only 0.07 percent of GDP higher by the fourth year (jackknife standard error 0.20), against holdings of 0.5 percent of GDP in the year before the opening, so the net inflow, which subtracts those purchases, is 3.2 percent of GDP higher by the fourth year as well (jackknife standard error 1.2).

**The revaluation adjustment.** Let $L_{c,t}$ be the dollar value of foreign holdings of domestic portfolio equity at the end of year $t$, $Y_{c,t}$ dollar GDP, and $r_{c,t}$ the dollar price return of the country's S&P/IFC Global index over year $t$. The return excludes dividends, which are income rather than

revaluation. The observed position is $F_{c,t} = 100\, L_{c,t}/Y_{c,t}$, as in equation (44). The revaluation in year $s$ is the return on the holdings at the start of the year, and the change in holdings net of revaluation is the rest of the change,

$$V_{c,s} = r_{c,s}\, L_{c,s-1}, \qquad P_{c,s} = (L_{c,s} - L_{c,s-1}) - V_{c,s} = L_{c,s} - (1 + r_{c,s})\, L_{c,s-1}. \tag{76}$$

For the cohort that opens in year $g$, with base year $b = b(g) = g - 1$, we rebuild every country's dollar position from its observed holdings in the base year,

$$\tilde{L}^{\,b}_{c,t} = \begin{cases} L_{c,b} + \sum_{s=b+1}^{t} P_{c,s}, & t > b, \\ L_{c,b}, & t = b, \\ L_{c,b} - \sum_{s=t+1}^{b} P_{c,s}, & t < b, \end{cases} \tag{77}$$

and divide it by the same GDP as the observed position,

$$\tilde{F}^{\,b}_{c,t} = 100\, \frac{\tilde{L}^{\,b}_{c,t}}{Y_{c,t}}, \tag{78}$$

so that $\tilde{F}^{\,b}_{c,b} = F_{c,b}$. Equation (42) is then applied cell by cell with $y_{c,t} = \tilde{F}^{\,b(g)}_{c,t}$ for the cohort and for its comparison countries alike, all rebuilt from the cohort's base year, and the cells are aggregated by equation (43). Since $L_{c,t} - L_{c,b} = \sum_{s=b+1}^{t} (P_{c,s} + V_{c,s})$,

$$F_{c,t} - \tilde{F}^{\,b}_{c,t} = \frac{100}{Y_{c,t}} \sum_{s=b+1}^{t} V_{c,s} \quad (t > b), \qquad F_{c,t} - \tilde{F}^{\,b}_{c,t} = -\frac{100}{Y_{c,t}} \sum_{s=t+1}^{b} V_{c,s} \quad (t < b), \tag{79}$$

so the difference between the two estimated paths in Figure 5 is the difference in differences of the revaluation since the base year, in percent of current GDP. With $r_{c,s} = 0$ in every year the rebuilt position equals the observed one. The adjustment requires the index return in every year between the base year and year $t$, which restricts it to the country-years with an index return, and neither the positions nor $P_{c,s}$ are winsorized. It rests on three assumptions: foreigners' holdings earn the price return of the index, while in practice foreigners held mostly the large investable stocks; purchases made during year $s$ are not revalued in that year; and exchange-rate movements are part of the revaluation, since the return is in dollars, as the positions are.

**Revaluation and zero positions.** In the 13 openers with an index return in the year before the opening, foreign holdings are 3.7 percentage points of GDP higher by the fourth year as measured and 2.0 net of the revaluation since the base year (jackknife standard errors 1.2 and 0.8), so about half of the rise is revaluation (jackknife standard error of the share 0.12). Measured as flows, with the yearly inflow net of revaluation $100\, P_{c,s}/Y_{c,s}$ winsorized like the other flows, removing the revaluation lowers the cumulated inflow in the fourth year from 5.0 to 3.0 percent of GDP (jackknife standard errors 1.5 and 1.0), and the average yearly inflow over years 0 to 4 from 0.93 to 0.56 percent

of GDP (standard errors 0.24 and 0.14), so about 40 percent of the inflow is revaluation. Net of revaluation, the path of foreign holdings before the opening is flat, at 0.07 percentage points of GDP in year −5 against −0.26 as measured, so the rise in foreign holdings before the opening is revaluation. In this sample the fourth year rests on six openers, and in the last years of the panel the comparisons rest on two comparison countries. On the full sample, excluding the countries whose recorded positions are zero throughout raises the average position and flow over years 0 to 4 from 1.5 and 0.6 percent of GDP (Table C.5, column (1)) to 1.7 and 0.7 percent (standard errors 0.5 and 0.2).

**Other external positions.** Figures C.1 to C.5 apply the specification of Figure 4 to the other external positions, each in percent of that year's GDP, and Figure C.6 shows the total net capital inflow, minus the current account, summed from 1981 with each year's inflow in percent of that year's GDP. By the fourth year after the opening, relative to the countries not yet open, net foreign assets are 28.3 percentage points of GDP higher (jackknife standard error 8.2), debt liabilities 29.4 lower (9.4), the net FDI position 2.0 higher (3.8), FDI liabilities 1.7 lower (3.8), and reserves 2.9 higher (3.4), while the cumulated total net capital inflow is 1.2 percent of GDP lower (8.3). Before the opening, none of these differs from the comparison countries (joint p-values between 0.17 and 0.98). Net foreign assets and debt were large positions, net foreign liabilities of 39 percent and debt liabilities of 51 percent of GDP in the year before the opening, so the faster growth of the openers' dollar GDP, 57 log points more than in the comparison countries by the fourth year (jackknife standard error 13), moves their ratios mechanically. With each country's GDP fixed at its 1980–1994 average, so that only the positions move, net foreign assets are 9.3 percentage points higher (8.5), debt liabilities 2.0 lower (8.4), the net FDI position 2.2 lower (4.7), FDI liabilities 2.7 higher (4.5), and reserves 9.0 higher (4.5), while foreign equity holdings are 4.5 higher (1.8). In dollars, the openers therefore borrowed about as much as the comparison countries and accumulated more reserves. The fall in their debt relative to GDP and the rise in their net foreign assets reflect their faster GDP growth, and also the debt crises of several comparison countries in these years, Côte d'Ivoire, Jordan, Nigeria, and Trinidad and Tobago, whose debt rose relative to a stagnant or falling dollar GDP. Summing the yearly reserve and FDI flows instead, each divided by that year's GDP, gives 6.9 percent of GDP for reserve accumulation (3.7) and 0.4 for the net FDI inflow (4.2).

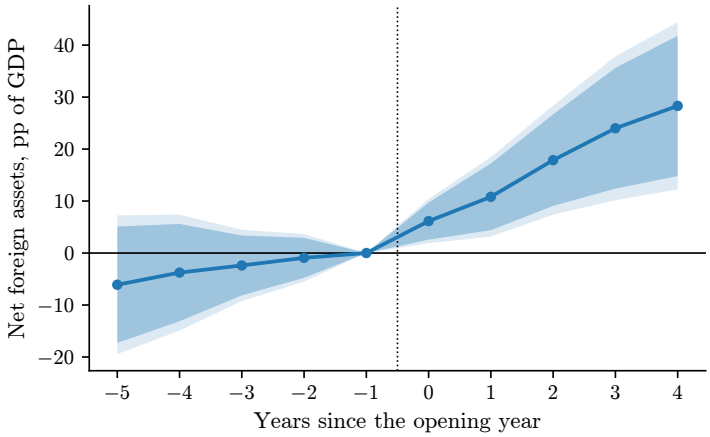



Figure C.1: Net foreign assets around equity market openings

*Notes:* Event-time estimates $\hat{\theta}_k$ of equation (43) for net foreign assets, the net international investment position (total foreign assets minus total foreign liabilities), in percent of that year's GDP, in the specification of Figure 4: changes from the year before the opening, relative to the countries not yet open, with 90 and 95 percent leave-one-country-out jackknife bands. Positions are not winsorized. Same 364 country-years, 26 countries, and 19 openers as Figure 4. Data: External Wealth of Nations (Lane and Milesi-Ferretti, 2007).

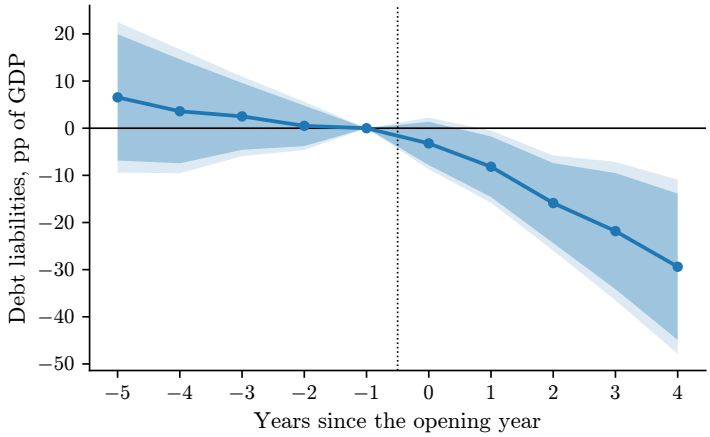



Figure C.2: Debt liabilities around equity market openings

*Notes:* Event-time estimates $\hat{\theta}_k$ of equation (43) for debt liabilities, portfolio debt and other investment, in percent of that year's GDP, in the specification of Figure 4: changes from the year before the opening, relative to the countries not yet open, with 90 and 95 percent leave-one-country-out jackknife bands. Positions are not winsorized. Same 364 country-years, 26 countries, and 19 openers as Figure 4. Data: External Wealth of Nations (Lane and Milesi-Ferretti, 2007).

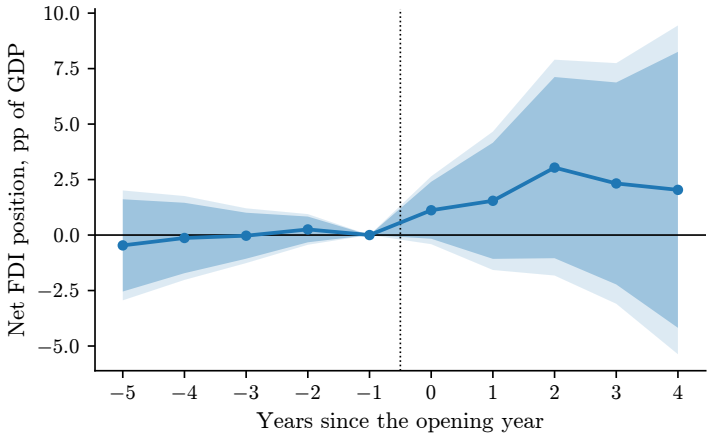



Figure C.3: The net FDI position around equity market openings

*Notes:* Event-time estimates $\hat{\theta}_k$ of equation (43) for the net FDI position, FDI assets minus FDI liabilities, in percent of that year's GDP, in the specification of Figure 4: changes from the year before the opening, relative to the countries not yet open, with 90 and 95 percent leave-one-country-out jackknife bands. Positions are not winsorized. Same 364 country-years, 26 countries, and 19 openers as Figure 4. Data: External Wealth of Nations (Lane and Milesi-Ferretti, 2007).

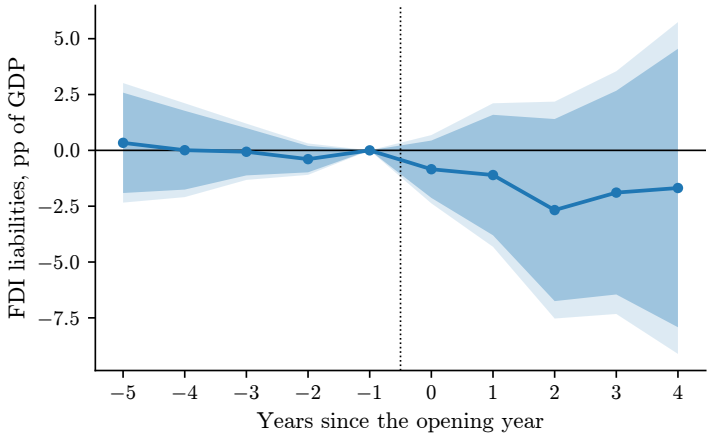



Figure C.4: FDI liabilities around equity market openings

*Notes:* Event-time estimates $\hat{\theta}_k$ of equation (43) for FDI liabilities, in percent of that year's GDP, in the specification of Figure 4: changes from the year before the opening, relative to the countries not yet open, with 90 and 95 percent leave-one-country-out jackknife bands. Positions are not winsorized. Same 364 country-years, 26 countries, and 19 openers as Figure 4. Data: External Wealth of Nations (Lane and Milesi-Ferretti, 2007).

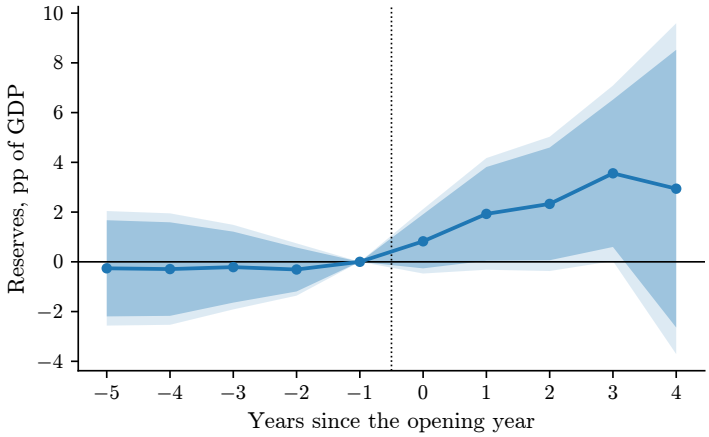



Figure C.5: Foreign exchange reserves around equity market openings

*Notes:* Event-time estimates $\hat{\theta}_k$ of equation (43) for foreign exchange reserves excluding gold, in percent of that year's GDP, in the specification of Figure 4: changes from the year before the opening, relative to the countries not yet open, with 90 and 95 percent leave-one-country-out jackknife bands. Positions are not winsorized. Same 364 country-years, 26 countries, and 19 openers as Figure 4. Data: External Wealth of Nations (Lane and Milesi-Ferretti, 2007).

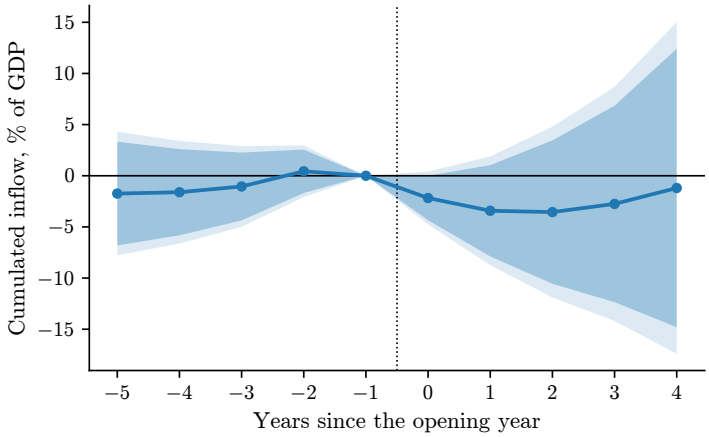



Figure C.6: The cumulated total net capital inflow around equity market openings

*Notes:* Event-time estimates $\hat{\theta}_k$ of equation (43) for the total net capital inflow, minus the current account, summed from 1981 with each year's inflow in percent of that year's GDP, in the specification of Figure 4: changes from the year before the opening, relative to the countries not yet open, with 90 and 95 percent leave-one-country-out jackknife bands. For $k \geq 0$ an estimate is the extra inflow over years 0 to $k$, and for $k \leq -2$ it is minus the extra inflow over years $k+1$ to $-1$. Same 364 country-years, 26 countries, and 19 openers as Figure 4. Data: External Wealth of Nations (Lane and Milesi-Ferretti, 2007).

## C.4 Equity valuations

Table C.2 reports average DiD estimates for the three valuation outcomes, dollar price index, dollar total-return index, and dividend yield, over different event windows under different reference months. Figure C.7 plots the event-time estimates using month −6 as the reference. Both use the

same sample of 2,531 country-month observations.

The pre-opening estimates suggest that part of the valuation adjustment occurs in anticipation of liberalization. At month $-7$, the estimated index is 20 percent below its level at month $-1$, indicating substantial appreciation before the official opening. The post-opening increases measured from month $-1$ are therefore additional to these earlier gains. Using month $-6$ as the reference captures more of the anticipatory appreciation alongside the subsequent post-opening increase.

Table C.2: Equity price changes: average DiD estimates over event windows, monthly

| | Pre-opening $-24$ to $-13$ (1) | Run-up $-8$ to 0 (2) | First year 0 to 11 (3) | Post-opening 0 to 23 (4) |
|---|---|---|---|---|
| *Panel A: base month* $-6$*, comparisons with* $G_c > \max(g,t)+5$ | | | | |
| Dollar price index, 100 ln | 16.6 | 17.9* | 40.0** | 51.4*** |
| | (17.0) | (10.5) | (16.5) | (17.9) |
| Dollar total-return index, 100 ln | 17.6 | 18.9* | 39.8** | 49.7*** |
| | (17.2) | (10.9) | (16.8) | (17.8) |
| Dividend yield, pp | $-0.68$ | $-0.53$ | $-1.18$ | $-1.13$ |
| | (1.09) | (0.90) | (1.07) | (1.03) |
| *Panel B: base month* $-1$*, comparisons with* $G_c > \max(g,t)$ | | | | |
| Dollar price index, 100 ln | $-1.4$ | 20.0** | 21.6 | 38.8* |
| | (15.5) | (9.9) | (15.8) | (20.1) |
| Dollar total-return index, 100 ln | $-1.1$ | 21.0** | 20.8 | 36.6* |
| | (16.1) | (10.3) | (15.6) | (19.7) |
| Dividend yield, pp | $-0.37$ | $-0.52$ | $-0.86$ | $-1.08$ |
| | (0.85) | (0.92) | (0.97) | (1.09) |

*Notes:* The table shows the average monthly difference-in-differences estimates $\hat{\theta}_k$ of equation (43) over different event windows. Columns (1), (3), and (4) are their averages over months $-24$ to $-13$, 0 to 11, and 0 to 23. Column (2) is a change rather than an average: $\hat{\theta}_0 - \hat{\theta}_{-8}$, the change in the estimate over the eight months to the official liberalization month. Panel A: base month $-6$; a country is a comparison for cohort $g$ in month $t$ only if its own opening comes more than five months after the later of $g$ and $t$, $G_c > \max(g,t)+5$. Panel B: base month $-1$; a comparison country must not have opened by the later of $g$ and $t$, $G_c > \max(g,t)$. Leave-one-country-out jackknife standard errors in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels, based on the jackknife standard errors. All rows use the same 2,531 country-months, 17 countries, and 15 openers.

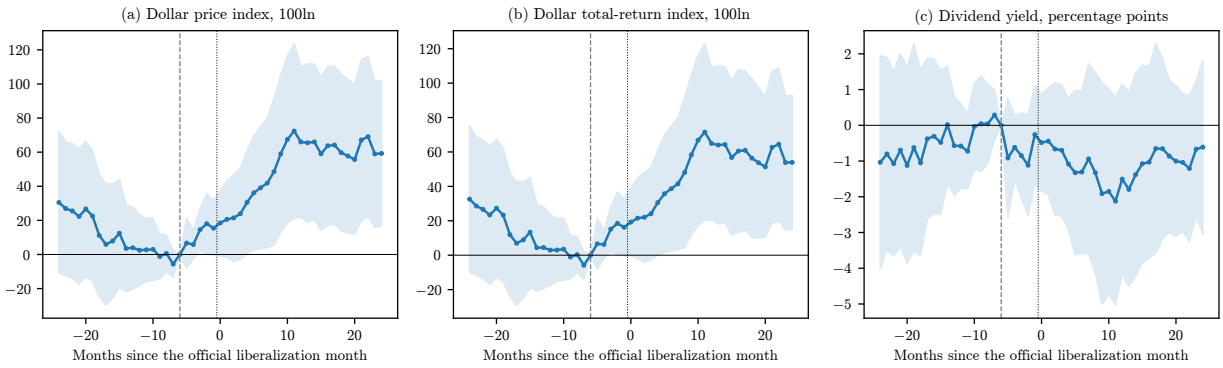



Figure C.7: Equity valuations around equity market openings, monthly, base month −6

*Notes:* Event-time effects $\hat{\theta}_k$ of equation (43) by month since the official liberalization month (dotted line), with the base month six months before it (dashed line) and comparison countries restricted to those whose own opening comes more than five months after the later of the cohort's opening month and the current month, $G_c > \max(g, t)+5$, with pointwise 95 percent confidence intervals from leave-one-country-out jackknife standard errors. Panel (a): 100 ln of the S&P/IFC Global price index in current US dollars. Panel (b): the same for the total-return index. Panel (c): the dividend yield of the index, in percentage points. All panels use the same 2,531 country-months, January 1980 to December 1994, in 17 countries, of which 15 open between September 1987 and June 1993.

**Annual paths from the monthly series.** Figure C.8 applies the annual design of Section 5.2, with the year of the official month as the cohort and the year before it as the base year, to the December values of the monthly series, on 217 country-years in 17 countries with 15 openers. The dollar index rises by 116 percent in the year of the opening (95 percent confidence interval 49 to 212 percent) and by 165 percent on average over years 0 to 4, and the dividend yield falls by 1.7 percentage points in the year of the opening (jackknife standard error 1.3).

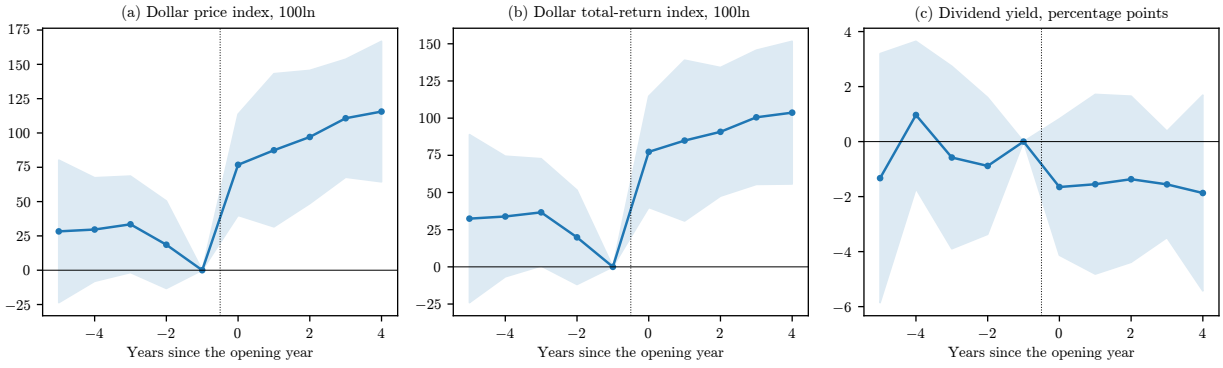



Figure C.8: Equity valuations around equity market openings, annual paths from the monthly series

*Notes:* Event-time effects $\hat{\theta}_k$ of equation (43) by year since the opening year, relative to the year before it, with pointwise 95 percent confidence intervals from leave-one-country-out jackknife standard errors. Outcomes are the December values of the monthly series of Figure C.7. Panel (a): 100 ln of the S&P/IFC Global price index in current US dollars. Panel (b): the same for the total-return index. Panel (c): the dividend yield of the index, in percentage points. All panels use the same 217 country-years, 1980–1994, in 17 countries, of which 15 open between 1987 and 1993.

## C.5 Fact 3: episodes and specifications

**Episodes.** The 17 episodes of Fact 3 are Argentina, Brazil, Chile, Colombia, Greece, India, Jordan, Korea, Malaysia, Mexico, Nigeria, the Philippines, South Africa, Thailand, Turkey, Venezuela, and Zimbabwe. Thirteen of them had a banking or currency crisis within ten years of the opening. Dropping any single episode leaves the slope between −0.199 and −0.159, and the most influential single episode is Nigeria.

**Robustness.** Table C.3 reports the correlation between re-pricing and subsequent TFP under different measurement windows for TFP growth and with different controls.

Table C.3: The re-pricing and later TFP growth: robustness

| | Episodes | Slope | s.e. | Perm. $p$ |
|---|---|---|---|---|
| *Panel A: the outcome* | | | | |
| Baseline: TFP growth, years +1 to +10 | 17 | −0.180 | 0.043 | 0.003 |
| TFP growth, years +1 to +5 | 17 | −0.151 | 0.049 | 0.015 |
| TFP growth, years +1 to +6 | 17 | −0.133 | 0.032 | 0.008 |
| TFP growth, years +1 to +11 | 17 | −0.169 | 0.045 | 0.003 |
| TFP growth from the opening year, years 0 to +10 | 17 | −0.147 | 0.043 | 0.004 |
| TFP growth relative to the world median | 17 | −0.166 | 0.041 | 0.005 |
| Crisis and post-crisis years left out | 14 | −0.188 | 0.037 | 0.001 |
| *Panel B: controls* | | | | |
| TFP growth in the previous decade | 17 | −0.179 | 0.053 | 0.004 |
| Date of the opening | 17 | −0.158 | 0.054 | 0.009 |
| High-inflation stabilization program | 17 | −0.186 | 0.052 | 0.002 |
| Inflation above 40 percent | 17 | −0.217 | 0.055 | 0.002 |
| Capital account opening, years −1 to +10 | 17 | −0.185 | 0.047 | 0.001 |
| Financing expansion, years −1 to +4 | 11 | −0.181 | 0.027 | 0.000 |

*Notes:* Slopes $b$ of equation (47), in percentage points of average annual TFP growth per ten units of the re-pricing $\Delta_c$, with small sample robust standard errors and permutation p-values from 20,000 reassignments of the re-pricing across episodes. Panel A changes the outcome: average annual growth from year +1 to year +5, to year +6, and to year +11; growth from the opening year to ten years later; growth minus the median growth of all Penn World Table countries over the same years; and the average of annual growth rates leaving out every year that Laeven and Valencia (2020) record as a banking or currency crisis and the year after it, for the episodes with at least five years left. Panel B adds one control at a time: average annual TFP growth over the ten years to the opening year; the date of the official month; an indicator of a program to stabilize high inflation within 24 months of the official month, as dated in Table A1 of Henry (2002), which covers Argentina, Brazil, and Mexico; an indicator of consumer price inflation above 40 percent in the year before, of, or after the official month, which covers Argentina, Brazil, Mexico, Nigeria, Turkey, Venezuela, and Zimbabwe; the change in the normalized capital account openness index of Chinn and Ito (2006) from the year before the opening to ten years after it, which measures the capital account liberalization that followed the equity opening; and 100 times the change in $\ln(1+\xi)$ from the year before the opening to year +4, the financing expansion measured in Section 6.3, which exists for 11 of the 17 episodes; its own coefficient is −0.026 (0.014) percentage points a year per log point.

Table C.4 reports the relation between the increase in the number of listed firms and subsequent TFP growth under different specifications and controls. The slope is negative in every row. The slope becomes more negative when the listings change is measured over a longer window: it is

−0.108 for years −1 to +3 and −0.189 for years −1 to +6. It is smaller in magnitude when TFP growth is measured to year +8 or to year +12. The slope changes little with prior TFP growth, high inflation, or initial income, and it becomes steeper with the stabilization indicator (−0.184). It falls to −0.068 when we control for the year of the opening, as later openers had smaller increases in listings and faster TFP growth afterward, and to −0.102 when we control for the number of listed companies in year −1. On the 15 episodes with both measures, the re-pricing slope barely changes when the listings change is added (−0.217 alone and −0.208 with it, per ten log points), while the listings slope falls to −0.121. Omitting one episode at a time leaves the baseline slope between −0.178 (without Portugal) and −0.086 (without Indonesia).

Table C.4: Listings growth and later TFP growth: specifications

| | Episodes | Slope | s.e. | Perm. $p$ |
|---|---|---|---|---|
| *Panel A: timing* | | | | |
| Baseline: listings, years −1 to +4; TFP growth, years +4 to +10 | 20 | −0.132 | 0.077 | 0.080 |
| Listings, years −1 to +3; TFP growth, years +3 to +10 | 21 | −0.108 | 0.075 | 0.108 |
| Listings, years −1 to +5; TFP growth, years +5 to +10 | 20 | −0.117 | 0.068 | 0.047 |
| Listings, years −1 to +6; TFP growth, years +6 to +10 | 20 | −0.189 | 0.073 | 0.006 |
| Listings from the opening year, years 0 to +4 | 22 | −0.129 | 0.098 | 0.286 |
| TFP growth, years +4 to +8 | 20 | −0.072 | 0.048 | 0.328 |
| TFP growth, years +4 to +12 | 20 | −0.094 | 0.058 | 0.137 |
| TFP growth relative to the world median | 20 | −0.116 | 0.078 | 0.096 |
| Crisis and post-crisis years left out | 13 | −0.139 | 0.169 | 0.275 |
| *Panel B: controls* | | | | |
| TFP growth in the previous decade | 20 | −0.130 | 0.076 | 0.077 |
| Year of the opening | 20 | −0.068 | 0.089 | 0.316 |
| High-inflation stabilization program | 20 | −0.184 | 0.104 | 0.030 |
| Inflation above 40 percent | 20 | −0.142 | 0.088 | 0.092 |
| Capital account opening, years −1 to +10 | 20 | −0.124 | 0.058 | 0.077 |
| Listings per million people in year −1 | 20 | −0.102 | 0.097 | 0.213 |
| GDP per capita in year −1 | 20 | −0.139 | 0.103 | 0.046 |
| Re-pricing $\Delta_c$ | 15 | −0.121 | 0.068 | 0.288 |

*Notes:* Slopes of equation (47) with $\Delta_c^\ell$ and $g_c^\ell$ of equation (48) in place of $\Delta_c$ and $g_c$, in percentage points of average annual TFP growth per ten log points of the change in listed companies per million people, with HC3 standard errors and permutation p-values from 20,000 reassignments of the listings change across episodes. Panel A changes the timing: the listings change from year −1 to year +3, +5, or +6, with TFP growth from that year to year +10; the listings change from the opening year to year +4; average annual TFP growth from year +4 to year +8 and to year +12; growth minus the median growth of all Penn World Table countries over the same years; and the average of annual growth rates leaving out every year that Laeven and Valencia (2020) record as a banking or currency crisis and the year after it, for the episodes with at least four of the six years left. Panel B adds one control at a time: average annual TFP growth over the ten years to the opening year; the opening year; an indicator of a program to stabilize high inflation within 24 months of the official month, as dated in Table A1 of Henry (2002), which covers Argentina, Brazil, and Mexico; an indicator of consumer price inflation above 40 percent in the year before, of, or after the opening, which covers Argentina, Brazil, Mexico, Nigeria, and Turkey; the change in the normalized capital account openness index of Chinn and Ito (2006) from year −1 to year +10; 100 ln of listed companies per million people in year −1; 100 ln of real GDP per capita at purchasing power parity in year −1, from the Penn World Table; and the re-pricing $\Delta_c$ of equation (45), on the 15 episodes with both measures.

## C.6 Aggregate TFP and real GDP

Figure C.9 shows the path of aggregate TFP and how it depends on the small comparison group. On all openers, aggregate TFP is 3.5 percent higher over years 0 to 4 than in the year before the opening (standard error 1.5, jackknife 1.6), and 2.0 percent higher than over years −5 to −2 (standard error 2.6, jackknife 2.8), because TFP was already 1.5 percent above its year −1 level in those earlier years. Dropping the three oil exporters in the panel (Nigeria, Trinidad and Tobago, and Venezuela) raises the change relative to years −5 to −2 to 3.0 percent (standard error 2.8); dropping Jordan, one of the seven comparison countries, lowers it to 0.0 (1.9); and dropping all four gives 0.6 (1.9), while the average relative to the year before the opening stays between 2.5 and 3.6 percent. On the countries of Fact 1 the average over years 0 to 4 is 4.0 percent (jackknife 1.6), and on the countries of Fact 2, where only two comparison countries remain in the late cells, it is 2.5 percent (jackknife 3.2), with a change relative to years −5 to −2 of −0.3 (jackknife 7.1). Table C.5 reports the annual average DiD estimates, unconditional and conditional on pre-opening income, using for the conditional estimates the outcome-regression version of the estimator, which adjusts the change among the comparison countries for log real GDP per capita in the year before the opening. With the log of real GDP per capita, the average for aggregate TFP is 3.6 percent (jackknife standard error 1.6); adding the TFP level relative to the United States gives 2.6 percent (1.6).

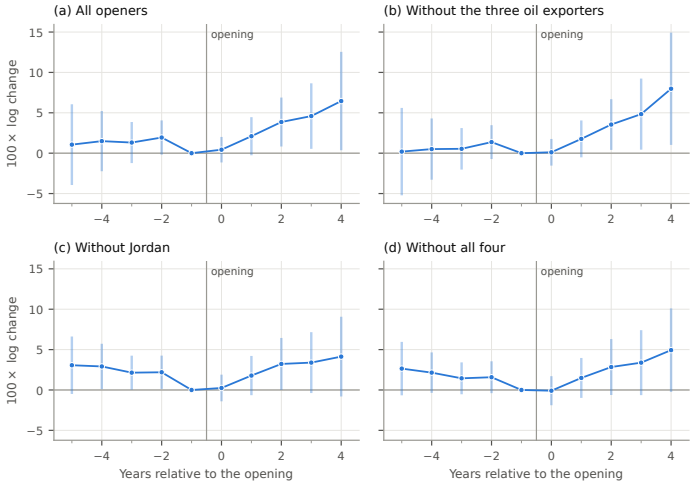



Figure C.9: Aggregate TFP around equity market openings

*Notes:* Event-time effects $\hat{\theta}_k$ of equation (43) with pointwise 95 percent confidence intervals from the influence-function standard errors, clustered by country; year −1 is the reference. Comparison countries are the countries not yet open; 1980–1994. The outcome is 100 ln TFP, with TFP at constant national prices. Each panel is a separate sample. Panel (a): all openers with TFP data, 405 country-years, 27 countries, 20 treated. Panel (b): without Nigeria, Trinidad and Tobago, and Venezuela, 360 country-years, 24 countries, 19 treated. Panel (c): without Jordan, 390 country-years, 26 countries, 20 treated. Panel (d): without all four, 345 country-years, 23 countries, 19 treated. Data: Penn World Table (Feenstra et al., 2015).

**Real GDP.** On the same 405 country-years, real GDP rises clearly after the opening, and by more than aggregate TFP (Figure C.10). Relative to the countries not yet open, the openers' real

GDP is 6.4 percent higher over years 0 to 4 than in the year before the opening (standard error 2.3, jackknife 2.4), and 6.4 percent higher than over years −5 to −2 (standard error 3.4, jackknife 3.6). Real GDP per capita is 10.0 and 11.9 percent higher (standard errors 2.4 and 3.8). Without Jordan, real GDP is 6.4 and 5.8 percent higher, while aggregate TFP is 2.6 percent higher than in the year before the opening and unchanged relative to years −5 to −2 (standard error 1.9). The difference between real GDP and aggregate TFP, the contribution of capital and labor in the Penn World Table's accounting, is 2.9 percent over years 0 to 4 (standard error 2.1).

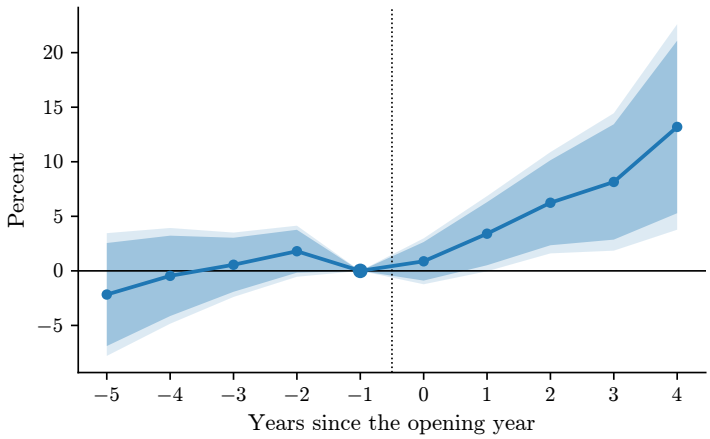



Figure C.10: Real GDP around equity market openings

*Notes:* Event-time effects $\hat{\theta}_k$ of equation (43) for 100 ln of real GDP at constant national prices, by year since the opening, relative to the year before it, with 90 and 95 percent leave-one-country-out jackknife bands. Comparison countries are the countries not yet open. Same 405 country-years, 27 countries, and 20 openers as panel (a) of Figure C.9, 1980–1994. Data: Penn World Table (Feenstra et al., 2015).

Table C.5: External positions and aggregate TFP: average DiD estimates, conditional on pre-opening log GDP per capita

| | Post-opening average | | Post minus pre | |
|---|---|---|---|---|
| | Unconditional (1) | Conditional (2) | Unconditional (3) | Conditional (4) |
| Foreign holdings of domestic equity | 1.49*** (0.46) | 1.45*** (0.49) | 1.77*** (0.51) | 1.73*** (0.54) |
| Foreign portfolio equity inflow | 0.63*** (0.20) | 0.60*** (0.22) | 0.68*** (0.19) | 0.65*** (0.21) |
| Total net capital inflow | 0.01 (1.33) | 0.31 (1.35) | −0.74 (1.47) | −0.45 (1.68) |
| Aggregate TFP, 100 ln TFP | 3.49** (1.59) | 3.58** (1.59) | 2.04 (2.78) | 2.49 (2.70) |
| adding the TFP level | | 2.63 (1.62) | | 1.77 (3.11) |

*Notes:* Averages of the event-time effects of equation (43): post-opening averages over years 0 to 4 relative to the year before the opening, and their difference from the pre-opening averages over years −5 to −2. Columns (1) and (3) are unconditional. Columns (2) and (4) are conditional on the log of real GDP per capita in the year before the opening (outcome-regression estimator); the last row also conditions on the log of the TFP level relative to the United States at current purchasing power parities. Leave-one-country-out jackknife standard errors in parentheses. *, **, and *** denote significance at the 10, 5, and 1 percent levels, based on these standard errors. External positions: 364 country-years, 19 treated countries. Aggregate TFP: 405 country-years, 20 treated countries. The doubly robust estimator would also fit a propensity score in every comparison, which cannot be estimated where only two or three countries have not yet opened.

# D Quantitative Analysis: Data, Solution, and Further Results

This appendix provides additional information regarding the calibration discussed in Section 6. It defines every moment and the assumptions behind the measures, describes how we compute the recursive equilibrium and solve the calibration, reports the full fit, including the changes at the openings, reports the sensitivity of the calibration, and decomposes the effects of liberalization.

## D.1 Construction of the moments

The calibration uses the 14 countries that opened their equity markets between 1987 and 1993: Argentina, Brazil, Chile, Colombia, India, Korea, Malaysia, Mexico, Pakistan, the Philippines, Thailand, Turkey, Venezuela, and Zimbabwe. Table D.1 documents the sources and construction of the empirical moments in Tables 2 and D.2, together with their model counterparts. Unless otherwise stated, empirical moments are calculated using cross-country medians of country averages over years $-5$ to $-1$. For financing capacity and foreign shares of listed outside equity, we use single-year observations in year $-1$ as the pre-opening value and in year $+4$ as the post-opening value for the 13 countries [6]. The private-sector bond position is calculated using the ten countries that report their private external debts, and index-return volatility is calculated using data from seven countries. For moments that require more than one data series, we first construct the country-by-country moments before taking the cross-country median. The only exception is the unlevered price-to-book ratio, which is computed by first taking the cross-country median of the series and then computing the unlevered price-to-book ratio, to minimize the influence of outliers.

One important feature of the model is that we distinguish between equities held by the insiders and outsiders. (Dahlquist et al., 2003) document that a substantial portion of the listed companies' shares are held by insiders, which makes the market capitalization a poor proxy for the outside equity. We therefore measure outside equity by multiplying market capitalization by one minus the closely held share reported for December 1997 (Dahlquist et al., 2003, Table 1). Since the ownership data is only available for 1997, we assume that the insider shares remain constant over time.

**The financing capacity $\xi$.** To calibrate $\xi$, we sum up three types of outside finance for private capital: private credit, the outsider share of the book value of listed equity, and direct investment liabilities. We first calculate the ratio between total outside financing and total capital stock $x_c$ and then back out $\xi = x_c/(1-x_c)$. $x_c$ is a proxy for aggregate outside financing that the corporate sector raises, as private credit includes credit to households, while unlisted outside equity and informal credit are not included. The transformation $\xi = x_c/(1-x_c)$ also magnifies measurement error when the outside share $x$ is close to one. The calibrated share $x_c$ exceeds one for Zimbabwe in year $-1$ and for Malaysia in year $+4$, making calibrating $\xi$ invalid. Section D.5 measures the financing capacity $\xi$ using four other measures and reports the sensitivity of the calibration. The

[6] Zimbabwe is excluded as its capital stock data seems extremely small relative to the country's total credit.

effect of financing expansion on TFP remains positive under each measure.

**The foreign share $\omega$.** We calibrate $\omega$ as foreign investors' portfolio holdings of domestic equity divided by the market capitalization of listed equity held by outside investors. Although the model represents outside finance through an equity index, we interpret $\omega$ more broadly as the foreign share of the financing pool measured by $\xi$. This interpretation assumes a similar extent of foreign participation across financing instruments. Applying the measured ownership share to all outside finance gives foreign holdings close to those observed in the data. Before liberalization, outside finance in the model amounts to 53 percent of output, and $\omega = 0.094$ implies foreign claims worth 5.0 percent of output. In the data, foreign portfolio equity and direct investment liabilities together amount to 5.7 percent of output.

**The premium on operating profits.** The model's one-period equity structure affects both the calibration and its ability to match equity-return volatility. At the end of every period, each firm distributes its operating profits and undepreciated capital, and equity carries no claim on future profits. The capital repayment is treated as a risk-free asset and accounts for 87.9 percent of index value in the calibrated benchmark. In other words, operating profits are the only risky part of the firm value, and they account for a small part of the total valuation of the firm. This large safe component reduces both the equity premium and return volatility. Equation (22) implies an index premium of only 0.82 percentage points. The absence of a risky continuation value is therefore an important structural reason for the model's low equity-return volatility. Listed equity, however, represents a continuing claim whose resale value depends on future profits and discount rates. In the calibration, we assume that the empirical cost of equity is a proxy for $\mathbb{E}[R^D]$. This assumption abstracts from the difference in claim duration. Section D.4 examines the sensitivity of the calibration results to adjusting the empirical premium for listed firms' leverage.

Table D.1: Construction of the moments in Tables 2 and D.2

| Moment | Source | Construction | $N$ | Model counterpart |
|---|---|---|---|---|
| *Conditions of the joint calibration* (49) | | | | |
| Premium on operating profits, $\mathbb{E}[R^D] - R^{f*}$ (levered, implied) | IFCG, PWT, FRED, WDI | Dividend yield plus the past five-year average of real GDP growth, less the real one-year US Treasury yield | 14 | $R^{f*}(\lambda - 1)$ |
| Net foreign bond position / output | EWN | Debt assets plus reserves excluding gold, less debt liabilities, over GDP | 14 | NFB/$\mathbb{E}[Y]$, (49b) |
| Stationarity of capitalist wealth | None | A restriction of the calibration | | $\mathbb{E}[W' \mid W_{\text{pre}}] = W_{\text{pre}}$, (41) |
| *Statistics that set parameters directly* | | | | |
| Volatility of annual log dividends | IFCG, WDI | Real US-dollar dividends: annual log changes before year 0, demeaned by country and pooled; robust standard deviation | 13 | $\alpha\sigma$, the standard deviation of $\ln D$ at a given wealth |

*Continued on the next page*

*Table D.1, continued*

| Moment | Source | Construction | $N$ | Model counterpart |
|---|---|---|---|---|
| Outside finance over private capital, $\xi/(1+\xi)$, year $-1$ | GFDD, IFC Factbook, EWN, ICSD, Dahlquist et al. (2003) | Private credit, plus the outsider share times listed book equity, plus FDI liabilities, over the private capital stock; year $-1$ | 13 | $\xi/(1+\xi)$, with $\xi$ the median of $x_c/(1-x_c)$ |
| Foreign share of listed outside equity, year $-1$ $(=\omega)$ | EWN, IFC Factbook, Dahlquist et al. (2003) | Portfolio equity liabilities over the outsider share times market capitalization; year $-1$ | 13 | $\omega$ |
| Foreign share of listed outside equity, year $+4$ $(=\omega_{\text{post}})$ | As above | Year $-1$ median plus the median change from year $-1$ to year $+4$ | 13 | $\omega_{\text{post}}$, the value the opening sets |
| *Untargeted moments of the pre-opening economy* | | | | |
| Capital / output | PWT | Capital stock over GDP at current national prices | 14 | $K/\mathbb{E}[Y]$, (80) |
| Replacement investment / output, $I/Y - g\,K/Y$ | PWT | Investment over GDP, less real GDP growth times capital over output | 14 | $\delta K/\mathbb{E}[Y]$ |
| Implied depreciation rate (data) against the set $\delta$ (model) | PWT | Replacement investment over capital over output, country by country | 14 | $\delta$, a parameter set directly |
| Price-to-book, unlevered, $(E+D)/(B+D)$ | IFC Factbook, Compustat Global | Index price-to-book ratio, unlevered with listed firms' equity share of market value; two cross-country medians | 14 | $\bar{q}$, (80) |
| Foreign equity / output, year $-1$, with direct investment | EWN | Portfolio equity plus FDI liabilities, over GDP; year $-1$ | 14 | $\omega\xi W^P/\mathbb{E}[Y]$ |
| Net foreign assets / output, including FDI | EWN | Net international investment position excluding gold, over GDP | 14 | $(W-K)/\mathbb{E}[Y]$, (38) |
| *Untargeted changes at the opening* | | | | |
| Change in foreign equity / output, year $-1$ to $+4$, with direct investment | EWN | Portfolio equity plus FDI liabilities over GDP, year $+4$ less year $-1$, country by country | 14 | Change in $\omega\xi W^P/\mathbb{E}[Y]$ |
| Change in net foreign bonds / output, year $-1$ to $+4$, all sectors | EWN | Year $+4$ less year $-1$, country by country | 14 | Change in NFB$/\mathbb{E}[Y]$ |
| Change in the dividend yield, event study (data) or $\mathbb{E}[D]/p$, the model's dividend yield (model) | IFCG | Callaway–Sant'Anna estimate, months 0 to 23 compared to month $-6$ (Section 5.3) | 15 | Change in $\mathbb{E}[D]/p$, the model's dividend yield, at a fixed world rate |
| TFP, years 0 to $+4$ compared to year $-1$ or compared to years $-5$ to $-2$ | PWT | Callaway–Sant'Anna estimates for $100\ln$ TFP (Appendix C.6) | 20 | Change in $100\ln$ TFP, (31) |
| *Untargeted dynamic checks: the simulated pre-opening economy* | | | | |
| Premium on operating profits, ergodic mean | IFCG, PWT, FRED, WDI | The premium target | 14 | Mean of $R^{f*}(\lambda-1)$ over simulated path-years |
| Net foreign bond position / output, ergodic mean | EWN | The bond target | 14 | Mean of NFB$/\mathbb{E}[Y]$ over simulated path-years |

*Continued on the next page*

*Table D.1, continued*

| Moment | Source | Construction | $N$ | Model counterpart |
|---|---|---|---|---|
| Volatility of annual changes in log dividends, simulated | IFCG, WDI | The dividend volatility above | 13 | Standard deviation of the simulated annual changes in $\ln(D\tilde{K})$ |
| First autocorrelation of annual changes in log dividends | IFCG, WDI | The pooled changes of the dividend volatility: first autocorrelation over consecutive years within a country | 13 | First autocorrelation of the simulated annual changes in $\ln(D\tilde{K})$ |
| *Scope of the one-period firm, reported without a target* | | | | |
| Excess return of the index | IFC Factbook, FRED | US-dollar total return of the index less the one-year US Treasury yield, both nominal | 14 | $\mathbb{E}[R^e] - R^{f*}$, (22) |
| Volatility of the index return | GFDD | Average 360-day volatility of the national stock market index | 7 | $\mathrm{sd}(R^e)$, (21) |
| Volatility of annual changes in log GDP | PWT | Annual log changes of real GDP from 1970 to year $-1$, demeaned by country and pooled; robust standard deviation | 14 | Standard deviation of the simulated annual changes in $\ln Y$ |

*Notes:* $N$ reports the number of countries whose data is used to compute each empirical moment. For the event-study estimates, $N$ only counts the liberalizing countries. We define year 0 as the calendar year when a country implemented the official liberalization reported by Bekaert et al. (2005). Unless specified otherwise, we first average each country's observations over years $-5$ to $-1$ and then report the cross-country median. We calculate dividend and output volatility from annual log changes pooled across countries after subtracting each country's mean change. *Sources.* IFCG is the country's S&P/IFC Global index (Datastream; end-of-year values and monthly values are used in the event study); the dividend yield is the past twelve-month US-dollar dividends over the price. IFC Factbook refers to the IFC Emerging Stock Markets Factbook. EWN refers to the External Wealth of Nations database (Lane and Milesi-Ferretti, 2007). PWT refers to the Penn World Table 11 (Feenstra et al., 2015). GFDD is the World Bank's Global Financial Development Database, and ICSD refers to the IMF Investment and Capital Stock Dataset. The one-year US Treasury constant-maturity yield (GS1) is obtained from the FRED database, and US consumer price inflation is obtained from the WDI database.

## D.2 Computing the recursive equilibrium and solving the calibration

**Analytical simplification.** Let $\pi^*$ and $b^*$ denote the empirical targets for the premium on operating profits and the net foreign bond position relative to output, and define $\lambda^* = 1 + \pi^*/R^{f*}$. We take $(\alpha, \delta, \eta, R^{f*}, \xi, \omega)$ as given from the calibration described in Section 6.1. The dividend-innovation volatility target $s_D$ implies $\sigma = s_D/\alpha$, and we compute $\mu = -\alpha\sigma^2/2$ in order to normalize $\mathbb{E}[e^{\alpha\varepsilon}] = 1$.

For a candidate occupational wedge $\Delta$, the cutoff identity (28), aggregation, and the bond-position give the candidate capital-output ratio and wealth relative to expected output:

$$k(\Delta) \equiv \frac{K}{\mathbb{E}[Y]} = \frac{\alpha(\eta-1)}{\eta\lambda^*\left[R^{f*} - (1-\delta) + R^{f*}\Delta\right]}, \tag{80}$$

$$w(\Delta) \equiv \frac{W}{\mathbb{E}[Y]} = b^* + \frac{1 + (1-\omega)\xi}{1+\xi}\, k(\Delta). \tag{81}$$

The first expression is derived by combining the cutoff identity with $\mathbb{E}[D]\mathrm{TFP}^{1/\alpha}K = \alpha\mathbb{E}[Y]$ and $\mathrm{TFP}^{1/\alpha} = \eta\bar{z}/(\eta-1)$. The second follows from $\mathrm{NFB} = W - K + \omega\xi W^P$ and $K = (1+\xi)W^P$.

These ratios determine the candidate cutoff, productivity, expected output, and wealth:

$$\bar{z}(\Delta) = \left[\frac{(1+\xi)w(\Delta)}{k(\Delta)}\right]^{1/\eta}, \qquad \mathrm{TFP}(\Delta) = \left[\frac{\eta\bar{z}(\Delta)}{\eta-1}\right]^{\alpha}, \tag{82}$$
$$y(\Delta) = \left[\mathrm{TFP}(\Delta)k(\Delta)^{\alpha}\right]^{1/(1-\alpha)}, \qquad W_{\mathrm{pre}}(\Delta) = w(\Delta)y(\Delta),$$

where $y(\Delta)$ denotes candidate expected output. We then replace benchmark wealth by the wedge in the numerical algorithm.

**Computing the recursive equilibrium.** For a given set of parameters, we solve for the continuation function $\bar{\Lambda}(W) = \mathbb{E}_z[v(z;W)]$, where $v(z;W)$ is specified in the individual value function (Appendix A.1). We describe the algorithm for $\gamma > 1$, with $\beta(R^{f*})^{1-\gamma} < 1$.

Given a candidate continuation function $\bar{\Lambda}(W)$, the common saving rate across all capitalists $S_{\bar{\Lambda}}(H)$ and the corresponding valuation weight satisfy

$$\frac{S_{\bar{\Lambda}}(H)}{1-S_{\bar{\Lambda}}(H)} = \left[\beta\bar{\Lambda}\big(S_{\bar{\Lambda}}(H)H\big)\right]^{1/\gamma}, \qquad L_{\bar{\Lambda}}(H) = \left[1-S_{\bar{\Lambda}}(H)\right]^{-\gamma}. \tag{83}$$

Here $H$ denotes aggregate cash on hand after the shock is realized and before consumption and saving. The equilibrium allocations (including occupational, production, and investment choices) at current wealth $W$ determine $H(W,\varepsilon)$ through (41). Cash on hand is uncertain before the shock is realized but it is known when capitalists choose consumption and saving. Given a candidate continuation function $\bar{\Lambda}(\cdot)$, the deterministic saving policy $S_{\bar{\Lambda}}(\cdot)$ defined by (83) specifies a saving fraction for each realization of $H$. Next-period aggregate wealth is therefore $W' = S_{\bar{\Lambda}}(H)H$, and the associated valuation weight is $L_{\bar{\Lambda}}(H) = [1-S_{\bar{\Lambda}}(H)]^{-\gamma}$.

Let $\mathcal{A}$ denote the collection of period allocations across wealth states, and let $R^{o*}_{\mathcal{A}}(z;W,\varepsilon)$ denote the gross return on a capitalist's wealth under the occupation and financial choices in $\mathcal{A}$. For fixed allocations, define the continuation operator

$$\mathcal{T}_{\mathcal{A}}[\bar{\Lambda}](W) = \mathbb{E}_{z,\varepsilon}\Big[L_{\bar{\Lambda}}\big(H_{\mathcal{A}}(W,\varepsilon)\big)\big(R^{o*}_{\mathcal{A}}(z;W,\varepsilon)\big)^{1-\gamma}\Big]. \tag{84}$$

The expectation takes the physical distributions of productivity and the aggregate shock. A recursive equilibrium requires both optimal period allocations given the continuation function and a fixed-point continuation function, i.e., $\bar{\Lambda} = \mathcal{T}_{\mathcal{A}}[\bar{\Lambda}]$.

We compute this fixed point on a wealth grid as follows.

1. *Initialize.* At the given risk aversion, solve the period equilibria with shock-independent valuation weights. These allocations do not depend on the continuation function. Holding them fixed, iterate (84), recalculating saving throughout, to obtain an initial continuation function.

2. *Update period allocations.* Given the current continuation function, solve the period equilibrium at each wealth node for the productivity cutoff and the price of the claim to operating

profits. The equations are Entry (25) and Pricing (11), where equity-market clearing (36) is imposed. At every candidate allocation, recompute cash on hand and the weights $L_{\bar{\Lambda}}(H)$. If investors' exposure reaches its upper bound, replace the interior portfolio condition by $\vartheta = 1$ and impose the associated optimality inequalities.

3. *Update the continuation function.* Hold the resulting one-period allocations $\mathcal{A}$ fixed and iterate $\mathcal{T}_{\mathcal{A}}[\bar{\Lambda}] \to \bar{\Lambda}$. Returns and cash on hand remain fixed during this inner iteration; saving rates and valuation weights are recalculated after every loop.

4. *Repeat and verify.* Repeat the allocation and continuation updates until the continuation function satisfies the stopping criterion below. Then recompute period equilibria using the final continuation function and check both the period-equilibrium conditions and the continuation-equation residual and convergence.

For $\gamma > 1$, we use a homotopy procedure to select a solution. In the first allocation update, we hold the candidate continuation coefficient function $\bar{\Lambda}(\cdot)$ fixed and solve a sequence of auxiliary period equilibria at each wealth node using the weights

$$\widetilde{\Lambda}_\tau(W, \varepsilon) = \left[1 - S_{\bar{\Lambda}}\big(H_\tau(W, \varepsilon)\big)\right]^{-\gamma\tau}, \qquad \tau \in [0, 1],$$

where $H_\tau(W, \varepsilon)$ is the cash on hand implied by the allocation being solved for at $\tau$. At $\tau = 0$, the weights equal one across all shock realizations. We gradually increase $\tau$ to one, using each solution as the initial guess for the next solve. At $\tau = 1$, the weights coincide with those required by the original model, and we retain the resulting allocation. In subsequent allocation updates, we solve directly with $\tau = 1$, using the preceding allocations as initial guesses. After the continuation coefficient function converges, we hold it fixed and repeat the homotopy procedure at every wealth node to verify that its endpoint reproduces the final allocations.

**Solving the joint calibration.** Let $\lambda_{\text{eq}}(W)$, $H_{\text{eq}}(W, \varepsilon)$, and $q_{\text{eq}}(\bar{z}_{\text{eq}}(W))$ denote the risk premium, aggregate cash on hand, and marginal producer's capital value computed in the recursive equilibrium. The conditional-mean aggregate wealth transition dynamic is

$$m_{\text{eq}}(W) \equiv \mathbb{E}\left[S\big(H_{\text{eq}}(W, \varepsilon)\big) H_{\text{eq}}(W, \varepsilon)\right], \tag{85}$$

where the expectation is over the distribution of the shock. The restriction $m_{\text{eq}}(W_{\text{pre}}) = W_{\text{pre}}$ characterizes a conditional-mean steady state for calibration. It is imposed at the average aggregate wealth while the realized aggregate wealth remains stochastic.

For each candidate $(\gamma, \beta, \Delta)$, we construct $W_{\text{pre}}(\Delta)$ from (82), compute the recursive equilib-

rium, and solve its period equilibrium directly at $W_{\text{pre}}(\Delta)$. In other words, we jointly solve

$$\lambda_{\text{eq}}\big(W_{\text{pre}}(\Delta)\big) = \lambda^*, \tag{86}$$

$$m_{\text{eq}}\big(W_{\text{pre}}(\Delta)\big) = W_{\text{pre}}(\Delta), \tag{87}$$

$$q_{\text{eq}}\big(\bar{z}_{\text{eq}}\big(W_{\text{pre}}(\Delta)\big)\big) - 1 = \Delta. \tag{88}$$

The first condition matches the premium, the second imposes conditional-mean stationarity, and the third makes the equilibrium wedge consistent with the wedge used in the analytical reconstruction. Once premium matching and wedge consistency hold, the equilibrium allocation agrees with the reconstruction and therefore also matches the bond-position target. The wedge condition is a consistency requirement of this reparameterization, rather than an additional empirical target. All three conditions are solved jointly because risk aversion, saving, and the continuation weights affect the equilibrium premium and allocation together.

**Numerical implementation.** We discretize $\ln \bar{\Lambda}$ on 41 nodes equally spaced in $\ln W$ on $[0.15, 5] \times W_{\text{ref}}$, where $W_{\text{ref}} = W_{\text{pre}}(\Delta_0)$ is computed from the initial wedge and remains fixed throughout the calibration. We interpolate by a cubic spline in $\ln W$ and extrapolate linearly in $\ln W$, restricting the lower extrapolation slope to $[0, 0.9\gamma]$. We impose the upper bound of Lemma A.1 on the continuation function value throughout the domain. Expectations are calculated using 40-node Gauss–Hermite quadrature for the aggregate shock and 40-node Gauss–Jacobi quadrature in $\bar{z}/z$ for producer productivity. We solve the scalar saving equation with safeguarded root finding and evaluate the state-dependent weights without a log-affine approximation.

Let $\{W_i\}$ denote the wealth grid. For a candidate continuation function $\bar{\Lambda}(\cdot)$ and fixed period allocations $\mathcal{A}$, define the continuation-equation residual as

$$r_{\text{cont}}(\bar{\Lambda}; \mathcal{A}) \equiv \max_i \left| \ln \frac{\mathcal{T}_{\mathcal{A}}[\bar{\Lambda}](W_i)}{\bar{\Lambda}(W_i)} \right|.$$

This residual compares the candidate continuation coefficient at each wealth node with the coefficient implied by one application of the continuation operator. Holding the period allocations fixed, the inner continuation iteration stops when its current function satisfies $r_{\text{cont}}(\bar{\Lambda}; \mathcal{A}) < 10^{-13}$.

Let $\bar{\Lambda}^{(k)}(\cdot)$ denote the continuation function returned by the inner iteration after allocation update $k$. We stop repeating the allocation and continuation function updates when

$$\max_i \left| \ln \bar{\Lambda}^{(k)}(W_i) - \ln \bar{\Lambda}^{(k-1)}(W_i) \right| < 10^{-10}.$$

This criterion measures the change between successive continuation functions. As a separate final check, we recompute period allocations $\mathcal{A}^{\text{fin}}$ using the final function $\bar{\Lambda}^{\text{fin}}(\cdot)$ and require $r_{\text{cont}}(\bar{\Lambda}^{\text{fin}}; \mathcal{A}^{\text{fin}}) < 10^{-9}$. The final check therefore evaluates the continuation equation using allocations recomputed from the final function.

For the calibration, we use a trust-region root finder in the transformed variables

$$u = \left( \ln(\gamma - 1),\ \ln \frac{\beta}{1-\beta},\ 10^3 \Delta \right)'.$$

For each candidate $u$, we let $W_p = W_{\text{pre}}(\Delta)$ and evaluate the following residual vector in the corresponding recursive equilibrium:

$$r_{\text{cal}}(u) = \begin{pmatrix} \ln \dfrac{\lambda_{\text{eq}}(W_p)}{\lambda^*} \\ \ln \dfrac{m_{\text{eq}}(W_p)}{W_p} \\ 10^3 \left[ q_{\text{eq}}\big(\bar{z}_{\text{eq}}(W_p)\big) - 1 - \Delta \right] \end{pmatrix}.$$

The first two entries measure proportional discrepancies in the premium and conditional-mean wealth conditions; the third measures wedge inconsistency in the scaled units used by the solver. We accept the calibration when $\|r_{\text{cal}}(u)\|_\infty < 10^{-11}$.

Given the calibrated parameters, we recompute the recursive equilibrium using the initialization and equilibrium-selection procedure described above. In this fresh solution, we directly verify the benchmark conditions by requiring

$$\max \left\{ \left| \frac{\lambda_{\text{eq}}(W_p)}{\lambda^*} - 1 \right|, \left| \left( \frac{\text{NFB}}{\mathbb{E}[Y]} \right)_{\text{eq}} (W_p) - b^* \right|, \left| \frac{m_{\text{eq}}(W_p)}{W_p} - 1 \right| \right\} < 10^{-8}.$$

We also evaluate $d \ln \bar{\Lambda}(W')/d \ln W'$ on a fine grid covering the interpolation and extrapolation regions and verify that it remains below $\gamma$. This is a sufficient condition that ensures that the saving equation is strictly increasing in the saving fraction. Together with the bounds used to bracket the root, it yields a unique saving rate. Finally, we require $\lambda_{\text{eq}}(W_i) \geq 1$ at every wealth node, ensuring that foreign investors are willing to hold the imposed ownership share at the computed prices.

### D.3 Model fit

Table D.2 reports the calibration conditions and compares targeted and untargeted empirical moments with their model counterparts, including moments computed from simulated pre-opening paths and the changes at the opening. Table D.3 reports additional variables' equilibrium values of the benchmark economy.

The model understates equity-return volatility, generating a value of 4.4 percent compared with 33.4 percent in the data. An important source of this shortfall is the one-period firm assumption. Equity pays current operating profits and undepreciated capital, where we assume that the value of undepreciated capital is risk-free. Consequently, only a small share of equity value is exposed to aggregate risk, limiting return volatility through the decomposition in (21). Allowing firms and their equity claims to persist across periods would introduce a continuation value that responds to changes in expected future profits and discount rates. We leave this extension and its implications

for equity-return volatility to future work.

**The changes at the opening.** The rows of Table D.2 on the changes at the opening compare the model's impact responses with changes in foreign holdings, the dividend yield, and aggregate TFP around the openings. These are comparisons of magnitudes under the measurement assumptions of Section 6.1. The empirical windows differ across outcomes, while the model receives the full measured changes in financing and ownership at date 0. The comparisons do not identify how much of the observed response was caused by either model channel.

**Foreign holdings.** Foreign holdings of domestic equity rise by 8.3 percentage points of output with the opening alone and 11.6 with both changes, compared with 5.4 in the data when direct investment is included. The comparison including direct investment is consistent with its inclusion in the financing measure. An important source of the discrepancy is the extrapolation from listed ownership to all outside finance. Median listed capitalization rises from 6 percent of output before the opening to 23 percent in year +4, while the model applies the measured foreign-share change to its entire outside-finance pool, worth 53 percent of output before the opening.

**The dividend yield.** The model's premium on operating profits falls by 0.42 percentage points with foreign risk sharing and 1.03 points with both changes, at a fixed world safe rate. The event study of Section 5.3 estimates a dividend-yield decline of 1.13 points over months 0–23 relative to month −6, with a standard error of 0.92. Under the maintained dividend-yield-plus-growth measure of the cost of equity, a yield change can be read as a change in the required return if expected growth is unchanged. Common movements in world rates are absorbed by the event study's comparison of countries in the same months.

The model responses are 38 and 91 percent of the point estimate, respectively, and lie 0.77 and 0.11 standard errors from it. These ratios compare magnitudes; they are not estimates of the channels' causal shares. The empirical estimate is too imprecise to distinguish the two model responses and does not differ from zero at the 5 percent level. The opening-only response also falls within the 5–75 basis-point range for the cost of capital reported by Bekaert and Harvey (2000). The model does not reproduce the equity-price revaluation documented by Henry (2000a): much of its one-period firm's value is undepreciated capital with a safe resale value, and entry keeps the marginal producer's value close to replacement cost.

**Productivity.** The empirical TFP measure is a Solow residual. Its model counterpart is TFP in (31), so the relevant model response is the change in $\ln \text{TFP}$: −0.15 percent with foreign risk sharing and +0.28 percent with both changes. These effects are small relative to the event-study point estimate of 3.49 percent over years 0–4 relative to year −1. The note to Table D.2 gives the analytic standard error, 1.48; the jackknife standard error used in the sensitivity analysis is 1.59.

The sensitivity analysis in Appendix C.6 gives a 95 percent confidence set of 0.44 to 6.53 percent under exact post-opening parallel trends, excluding both model responses. Allowing a post-opening

violation 7 percent as large as the largest pre-opening violation makes the set include zero; allowing one quarter as much yields a set from $-1.37$ to 7.84 percent, containing both model responses. The average post-opening change also reflects other reforms and shocks, so it does not provide a precise test of either mechanism.

**Scope of the comparison.** The small responses of equity values and of the premium reflect the safe resale value of undepreciated capital, discussed in Appendix D.3, and diminishing returns damp the cutoff's response by $1/(1+e_D) = 0.43$ before TFP moves by $\alpha$ times the cutoff.

Table D.2: Targeted and untargeted moments

| Moment | Data | Model |
|---|---|---|
| *Targeted, and inferred from financing and ownership measures* | | |
| Volatility of annual changes in log dividends | 0.322 | 0.322 |
| Premium on operating profits, $\mathbb{E}[R^D] - R^{f*}$ (levered, implied) | 0.0675 | 0.0675 |
| Outside finance over private capital, $\xi/(1+\xi)$, year $-1$ | 0.225 | 0.225 |
| Foreign share of listed outside equity, year $-1$ $(= \omega)$ | 0.094 | 0.094 |
| Net foreign bond position / output | $-0.267$ | $-0.267$ |
| Stationarity of capitalist wealth, $\mathbb{E}[W' \mid W_{\text{pre}}]/W_{\text{pre}}$ | | 1.000 |
| *Set by the opening* | | |
| Foreign share of listed outside equity, year $+4$ $(= \omega_{\text{post}})$ | 0.249 | 0.249 |
| *Not targeted* | | |
| Capital / output | 2.471 | 2.351 |
| Replacement investment / output, $I/Y - g\,K/Y$ | 0.153 | 0.141 |
| Implied depreciation rate (data) against the set $\delta$ (model) | 0.058 | 0.060 |
| Price-to-book, unlevered, $(E+D)/(B+D)$ | 1.193 | 1.047 |
| Foreign equity / output, year $-1$, with direct investment | 0.0566 | 0.0499 |
| Change in foreign equity / output, year $-1$ to $+4$, with direct investment | 0.054 | 0.083 |
| with both $\omega$ and $\xi$ adjusted | 0.054 | 0.116 |
| Net foreign assets / output, including FDI (data: net IIP) | $-0.298$ | $-0.316$ |
| Change in net foreign bonds / output, year $-1$ to $+4$, all sectors | 0.127 | 0.053 |
| Change in the dividend yield, event study (data) or $\mathbb{E}[D]/p$, the model's dividend yield (model) | $-0.0113$ | $-0.0042$ |
| with both $\omega$ and $\xi$ adjusted | $-0.0113$ | $-0.0103$ |
| Change in TFP, years 0 to $+4$ compared to year $-1$ | 3.49 | $-0.15$ |
| with both $\omega$ and $\xi$ adjusted | 3.49 | 0.28 |
| Change in TFP, years 0 to $+4$ compared to years $-5$ to $-2$ | 2.04 | $-0.15$ |
| *Untargeted dynamic checks: the simulated pre-opening economy* | | |
| Premium on operating profits, ergodic mean | 0.0675 | 0.0683 |
| Net foreign bond position / output, ergodic mean | $-0.267$ | $-0.226$ |
| Volatility of annual changes in log dividends, simulated | 0.322 | 0.446 |
| First autocorrelation of annual changes in log dividends | 0.183 | $-0.498$ |
| *Scope of the one-period firm, reported without a target* | | |
| Excess return of the index | 0.119 | 0.008 |
| Volatility of the index return | 0.334 | 0.044 |
| Volatility of annual changes in log GDP (data) or log output (model, simulated) | 0.031 | 0.446 |

*Notes:* Unless otherwise specified, the Data column reports medians across the opening countries of country averages over years $-5$ to $-1$. Table D.1 documents construction, sample coverage, and exceptions. Variables reported in levels are evaluated in the pre-opening equilibrium at wealth $W_{\text{pre}}$. Changes are measured as the difference between the post-opening and pre-opening period equilibria, both evaluated at that same wealth level. For simulated variables, we use 2,000,000 simulated path-years of the pre-opening economy. In the dynamic-check panel, the Data column reports the empirical calibration targets; the Model column reports statistics computed from simulated paths. Risk premium and dividend-yield changes are in decimal units, so 0.01 denotes one percentage point. For external positions, which are ratios to output, the convention is that 0.01 denotes 1 percent of output. Changes in TFP are in units of $100[\ln X_{\text{post}} - \ln X_{\text{pre}}]$, so an entry of 1 denotes approximately a 1 percent increase. The rows marked 'with both $\omega$ and $\xi$ adjusted' apply the two changes together; their data cells repeat the estimate of the row above. The TFP rows compare the event-study estimates of Figure C.9 for the Penn World Table's TFP, whose analytic standard errors are 1.48 and 2.59 percent, with the change in $100 \ln$ TFP on impact; the jackknife standard error used in the sensitivity analysis of Appendix C.6 is 1.59.

Table D.3: Model objects of the benchmark economy

| Model object | Value |
|---|---|
| *Model only* | |
| Share of capitalists who produce, $\bar{z}^{-\eta}$ | 0.896 |
| Investors' equity share $\theta$ | 2.256 |
| Investors' exposure $\vartheta = \theta\bar{\psi}$ | 0.273 |
| Average risky share of producers' equity $\bar{\psi}$ | 0.121 |
| Risky share of the marginal producer's equity $\psi(\bar{z})$ | 0.086 |
| Occupational wedge $\Delta$, $q(\bar{z}) = 1 + \Delta$ | 0.0037 |
| Risk premium $\lambda = \mathbb{E}[R^D]/R^{f*}$ | 1.0659 |
| Risk-premium term of the cutoff, $\ln \lambda$ | 0.0638 |
| Wedge term of the cutoff, $\ln\left(1 + \Delta R^{f*}/(r^* + \delta)\right)$ | 0.0440 |
| Lower edge of the interior region, $W_{\text{crit}}/W_{\text{pre}}$ | 0.272 |
| Sharpe ratio of the index | 0.187 |
| *Model only: saving* | |
| Saving rate at $W_{\text{pre}}$, mean of $S(H)$ over the shock | 0.917 |
| Expected cash on hand over wealth, $\mathbb{E}[H]/W_{\text{pre}}$ | 1.090 |
| Discount factor $\beta$ | 0.905 |

*Notes:* Model objects of the equilibrium at pre-opening wealth $W_{\text{pre}}$. The saving rate is the mean of $S(H)$ over the shock at $W_{\text{pre}}$. Since $W_{\text{pre}}$ is a conditional-mean steady state, capitalists there consume the expected net return on wealth, which is $\mathbb{E}[H]/W_{\text{pre}} - 1$.

## D.4 Sensitivity of the calibration

Table D.4 compares the benchmark with six alternative calibrations. We change one input or measurement convention at a time and re-solve (49) and report the resulting parameters and the impact of liberalization on TFP and the risk premium.

The first two exercises revisit the risk premium and bond-position mappings in Section 6.1. To adjust for listed firms' leverage, we multiply each country's risk premium by its firms' equity share of total market value. The median adjusted risk premium is 3.8 percent, and this lowers the calibrated $\gamma$ to 1.52, and liberalization reduces TFP by 0.12 percent on impact. Since the model has no government, the benchmark attributes the country's total bond holdings to investors. If we target the private net safe position of +4.8 percent of output instead, the calibrated investors' equity holdings account for a 91% share of their wealth, and their exposure to aggregate risk falls from 27.3% to 10.7%. Matching the unchanged premium of 6.75 percent raises $\gamma$ to 10.8 and lowers $\beta$ to 0.796. Liberalization then reduces the premium by 1.21 percentage points, lowers the cutoff by 0.56 percent, and lowers TFP by 0.19 percent. The risk-premium term accounts for 0.48 percentage points of the cutoff decline.

With $\eta = 2.5$ and $\eta = 4$, the damping factor $1/(1 + e_D)$ in (35) is 0.50 and 0.33, respectively, compared with 0.43 in the benchmark. The corresponding impact TFP declines are 0.16 and 0.12 percent, while the capital-output ratios are 2.07 and 2.68, compared to the empirical value of 2.47. With $\delta = 0.07$, the cutoff falls by 0.40 percent and TFP by 0.135 percent.

The last calibration removes the outsider-share adjustment from both financing-variable proxies: $\xi$ is computed using private credit plus all listed book equity excluding direct investment, while $\omega$

is computed using foreign portfolio equity divided by total market capitalization. As a result, $\omega$ rises from 0.036 to 0.117 after liberalization. The opening transfers a fraction 0.085 of domestic investors' initial outside-finance holdings to foreigners, compared with 0.171 in the benchmark, and reduces TFP by 0.08 percent.

Across the seven calibrations, liberalization lowers TFP by 0.08–0.19 percent on impact and 0.10–0.29 percent in the long run. The smallest declines occur without the outsider-share adjustment and the largest with the private bond-position target. A one-percentage-point reduction in the world risk-free rate lowers TFP on impact by between 0.27 and 0.87 percent.

Table D.4: Sensitivity of the calibration and the experiments

| | Benchmark | Unlevered premium $\mathbb{E}[R^D] - R^{f*} = 3.8$ percent, $(D/P + g - r^*) \times E/(E+D)$ | Private bond position net safe position 0.048 of output, ten countries | $\eta = 2.5$ | $\eta = 4$ | $\delta = 0.07$ | No outsider share, either side $\xi$ from credit and listed book; $\omega$ over market cap |
|---|---|---|---|---|---|---|---|
| Volatility of the shock $\sigma$ | 0.965 | 0.965 | 0.965 | 0.965 | 0.965 | 0.965 | 0.965 |
| Financing capacity $\xi$ | 0.290 | 0.290 | 0.290 | 0.290 | 0.290 | 0.290 | 0.295 |
| Risk aversion $\gamma$ | 2.695 | 1.516 | 10.783 | 1.897 | 3.808 | 1.870 | 3.091 |
| Capitalist wealth $W$ | 3.893 | 4.234 | 4.891 | 3.339 | 4.536 | 3.170 | 4.007 |
| relative to expected output, $W/\mathbb{E}[Y]$ | 2.035 | 2.155 | 2.446 | 1.763 | 2.360 | 1.768 | 2.081 |
| Foreign share of outside finance $\omega$ | 0.094 | 0.094 | 0.094 | 0.094 | 0.094 | 0.094 | 0.036 |
| after the opening $\omega_{\text{post}}$ | 0.249 | 0.249 | 0.249 | 0.249 | 0.249 | 0.249 | 0.117 |
| Discount factor $\beta$ | 0.9053 | 0.9221 | 0.7961 | 0.8888 | 0.9239 | 0.9020 | 0.9029 |
| Investors' equity share $\theta$ | 2.256 | 2.123 | 0.913 | 2.712 | 1.952 | 2.701 | 2.050 |
| Investors' exposure $\vartheta$ | 0.273 | 0.252 | 0.107 | 0.359 | 0.213 | 0.366 | 0.247 |
| Marginal producer's risky share $\psi(\bar{z})$ | 0.086 | 0.084 | 0.083 | 0.088 | 0.085 | 0.097 | 0.086 |
| Wedge $\Delta$ | 0.00373 | 0.00169 | 0.00023 | 0.00551 | 0.00249 | 0.00534 | 0.00317 |
| Risk premium $\lambda = \mathbb{E}[R^D]/R^{f*}$ | 1.0659 | 1.0373 | 1.0659 | 1.0659 | 1.0659 | 1.0659 | 1.0659 |
| Capital / output $K/\mathbb{E}[Y]$ (untargeted; data 2.471) | 2.351 | 2.474 | 2.450 | 2.074 | 2.684 | 2.078 | 2.367 |
| Tilt of the continuation weights $c = \ln \mathbb{E}^{\Lambda}[X]$ | 0.0144 | 0.0036 | 0.0653 | 0.0077 | 0.0236 | 0.0081 | 0.0176 |
| *Opening, $\omega \to \omega_{\text{post}}$* | | | | | | | |
| 100 $\Delta \ln$ TFP, on impact | −0.148 | −0.117 | −0.188 | −0.163 | −0.118 | −0.135 | −0.079 |
| Cutoff, 100 $\Delta \ln \bar{z}$, on impact | −0.445 | −0.351 | −0.564 | −0.489 | −0.354 | −0.404 | −0.237 |
| risk-premium term | −0.167 | −0.143 | −0.476 | −0.145 | −0.164 | −0.130 | −0.096 |
| wedge term | −0.278 | −0.208 | −0.087 | −0.343 | −0.191 | −0.274 | −0.141 |
| Premium on operating profits, change (pp) | −0.42 | −0.35 | −1.21 | −0.32 | −0.54 | −0.33 | −0.24 |
| Producers, 100 $\Delta \ln$ | 1.34 | 1.05 | 1.69 | 1.22 | 1.42 | 1.21 | 0.71 |
| 100 $\Delta \ln$ TFP, long run | −0.182 | −0.166 | −0.294 | −0.186 | −0.159 | −0.158 | −0.102 |
| *World safe rate 1 pp lower* | | | | | | | |
| 100 $\Delta \ln$ TFP, on impact | −0.36 | −0.52 | −0.87 | −0.33 | −0.33 | −0.27 | −0.43 |
| *Interior region and cap* | | | | | | | |
| $W_{\text{crit}}/W$, before the opening | 0.272 | 0.405 | – | 0.392 | 0.179 | 0.424 | 0.231 |
| Cap check: smallest simulated $\lambda$ after the opening | 1.0040 | 1.0032 | 1.0068 | 1.0043 | 1.0033 | 1.0040 | 1.0051 |
| Grid guard: smallest $\lambda$ on the wealth grid | 1.0013 | 1.0010 | 1.0038 | 1.0013 | 1.0015 | 1.0013 | 1.0018 |

*Notes:* The first column reproduces the benchmark in Table 1; the remaining columns recalibrate $\sigma$, $\gamma$, $\beta$, and $W$ under alternative inputs or measurements. The immediate effect of liberalization fixes the wealth level at its calibrated pre-opening value; the long-run effects are evaluated at the post-opening conditional-mean steady state. TFP and cutoff changes are 100 times log changes, while premium changes are in percentage points. If $W < W_{\text{crit}}$, investors' exposure equals one and we do not have an interior equilibrium; $W_{\text{crit}}/W$ expresses this threshold relative to calibrated pre-opening wealth. The dash in the private-bond column indicates that no transition from exposure one to interior exposure was detected within the wealth grid. The grid guard row reports the minimum $\lambda$ across wealth nodes in the pre-opening economy, the post-opening economy, and the economy with the world rate one percentage point lower. Note that an interior equilibrium also requires $\lambda > 1$. The cap check row reports the minimum $\lambda$ along simulated post-opening paths. Values of at least one indicate that foreign investors are willing to fill the ownership cap.

### D.5 Decomposing the effects of liberalization

Starting from the calibrated pre-opening economy, we increase $\omega$ (foreign risk sharing), $\xi$ (financing capacity), or both. We measure their effects on impact, along each experiment's conditional-mean wealth path, or at its new conditional-mean steady state. The outcomes of interest are 100 times log changes for TFP, output, capital, wealth, and the number of producers, and percentage-point changes of the risk premium. For a given outcome and horizon, let $D_\omega$, $D_\xi$, and $D_{\omega\xi}$ denote the effects of the two separate changes and their joint effect. The Shapley decomposition assigns each change its separate effect plus half the interaction, $I = D_{\omega\xi} - D_\omega - D_\xi$:

$$\Phi_\omega = D_\omega + \tfrac{1}{2}I, \qquad \Phi_\xi = D_\xi + \tfrac{1}{2}I, \qquad \Phi_\omega + \Phi_\xi = D_{\omega\xi}. \tag{89}$$

We distinguish the immediate effects of foreign risk sharing and financing capacity from the subsequent effects of wealth adjustment. We describe the computed equilibrium responses and use the decomposition specified in (35). With $R^{f*}$, $\delta$, $\alpha$, $\eta$, and the shock distribution unchanged, let $u = R^{f*} - 1 + \delta$. The decomposition can be written as

$$(1 + e_D)\Delta \ln \bar{z} = \Delta \ln \lambda + \Delta \ln\left(1 + \frac{R^{f*}\Delta}{u}\right) + (1 - \alpha)\Delta \ln\big[(1 + \xi)W\big], \tag{90}$$

where $e_D = (\eta - 1)(1 - \alpha)$. Dividing the three terms by $1 + e_D$ gives the contributions of three components to the cutoff change: the risk premium, occupational wedge, and capital deepening. The corresponding TFP change is $\Delta \ln \text{TFP} = \alpha \Delta \ln \bar{z}$.

**Immediate effects of foreign risk sharing.** Holding equity supply fixed, the increase in foreign ownership would transfer the fraction $(\omega_{\text{post}} - \omega_{\text{pre}})/(1 - \omega_{\text{pre}}) \simeq 0.17$ of domestic investors' initial equity holdings to foreigners. In equilibrium, investors' equity holdings fall from 2.26 to 2.14 times their wealth, while their exposure $\vartheta$ falls by 5.6 percent. Entry increases the equity supply and offsets part of the foreign absorption. The risk-premium factor $\lambda$ falls from 1.0659 to 1.0617, and the occupational wedge $\Delta$ falls from 0.00373 to 0.00318. With a damping factor $1/(1 + e_D) \simeq 0.43$, the risk-premium and wedge terms contribute declines of 0.17 and 0.28 percent to the cutoff, respectively, yielding a total cutoff decline of 0.45 percent and a TFP decline of 0.15 percent.

Before liberalization, investors' risk exposure is 0.273, which is larger than the marginal producer's risky share of 0.086. Liberalization narrows this gap and lowers the wedge. Although the wedge is small in valuations, it enters the cutoff condition relative to the user cost of capital, 8.5 percent, making its contribution to selection appreciable.

**Immediate effects of financing capacity.** The financing expansion raises capital per producer, $(1 + \xi)W_{\text{pre}}$, by 6.83 log points. The capital deepening raises the cutoff by 1.95 percent, while the risk-premium and wedge lower it by 0.20 and 0.42 percent. TFP rises by 0.45 percent and the number of producers falls by 4.01 percent. The risk premium falls from 1.0659 to 1.0610 as

exiting producers become investors and offset the increase in financing per remaining producer. Table D.5 reports the immediate responses and the exact decomposition of the cutoff change for four experiments: increased foreign risk sharing alone, expanded financing capacity alone, both changes together, and a financing expansion chosen to match the premium reduction generated by increased foreign risk sharing.

Table D.5: Entry, scale, and components of cutoff changes

| | Producers | Capital per | Contribution to $100\,\Delta \ln \bar{z}$ | | |
|---|---|---|---|---|---|
| | (%) | producer (%) | Risk premium | Wedge | Capital deepening |
| Opening | 1.34 | 0.00 | −0.17 | −0.28 | 0.00 |
| Financing expansion | −4.01 | 6.83 | −0.20 | −0.42 | 1.95 |
| Both changes | −2.48 | 6.83 | −0.41 | −0.72 | 1.95 |
| Financing expansion matched to the opening's premium fall | −3.33 | 5.72 | −0.17 | −0.36 | 1.64 |

*Notes:* The experiments of Table 3, evaluated on impact. Producer numbers and capital per producer, $(1+\xi)W$, are reported as 100 times log changes. The three contributions are the terms on the right-hand side of (90), divided by $1 + e_D$ and multiplied by 100. They sum to $100\Delta \ln \bar{z}$, or the TFP change in Table 3 divided by $\alpha$, up to rounding. The safe-rate contribution is zero.

**Alternative measures of the financing expansion.** Table D.6 applies the same opening magnitudes of $\omega$ but uses alternative measurements for financing expansions. The benchmark financing capacity change is measured as the median of country changes, $100\Delta \ln(1 + \xi) = 6.83$, as in Table 3. The first alternative takes the median financing level at each date. The next two alternatives measure without the outsider-share adjustment and count all listed book equity as outside finance, without and with direct investment, respectively. The last measure uses private credit alone.

Table D.6: Financing measures and the attribution of the joint impact effect

| | $100\Delta \ln(1 + \xi)$ | $100\Delta \ln \text{TFP}$ | | |
|---|---|---|---|---|
| | | $\omega$ part | $\xi$ part | Total |
| Benchmark | 6.83 | −0.16 | 0.43 | 0.28 |
| Change in the median level | 6.12 | −0.16 | 0.39 | 0.23 |
| No outsider-share adjustment, no FDI | 9.57 | −0.16 | 0.63 | 0.46 |
| No outsider-share adjustment, with FDI | 10.27 | −0.16 | 0.67 | 0.51 |
| Credit only | 1.70 | −0.15 | 0.10 | −0.05 |
| 14 openers | 6.83 | −0.16 | 0.43 | 0.28 |

*Notes:* All effects are evaluated at the calibrated pre-opening wealth level, with each alternative using its own continuation weights. The benchmark measure includes private credit, the outsider share of listed book equity, and direct investment. Alternative measures are defined in the text. The $\omega$ and $\xi$ contributions follow (89).

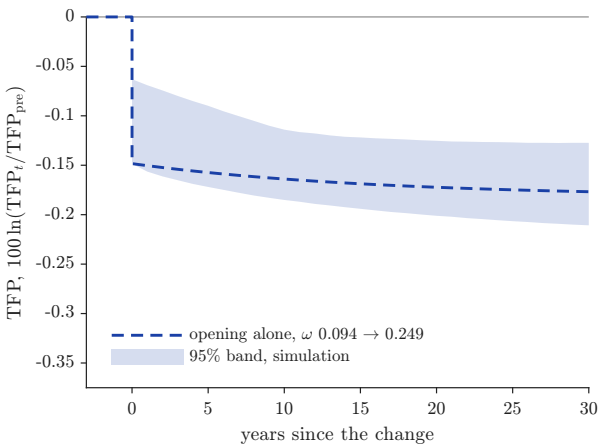
**Constructing transition paths and simulation bands.** For each experiment, we construct a conditional-mean wealth path by starting at $w_0 = W_{\text{pre}}$ and iterating

$$w_{t+1} = m(w_t), \qquad m(w) = \mathbb{E}[W_{t+1} \mid W_t = w]. \tag{91}$$

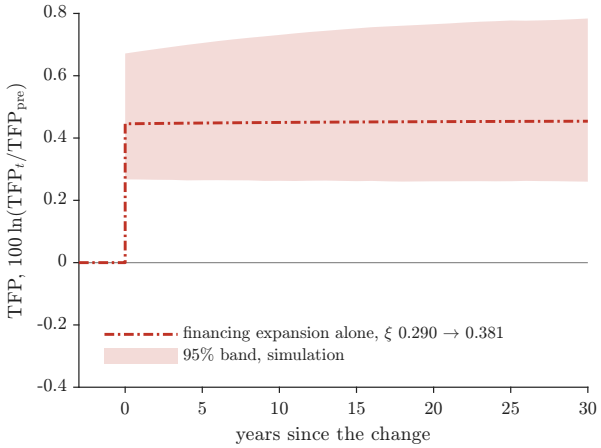
We evaluate $m$ from the wealth transition (41) in that environment, using the full distribution of the aggregate shock at each wealth level. Each experiment follows its own wealth dynamic, and its long-run outcome is evaluated at its own conditional-mean steady state. This deterministic sequence generally differs from the mean wealth path across stochastic histories since the wealth transition is nonlinear.

The shaded bands in Figure 8 and the figures below use 20,000 paired treatment and control histories with common shocks. Each history runs for 1,000 years at the pre-opening parameters before the experiments. Treatment and control therefore start from the same wealth within each pair, while initial wealth differs across pairs; the standard deviation of log wealth at year 0 is 0.15. At each date, the bands report the 2.5th and 97.5th percentiles of the treated-minus-control effects. They describe variation across shock histories conditional on calibrated parameters. Note that effects along the conditional-mean path do not equal the mean effects across simulated histories due to the nonlinearity of wealth dynamics.

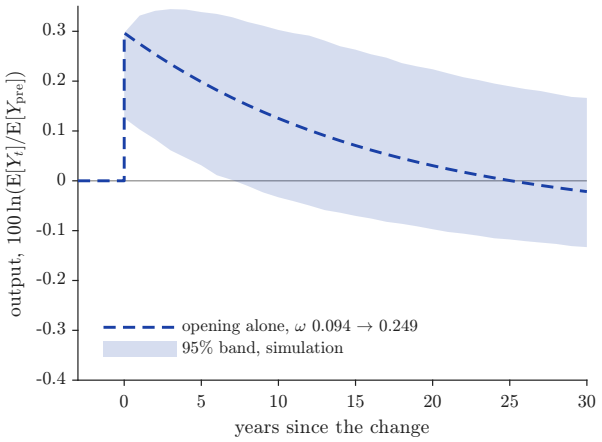
**Variation across shock histories.** The central 95 percent of simulated impact TFP effects ranges from $-0.15$ to $-0.06$ percent for the opening and from 0.27 to 0.67 percent for the financing expansion. These distributions reflect the different initial wealth levels generated by pre-opening shock histories. Figures D.1, D.2, and D.3 show the subsequent responses of productivity, output, wealth, producer numbers, prices, and capital.



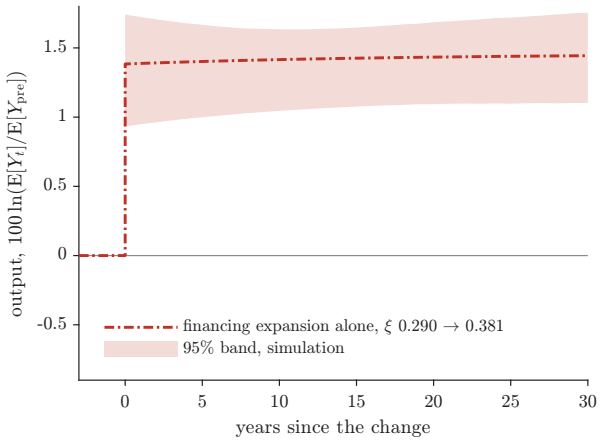

(a) TFP, the opening alone




(b) TFP, the financing expansion alone



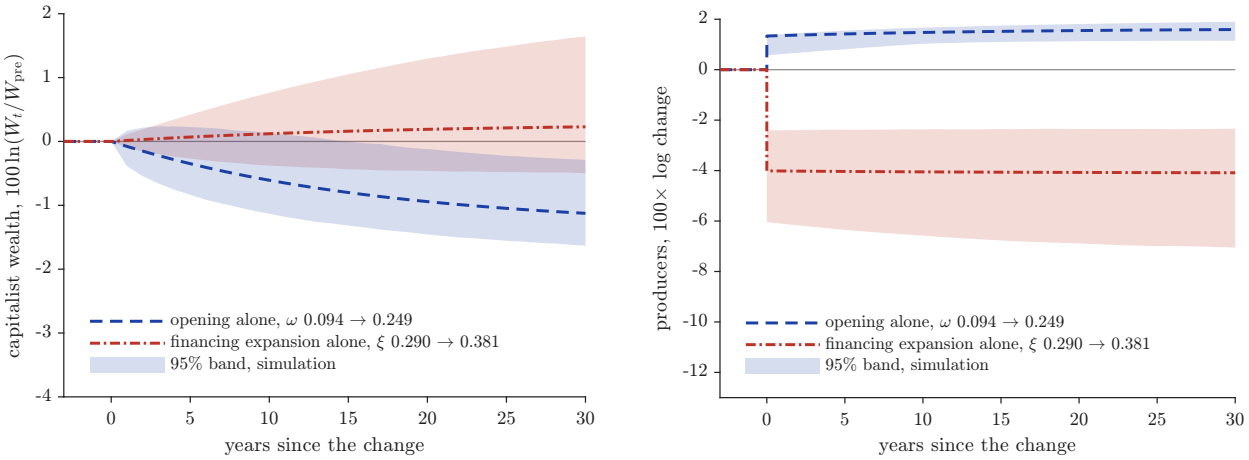

(c) Expected output, the opening alone




(d) Expected output, the financing expansion alone

Figure D.1: Each change on its own, over time

*Notes:* Conditional-mean paths and simulated bands are constructed as described in the text and Figure 8. Each panel shows one change on its own wealth path. For TFP and expected output, lines report 100 times log changes from the pre-opening conditional-mean steady state, while bands report percentiles of paired treatment-minus-control log differences multiplied by 100. Left column: the opening alone, $\omega$ from 0.094 to 0.249. Right column: the financing expansion alone, $\xi$ from 0.290 to 0.381.

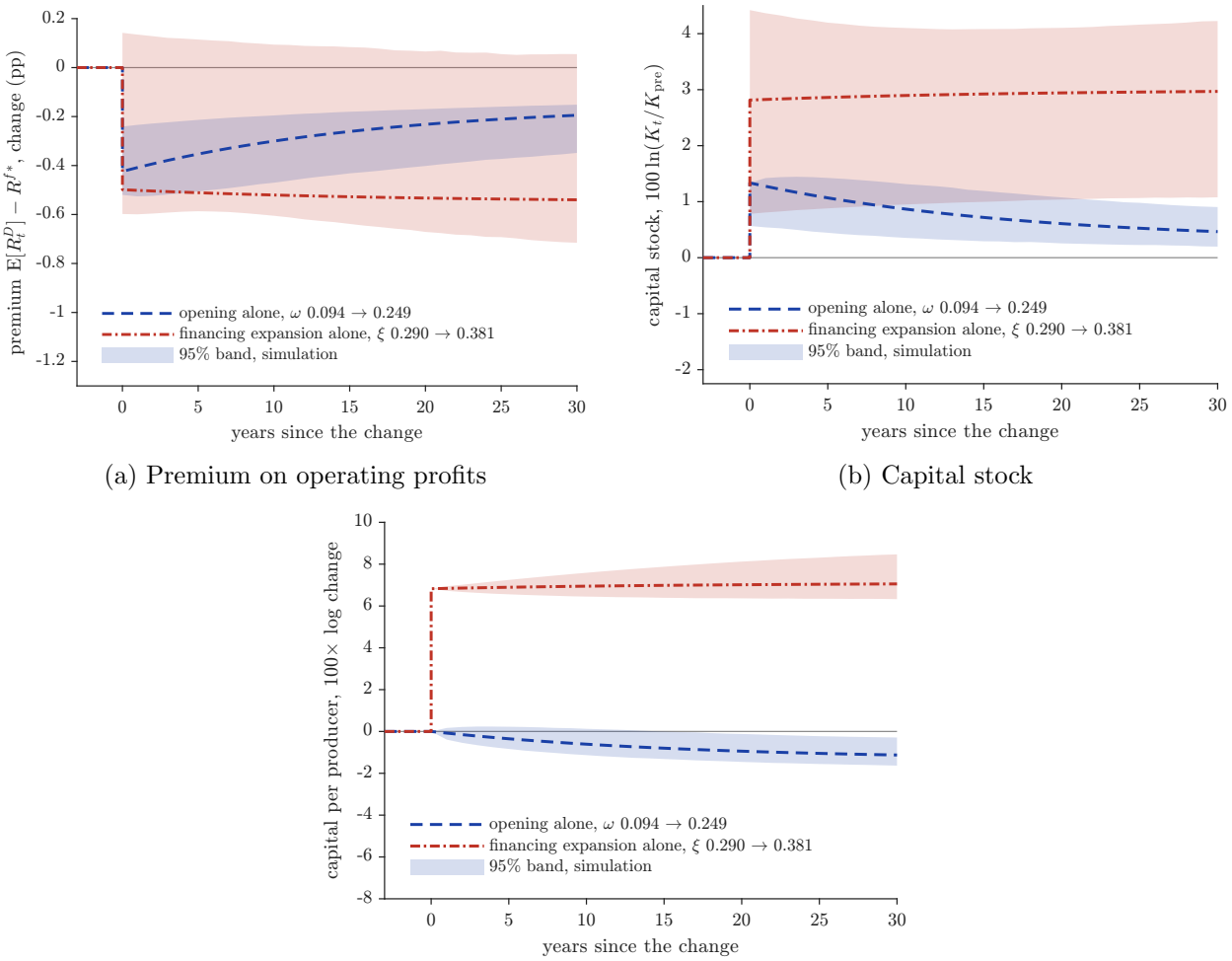



(a) Capitalist wealth

(b) Producers

Figure D.2: Wealth and the number of producers, one change at a time

*Notes:* Paths and bands follow Figure D.1. Dashed lines show the opening alone; dash-dot lines show the financing expansion alone. Capitalist wealth and producer numbers are expressed in 100 times log-change units, with the line and band reference points defined in Figure D.1.




(a) Premium on operating profits

(b) Capital stock

(c) Capital per producer

Figure D.3: Prices, capital, and scale, one change at a time

*Notes:* Paths, bands, and line styles follow Figure D.2. The premium is the change in $\mathbb{E}[R^D] - R^{f*}$ in percentage points. The capital stock and capital per producer, $(1+\xi)W_t$, are expressed in 100 times log-change units, with the line and band reference points defined in Figure D.1.

**Wealth adjustment after foreign risk sharing.** Along the conditional-mean path in Figure D.2(a), capitalist wealth converges to 1.35 percent below its pre-opening level. As wealth declines, the risk premium in Figure D.3(a) recovers most of its initial fall. On impact, the risk-premium channel accounts for a 0.17 percent decline in the cutoff, and the wedge channel accounts for another 0.28 percent decline. In the long run, the risk-premium channel contributes a decline of only 0.06 percent, while the wedge channel contributes a decline of 0.10 percent. Wealth adjustment contributes a further cutoff decline of 0.38 percent through capital deepening. The total long-run cutoff decline is 0.55 percent, as in Figure D.1(a). The capital stock in Figure D.3(b) remains 0.29 percent above its pre-opening value, compared to 1.34 percent on impact.

**Wealth adjustment after the financing expansion.** Capitalist wealth rises by 0.28 percent along the conditional-mean path, which is the dash-dot line in Figure D.2(a). The capital-deepening

channel contributes a 1.95 percent increase in the cutoff on impact and a 2.03 percent increase in the long run. In the long run, the risk-premium channel lowers the cutoff by 0.22 percent and the wedge channel lowers it by 0.45 percent. The net long-run effect is a 1.37 percent increase in the cutoff and a 0.46 percent increase in TFP (Figure D.1(b)). The capital stock in Figure D.3(b) is 3.00 percent above its pre-opening level.

**Joint effects over time.** With both changes (Figure 8), the expected output is 1.36 percent higher in the long run. The TFP increase contributes 0.23 percentage points of the output gain, and the remaining increase in output reflects a capital stock increase of 3.41 percent above its pre-opening level. Table D.7 reports the Shapley decomposition along the transition path. The interaction becomes more negative as wealth adjusts: in the long run, the opening's contribution is −0.21 percent, compared with its individual effect of −0.18 percent, while the financing contribution is 0.43 percent, compared with its individual effect of 0.46 percent.

Table D.7: Decomposition of the joint TFP effect along the wealth transition

| | $100\Delta \ln \text{TFP}$ | | | $W_t/W_{\text{pre}}$ |
|---|---|---|---|---|
| Year | $\omega$ part | $\xi$ part | Total | Both changes |
| 0 | −0.16 | 0.43 | 0.28 | 1.000 |
| 4 | −0.17 | 0.43 | 0.27 | 0.997 |
| 10 | −0.18 | 0.43 | 0.25 | 0.994 |
| 30 | −0.20 | 0.43 | 0.23 | 0.989 |
| Long run | −0.21 | 0.43 | 0.23 | 0.986 |

*Notes:* Contributions follow (89), with each experiment evaluated on its own conditional-mean wealth path from $W_{\text{pre}}$. Long-run effects use the corresponding post-intervention conditional-mean steady states. The last column reports wealth under both changes. Parts and totals are rounded separately.